

Income-Based Fines: Evaluating Rationales for Finnish Speeding Penalties

Aaron Payne^{*}(*Job Market Paper*) & Martti Kaila[†]

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Abstract

We evaluate rationales for income-based speeding fines. Such fines are famously applied in Finland, where a \$250 offense for a low-income speeder can cost \$100,000 for the rich. We consider four potential policy goals—externality mitigation, redistribution, equal compliance, and proportional punishment—using linked Finnish administrative data and an original survey. First, using a “job-loss” design, we find that marginal speeding costs *decrease* with offender income, leading us to reject externality mitigation as a rationale. Next, we assess the redistributive rationale. In standard models, indirect taxes aid redistribution only if the taxed action serves as a signal for earnings ability. We recover this signal by differencing speeding behavior with respect to causal effects of income on speeding, estimated from within-individual earnings variation and inheritance shocks. The estimated signal is *negatively* correlated with earnings, implying that Mirrleesian redistribution rationalizes a *lower* fine on the rich. Finally, we use our survey to assess whether preferences for equal compliance or proportional punishment are motivating rationales. Respondents trade off fixed and income-based speeding fine policies, where the latter vary in the induced across-income distribution of behavior and schedule “steepness.” Net of redistribution, respondents are on-average willing to forgo €144 million in government revenue to implement income-based policies. However, these valuations are insensitive to induced behavior or schedule-steepness, ruling out our candidate rationales. Given our findings, speeding fines should incorporate *some* income-dependence but need not feature the extremes in Finland. Residual support for income-dependence suggests that current policy reflects either unmodeled instrumental considerations or direct, non-instrumental preferences.

^{*}The Wharton School at the University of Pennsylvania; E-mail: aapayne@wharton.upenn.edu; corresponding author.

[†]Adam Smith Business School at the University of Glasgow; E-mail: martti.kaila@glasgow.ac.uk

1 Introduction

Governments use punishments to deter anti-social behavior, be it tax evasion, littering, or murder. In developed countries, the majority of offenses committed are punished with a fine. However, despite their ubiquity, there is substantial heterogeneity in how fines are determined. While fines commonly scale with the severity of the offense, several countries (e.g. Germany, Switzerland, Denmark, Sweden, and Finland) also scale some fines with income. In Finland, the oldest and most comprehensive income-based system, the rich can receive fines in excess of €100,000 for speeding offenses that would cost a few hundred euros for low-income drivers.¹

How can we rationalize income-based fine systems? More broadly, are they effective in achieving their economic and/or policy goals? In this paper, we present the leading economic and fairness-based rationales that have been put forth for these policies and test whether any of these motives can explain the Finnish speeding-fine system.

On the economic side, two frameworks are readily applicable to speeding fines. First, we consider the optimal penalty framework in Becker (1968), which prescribes that fines be set equal to the marginal social cost of crime. In this framework, income-based speeding fines can be rationalized if the marginal social cost of speeding increases with income. Second, we apply a Mirrleesian optimal redistribution framework to our setting. In this framework, the planner’s objective is to redistribute resources from high- to low-ability types, but the un-observability of type means the planner must turn to distortionary taxation. In the canonical Atkinson-Stiglitz result (Atkinson and Stiglitz, 1976), it is most efficient for the planner to use direct income taxes, rather than indirect commodity taxes, to redistribute resources. However, subsequent work (Saez, 2002a; Allcott et al., 2019; Ferey et al., 2024) shows that indirect taxes can lower the efficiency costs of redis-

¹For example, Andres Wiklöf was fined €121,000 for speeding in 2023; in 2002, Nokia executive Anssi Vanjoki received a speeding fine of €116,000.

tribution if the taxed behavior is a signal of earnings ability. Building on these results, we show that speeding fines should increase with income if the signaling value of speeding behavior increases with income, giving us a testable benchmark against which to assess redistributive rationales for income-based fines.

Fairness rationales have also featured prominently in the debate surrounding income-based fines. For example, official argumentation in the lead up to the implementation of Finland’s policy cited a desire to equalize the “impact” of fines on people of varying means (Lahti, 2021). This evidence suggests a deterrence objective, consistent with scholarly work citing general deterrence across incomes as an attractive theoretical basis for income-based systems (Kantorowicz-Reznichenko, 2013); however, it is unclear what formal notion of deterrence underlies these motivations. As we show in section 3, with log-utility over consumption, income-proportional fines simultaneously achieve two outcomes: equal marginal costs of speeding across incomes *and* equal speeding rates across incomes. Given that income-dependent speeding fines in Finland are both proportional to net-of-tax income *and* associated with across-income equality in speeding rates, we test whether the equal distribution of behavior (hereafter, “equal compliance”) or the proportional schedule itself is the instrumental objective of income-based fines.

To empirically evaluate each of these four rationales, we draw on linked Finnish administrative data and construct an original survey that we administer to a representative sample of Finnish residents.

To evaluate the Beckerian rationale, we measure the slope of marginal social speeding costs with respect to income using the injuries that result from accidents, as collected in accident report data. Linking this information to data from income-tax returns and crime histories, we leverage cross-sectional variation in rates of speeding-ticket receipt conditional on income to measure the marginal impact of speeding on accident-related

injuries across incomes. With respect to injuries, we find the slope of marginal speeding costs is *negative*: the marginal impact of speeding on accident-related injuries is *lower* at high incomes. Drawing on additional data, we show that this result is robust to controlling for non-speeding traffic-crime rates, car ownership status, annual driving distance (measured using odometer data), and vehicle mass. Evidence from the accident reports suggests that across-income differences in traffic density while driving explain our results. We translate this slope into a money-metric by measuring the cost of each injury using a “job-loss” design (von Wachter and Bender, 2006); this approach allows us to capture across-victim heterogeneity in costs and to separate fiscal externalities from net-income losses, which may be weighed differently in the planner’s objective. In particular, we compare the earnings trajectories of individuals injured in an accident to those of demographically-matched individuals from the general population to estimate the net-present-value (NPV) cost of each injury. We then apply welfare weights (Saez, 2002b) to the NPV net-income losses and re-estimate the slope of marginal speeding costs with respect to income. Regardless of the discount rate we apply to future injury costs or how we welfare-weight net-income losses, we find a consistently *negative* slope. In our preferred estimates, we find that the marginal social cost of speeding decreases by €7.6 (se=4.4) for every 10,000 € of income, well below our estimate of the overall slope of the Finnish speeding fine schedule (+12.4 €). We therefore reject the externality-based rationale.

Turning to the redistributive rationale, we measure the extent to which speeding behavior is a signal of earnings ability. As clarified by Allcott et al. (2019) and Ferey et al. (2024), this signal is identified by differencing observed across-income variation in speeding rates with respect to causal effects of earnings on speeding, where the latter are identified from variation in *labor supply*. With weak separability between labor supply and consumption (Allcott et al., 2019; Ferey et al., 2024), the earnings effect can be identified using any exogenous earnings variation. We reject this assumption in our setting; the earnings effect

is the sum of a labor supply effect (holding earnings fixed) *and* an income effect (holding labor supply fixed). To measure the labor supply effect, we use a yearly panel of income and speeding ticket receipt and leverage within-individual variation in months-employed (holding earnings fixed). We estimate an average labor-supply effect that is about 25% as large as the across-income gradient of speeding ticket receipt. Due to mis-measured labor supply, the coefficient on earnings in this regression is likely upward-biased, so we turn to an inheritance-shock approach to estimate the income effect. Building on Nekoei and Seim (2022), we exploit the differential timing of first parental deaths as an exogenous income shock; to address violations of the exclusion restriction, we also difference with respect to inheritance size, yielding a triple-differences approach. From this approach, we estimate an income effect that is roughly equal to the average labor supply effect. We then difference observed speeding behavior with respect to our earnings effects and find that the resulting earnings ability signal is *negatively* correlated with income, implying that Mirrleesian redistributive motives rationalize a *lower* fine for the rich. In our preferred estimate, fines should *decrease* by €12 for every €10,000 of income. Across various assumptions about the planner’s redistributive preferences, we find consistently negative fine gradients, leading us to reject the redistribution-based rationale for income-based fines.

Finally, to evaluate preferences for equal compliance and proportional punishment, we turn to our survey. In particular, we ask Finnish residents to make trade-offs between hypothetical income-dependent and fixed-fine policies using government revenue (our willingness-to-pay medium). We vary three key elements of the policy comparison: the distribution of behavior under the income-dependent policy, the “steepness” of the income-dependent fine schedule, and the behavior change induced by income-dependent fines. We hold total speeding fixed across policies to address safety-related confounds; to address redistributive motives, we difference WTP for income-dependent fines with respect to WTP for income-dependent transfers. Net of redistribution, 56% of respon-

dents have a positive WTP for income-dependent fines, with respondents on-average willing to forgo €144 million in government revenue to implement an income-based policy. However, we find relatively small and insignificant responses to the three randomized fine-policy characteristics described above; if anything respondents express a slight *distaste* for equal behavior and proportional fine schedules. To assess the robustness of our findings, we explore heterogeneity with respect to other randomized characteristics of the policy comparison and respondents' background characteristics; respondents consistently value income-based fines but remain unresponsive to randomized features of the comparison.

In summary, our results suggest that commonly cited economic and fairness-based rationales for income-based fines are inadequate to account for observed policy. Nonetheless, we find that Finnish residents have a strong residual preference for income-dependent fines that is orders of magnitude larger than the losses resulting from inefficiencies in corrective pricing and redistribution. Therefore, given our findings, optimal fines should incorporate *some* income-dependence, but need not exhibit the extremes of the Finnish system. Furthermore, to account for the popularity of income-based policies requires an appeal to alternative unmodeled fairness considerations (e.g. procedural fairness) or a richer economic model that incorporates, e.g., liquidity concerns and/or income-correlated mis-perceptions.

After discussing related literature and contributions, our paper proceeds as follows. First, section 2 discusses our setting: speeding fines in Finland. We then turn to a theoretical framework (section 3), which clarifies the empirical objects we plan to measure for assessing each of the four rationales for income-dependent fines. We briefly discuss data (section 4) before testing the externality- (5) and redistribution-based (6) rationales. We then turn to our survey (7), which we use to assess the fairness-based rationales of equal compliance and proportional punishment. Section 8 assesses our results jointly,

and section 9 concludes.

1.1 Related Literature

This project speaks to three main literatures. First, there is an extensive literature that explores the redistributive potential of commodity taxes in Mirrleesian (Mirrlees, 1971) models where income taxes are present. The benchmark result, due to Atkinson and Stiglitz (1976), shows that, under preference homogeneity, commodity taxes are inefficient (relative to income taxation) for redistributing resources. Saez (2002a) clarified conditions under which commodity this result breaks down, and more recent work (Allcott et al., 2019; Ferey et al., 2024) shows how to identify the redistributive value of commodity taxes for optimal tax design using the difference between observed across-income consumption variation and causal effects of earnings on consumption. While we build methodologically on this literature, our paper differs in two key ways. First, empirically, recent work that identifies earnings effects (Allcott et al., 2019; Ferey et al., 2024) plausibly relies on weak separability assumptions between labor supply and consumption, implying that earnings effects can be identified using any exogenous income variation. However, we test for and reject weak separability in our setting; as a result, our empirical approach involves estimating both labor supply and income effects on consumption. Second, rather than seeking to identify optimal tax rates, as in Allcott et al. (2019) for soda or Ferey et al. (2024) for savings, we invert the approach, using the optimal redistributive speeding fine as a benchmark against which to test rationales for a given policy.

In relation to this inversion approach, our paper also relates to the “inverse optimum” literature (Christiansen and Jansen, 1978; Bourguignon and Spadaro, 2012; Bargain et al., 2014; Hendren, 2020), which uses analytical results from optimal tax theory and estimates of behavioral elasticities to infer society’s revealed redistributive preferences. This approach has the benefit of removing the problem of selecting normative criteria, treating this as an empirical inference exercise. However, such an approach leaves the source

of society’s preferences opaque and can lead to results that contradict conventionally held assumptions (Lockwood and Weinzierl, 2016), without little ability to diagnose the problem. Our paper follows this literature in spirit but seeks to open the black box of society’s revealed preferences, enabling us to assess their normative consistency.

Finally, this work relates to a theoretical and experimental literature on domain-specific fairness. On the theoretical side, specific egalitarianism (Tobin, 1970) argues that specific goods should be equally allocated; recent experimental evidence provides support for this view in the domains of health care and legal aid (Caspi et al., 2024). In our setting, the fact that high-income individuals consume more speeding (a good allocated by the justice system) may be viewed as a violation of specific egalitarianism. Furthermore, in the context of real-stakes lab games, a wealth of lab evidence has examined under what conditions individuals favor equal allocations, which has given rise to numerous fairness theories (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000; Charness and Rabin, 2002; Engelmann and Strobel, 2004). However, work that explicitly links elicited domain-specific concerns to observed policy remains limited. Our paper makes progress on this front by taking as its object a policy (income-based speeding fines) that explicitly achieves equality of behavior and eliciting preferences for such equality among individuals subjected to the policy. This approach bears similarities to Rafkin and Soltas (2025), which uses experimentally elicited social preferences among landlords and tenants to conduct counterfactual analysis of rental assistance policies.

2 Setting: Speeding Fines in Finland

We now emphasize key features of our empirical setting. We use such features to inform our model and the rationales for income-dependence that we consider.

2.1 Day Fine System in Finland

In Finland, except for very minor offenses (e.g. minor littering), all financial are punishments are income-based. Each crime is punished with a fixed number of “day units.” The value of each unit I depends on an individual’s net-of-tax income according to the following formula:

$$I = \frac{y_m - 255}{60} - 3 \times D \quad (1)$$

where y_m is monthly net-of-tax income and D is an individual’s number of dependents. Other than a €255 basic-living deduction, fines are proportional to half-a-day’s net income, hence the “day fine” terminology.

Around the time of the Finnish day fine system’s implementation in 1921, official argumentation cited a desire to reduce the number of crimes punished by prison; the income-dependent feature was meant to equalize the “impact” of fines on people with varying means (Lahti, 2021). However, it is ambiguous whether the “impacts” refer to behavior (e.g. equal deterrence) or financial outcomes. Furthermore, the criminal code explicitly references “ability to pay” in detailing the procedure for calculating income-based fines.² However, it is ambiguous as to whether this desire is borne out of a domain-specific version of vertical equity, or is in fact confounded by general redistributive preferences.

2.2 Speeding Fines in Finland

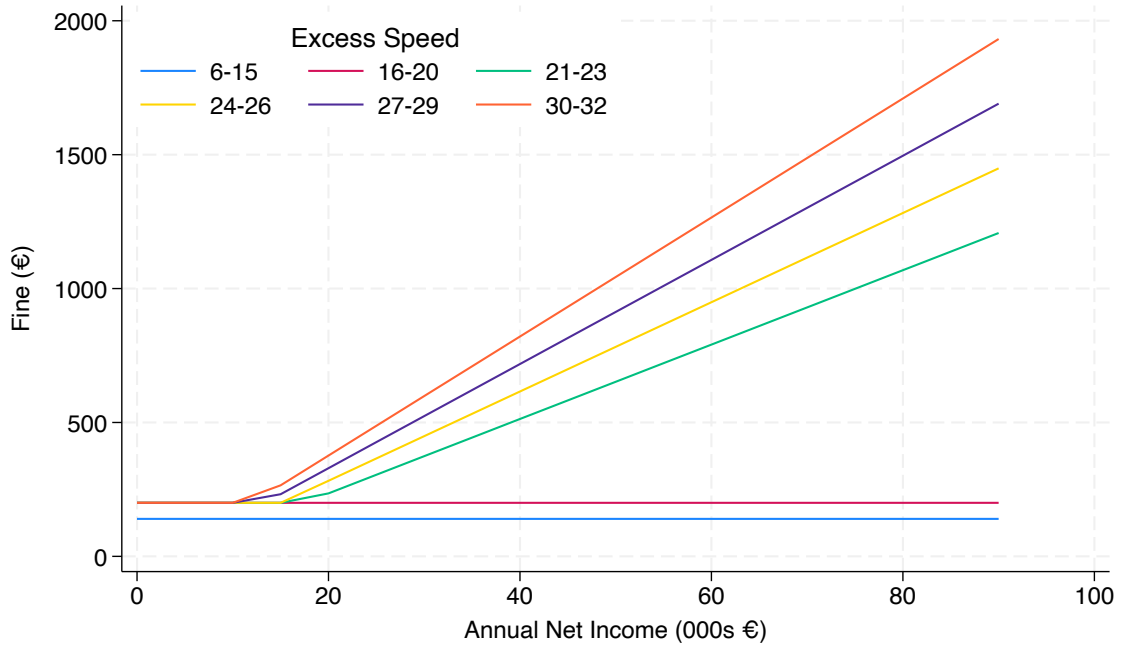
Not all speeding crime in Finland is subjected to income-based fines: drivers are given fixed fines of 140-200 euros for exceeding the limit by 7-20 km/h (minor traffic infractions) and receive income-dependent fines for exceeding the limit by more than 20 km/h.³ To visualize this schedule, figure 1 displays hypothetical speeding fines as a function of net-of-tax income for different levels of “excess speed” (*i.e.* the amount by which a person

²See statute 39/1889, chapter 2a section 2.

³Drivers are given grace for exceeding the limit by no more than 6 km/h.

exceeds the speed limit); we use a speed limit of 100 km/h to construct this figure.⁴ For “minor” speeding crime (7-20 km/h over the limit), fines are constant across incomes; however, for day-fine eligible speeds, fines are essentially proportional to net-of-tax income except for a fine-floor that holds at low incomes.⁵ The proportionality is shown explicitly in Appendix figure 18: above the fine floor, income-dependent fines constitute a fixed share of monthly net-of-tax income. We also note that there is no ceiling on the size of a fine in Finland; individuals can receive arbitrarily large fines at arbitrarily high incomes.

Figure 1: Hypothetical Fine Schedule (100 km/h limit)



Net-of-tax income consists of earned and capital income, as well as any transfers received, after the payment of taxes. We assume no dependents for the formula in equation 1.

Because of the income-dependent cutoff, there is a fine discontinuity at 121 km/h, the size of which grows with income.⁶ As shown in Kaila (2024), despite large discontinuities, individuals do not bunch, either in ticket data collected from speeding cameras or

⁴While the income-dependent cutoff (21 km/h over the limit) is constant across limits, a given excess speed will have larger fines at lower speed limits.

⁵This floor is inherited from the fixed-fine region of the fine schedule.

⁶In particular, holding income fixed in figure 1, the discontinuity is the difference between the red (“16-20”) line and the green (“21-23”) line.

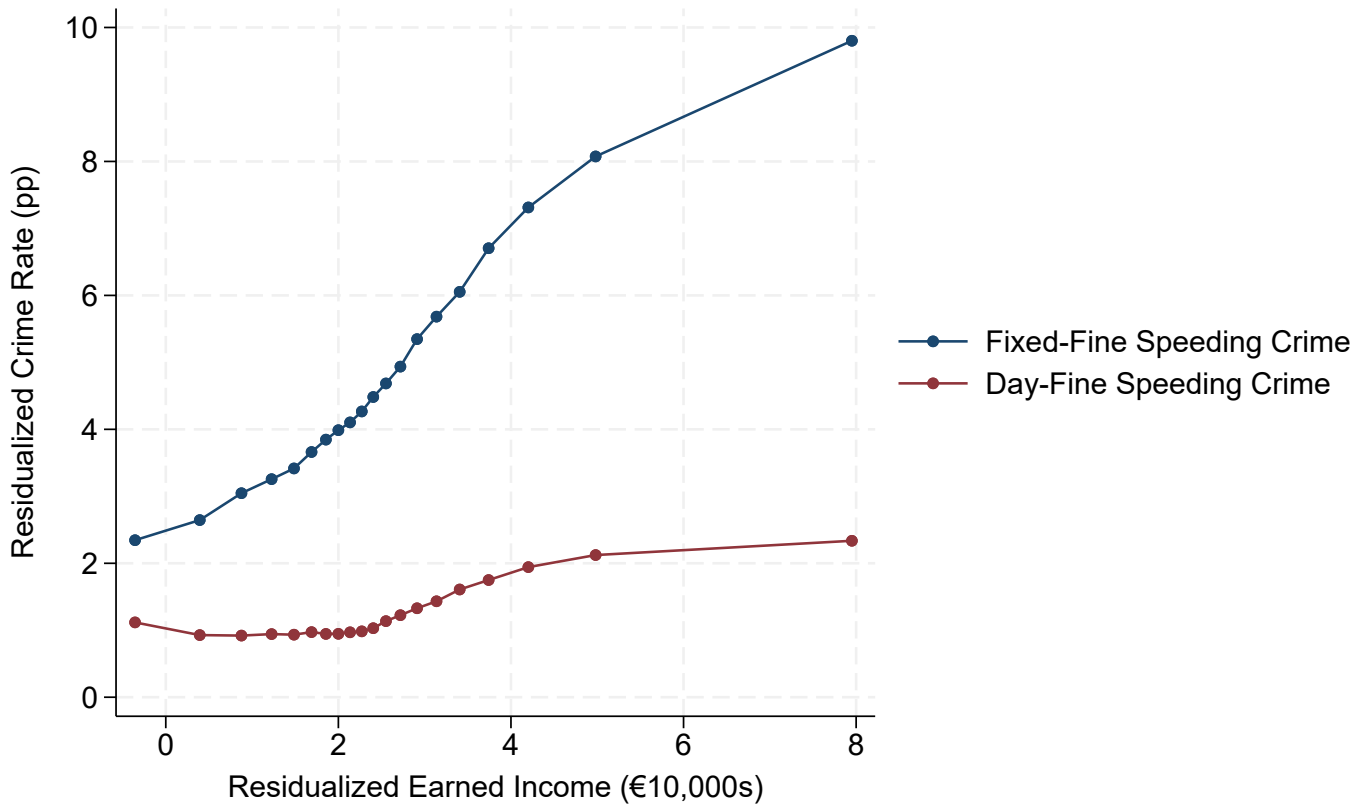
data from traffic monitoring stations. Furthermore, while Kaila (2024) shows using a regression-discontinuity-design that individuals who receive larger fines respond by *ex-post* reducing their likelihood of speeding (the extensive margin), there is no evidence that they respond on the *intensive-margin* (i.e. by lowering their speed to avoid the higher fine).

2.3 Relationship Between Income and Speeding

To motivate the potential to use speeding fines as a means for redistribution, figure 2 plots the relationship between income and the probability of receiving a speeding ticket in a given year using our crime-income panel described in section 4. We plot the relationship separately for the fixed- and day-fine regions of the fine schedule. We see a strong positive gradient for fixed-fine speeding crime: the fixed-fine speeding rate moves from around 2.5pp to 9pp as we move from the bottom to top earnings decile. While this relationship ostensibly provides a basis to use income-dependent speeding fines as a way to redistribute resources from high- to low-earners, such a possibility needs to be juxtaposed against the option of simply increasing income taxes on high earners, as we discuss in section 3.

We also use figure 2 to motivate key features of our survey and elicitation of fairness preferences. In particular, the day-fine region of the speeding-fine schedule is associated with fines that are proportional to net-of-tax income (equation 1) and a significantly compressed across-income distribution of behavior. While the latter is not necessarily causal, such an outcome could be an intended “impact” of income-based fines over which society may have fairness preferences. We disentangle the proportionality and equal-compliance channels using our survey (section 7).

Figure 2: Relationship Between Income and Crime



This figure shows the relationship between earnings and speeding crime rates using the entire population of Finnish adults from 2006-2019. We residualize using age-by-year fixed effects.

2.4 Typical Speeding Crime

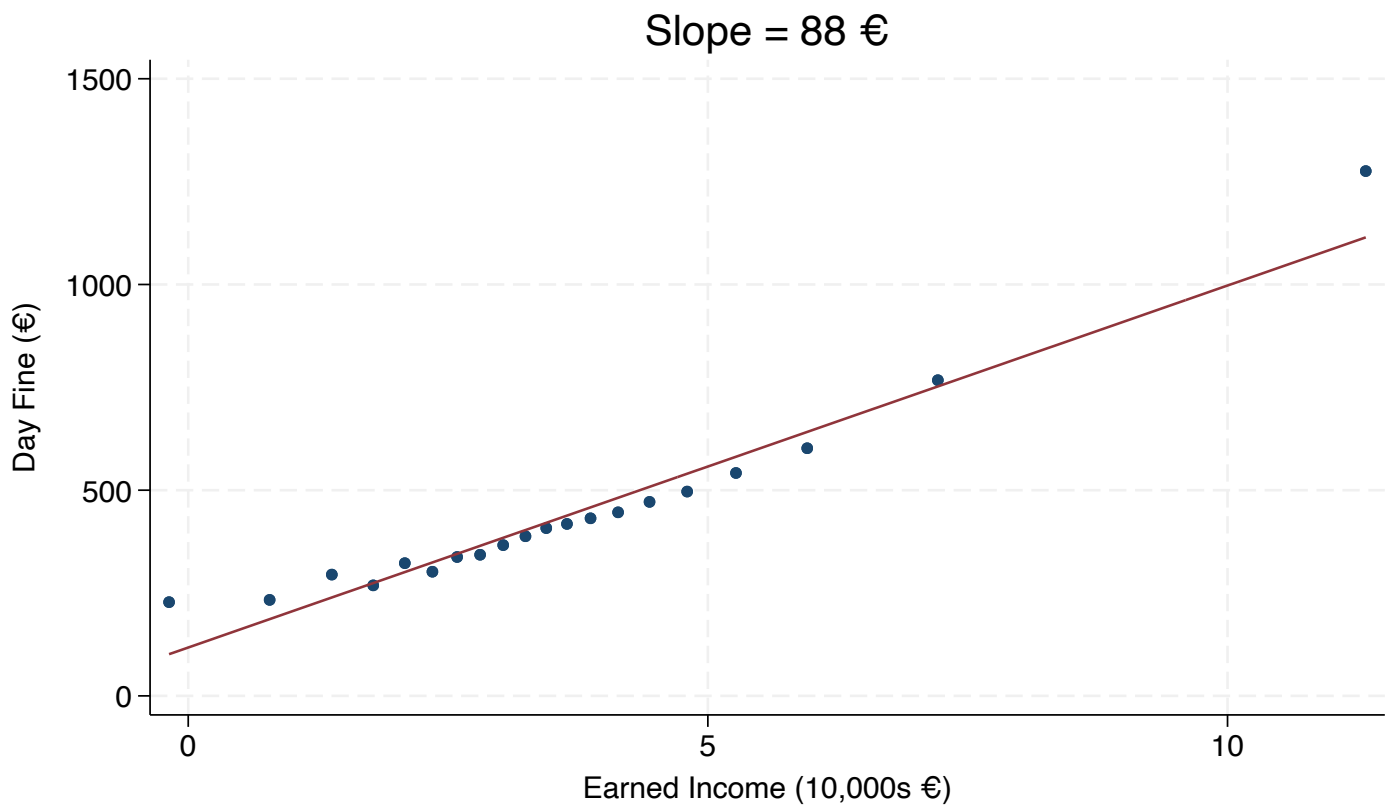
To simplify our analysis, we collapse all speeding into a single extensive-margin speeding crime. We justify this decision in two ways. First, there are no discernible intensive-margin responses to fine size (*i.e.* no bunching at fine discontinuities). Second, the typical speeding crime is similar across incomes. In particular, Appendix figure 19 shows, using our police ticketing data, that the average speed limit for a given speeding crime (panel ??) is uncorrelated with income. While there is a negative correlation between income and excess speed (panel ??), Appendix table 17 shows that most of this relationship can be explained by differential police-officer discretion across municipalities. We therefore view the typical speeding crime as approximately equivalent across incomes conditional on age and municipality. As a result, our key behavioral margin is the likelihood of ticket receipt.

To obtain a collapsed benchmark against which to test rationales for income-dependence, figure 3 uses our police speeding-ticket data to plot the relationship between earned income and fine size for all day-fine speeding crime, after residualizing on age.⁷ We measure an average income-gradient of 88 € per 10,000 € of earned income; the schedule exhibits some convexity at high incomes and is flatter at the bottom of the income distribution (reflecting the fine floor inherited from the fixed part of the schedule in figure 1). This slope represents a more stringent benchmark against which to test our hypothesized rationales.

However, we can also collapse all speeding crime (fixed- and day-fine regions) to obtain a weaker benchmark gradient, as shown in figure 4. For all speeding crime, the average fine increases by 12.4 € per 10,000 € of income. Even under this more modest gradient, the top income decile pays around €100 more on-average than the bottom income decile. We show that, even against this straw-man case, we cannot rationalize the Finnish speeding-

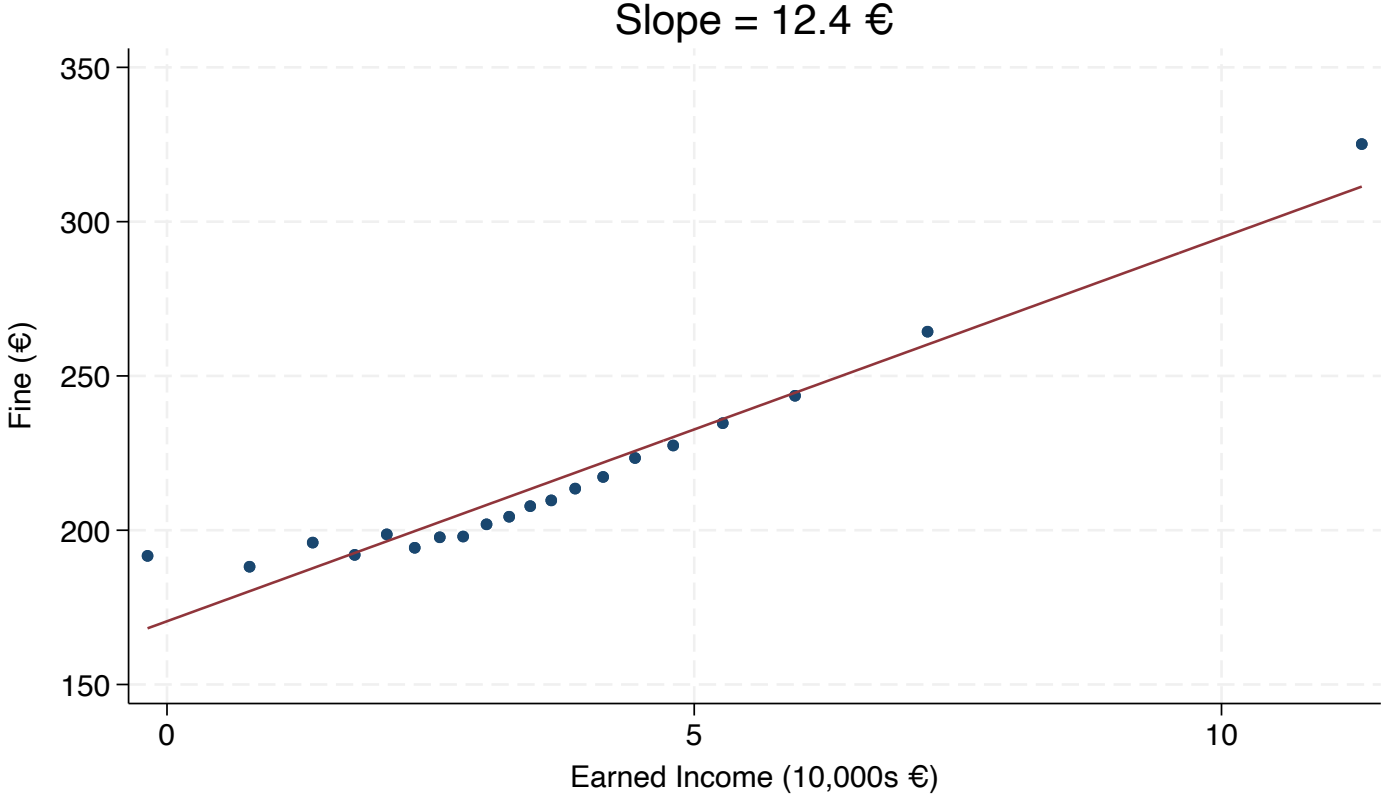
⁷While day fines in Finland are statutorily proportional to *after-tax* income (ignoring the fine floor, basic living deduction, and dependent deductions), they may not be proportional to *pre-tax* income due to the convexity of income taxes. However, other fine determinants can offset this convexity, including excess speed, speed limit, number of dependents, and income tax deductions.

Figure 3: Day Fine Slope



Binscatter based on police ticket data from 2018-2020. Income and the fine amount are residualized on age fixed-effects.

Figure 4: Fine Slope



Binscatter based on police ticket data from 2018-2020. Income and the fine amount are residualized on age fixed-effects.

fine system.

3 Theoretical Framework

In this section, we build a model of optimal fine-setting in the spirit of an optimal commodity tax framework (Atkinson and Stiglitz, 1976; Saez, 2002a; Kaplow, 2006; Allcott et al., 2019), which embeds a commodity tax decision within the redistributive income-tax framework of Mirrlees (1971). Our goal is to clarify the role of each rationale that we consider for income-dependence and set the stage for our empirical approach.

3.1 Setting

Individuals in the economy are indexed by a two-dimensional type $(\theta, \epsilon) \in (\Theta, E)$; θ indexes an individual's "earnings ability," *i.e.* their latent wage $w(\theta)$, while ϵ is a shock that governs the benefit of speeding.

There are two periods: in period 0, individuals make a labor supply choice $l(\theta)$ given their type and *ex-ante* uncertainty over ϵ ; this choice generates an earnings level $z(\theta) := w(\theta) * l(\theta)$ for each type. In period 1, ϵ is realized and individuals make an *extensive-margin* speeding decision given their period-0 earnings choice. We model speeding as the observed average speeding crime (*e.g.* driving 13 km/h over the limit). We justify this with our finding from section 2 that the "intensity" of speeding does not vary systematically with income and based on evidence from Kaila (2024) that there are no discernible intensive-margin responses to speeding fines. Our key behavioral margin is therefore the speeding rate at each earnings level.

In period 1, an individual with crime decision $s \in \{0, 1\}$, earnings z , earnings ability θ , disposable income c ,⁸ and shock ϵ receives *ex-post* utility $U(c, s * \epsilon, z; \theta)$. We assume that U is strictly increasing and concave in its first two arguments and strictly decreasing and concave in z . This assumption gives rise to a threshold $\epsilon^*(c, z; \theta)$ such that all type- θ z -earners with $\epsilon \geq \epsilon^*$ will choose to speed. We define $s(z; \theta)$ as the speeding rate for type θ (given earnings z). Given period 1 choices, individuals choose earnings $z(\theta)$ to maximize expected utility.

Taxes and Fines In the spirit of the Mirrlees model, the planner cannot observe individual types θ and therefore must levy taxes on choices z and s , giving rise to distortions.

We ignore any risk associated with speeding fines and assume that individuals are as-

⁸Disposable income is a mechanical function of s and z through the system of taxes and fines; we suppress this dependence for clarity of notation.

sessed fines based on their expected speeding rate $s(z; \theta)$. We justify this by noting that, unlike our one-shot static model, individuals in fact make many speeding decisions (e.g. every second they are driving). Therefore, by central-limit-theorem arguments, as the number of choice occasions grows large, the variance of fine payments will be negligible. Nonetheless, we highlight where risk will matter when we present our main results.

We consider a restricted system of taxes and fines that reflects our empirical setting: an individual earning z with speeding rate $s(z)$ pays income taxes $T(z)$ and speeding fines $\tau(z) * s(z) := (a + b * z) * s(z)$. When $b = 0$, fines are “fixed,” whereas $b > 0$ implies that fines scale with income. We refer to b as the “income-gradient” of the fine schedule.

Externalities Each individual’s speeding behavior imposes a cost on others; this cost can vary with the identity of the offender, e.g. because they interact with certain types of victims. We denote the dis-utility experienced by victims of type θ from others’ speeding as $E(\theta)$ and denote the imposed by offenders of type θ as $\bar{e}(\theta)$.⁹

Standard Planner’s Problem The planner chooses $T(\cdot)$, a , and b to solve:

$$\begin{aligned} \max_{T(\cdot), a, b} & \int_{\Theta} \alpha(\theta) G(V(\theta) - E(\theta)) dF_{\theta}(\theta) \\ \text{s.t.} & \int_{\Theta} [T(z(\theta)) + (a + b * z(\theta)) * s(z(\theta); \theta)] dF_{\theta}(\theta) \geq R \end{aligned} \tag{2}$$

where $G(\cdot)$ is a concave social welfare function, $\alpha(\cdot)$ is a Pareto-weighting function, and R is an exogenous revenue requirement. We make the following assumption.

Assumption 1. *At the optimum: (i) the nonlinear income tax $T(\cdot)$ is smooth; (ii) $z(\theta)$ is smooth and strictly increasing, and $s(z; \theta)$ is a smooth function of z and θ ; and (iii) individuals’ optima are unique and their first- and second-order conditions strictly hold.*

⁹We use overline notation to denote that the imposed cost is aggregated over victims.

Parts (i)-(iii) of assumption 1 allow us to use perturbation-based techniques to derive the optimal tax and fine schedule. Furthermore, this assumption allows us to express all heterogeneity with respect to z , rather than θ , at the optimum; in other words, the tax-and-fine system perfectly separates types. Therefore, in what follows, we suppress dependence on type.

3.2 Optimal Fines

Before discussing optimal fine schedules, we define several terms that appear in our formulas. First, the *social marginal welfare weight* for a z -earner $g(z)$ captures the social value of a marginal transfer to a z -earner at the optimum (relative to the value of public funds); these weights embed the planner's preferences for redistribution.¹⁰

We also define the following behavioral responses:

- Fine (semi-)elasticities: $s'_{\tau_s}(z) = \frac{\partial s(z)}{\partial \tau}$
- Causal earnings effects on speeding: $s'_{inc}(z) := \left. \frac{ds(z;\theta)}{dz} \right|_{\theta=\theta^{-1}(z)}$

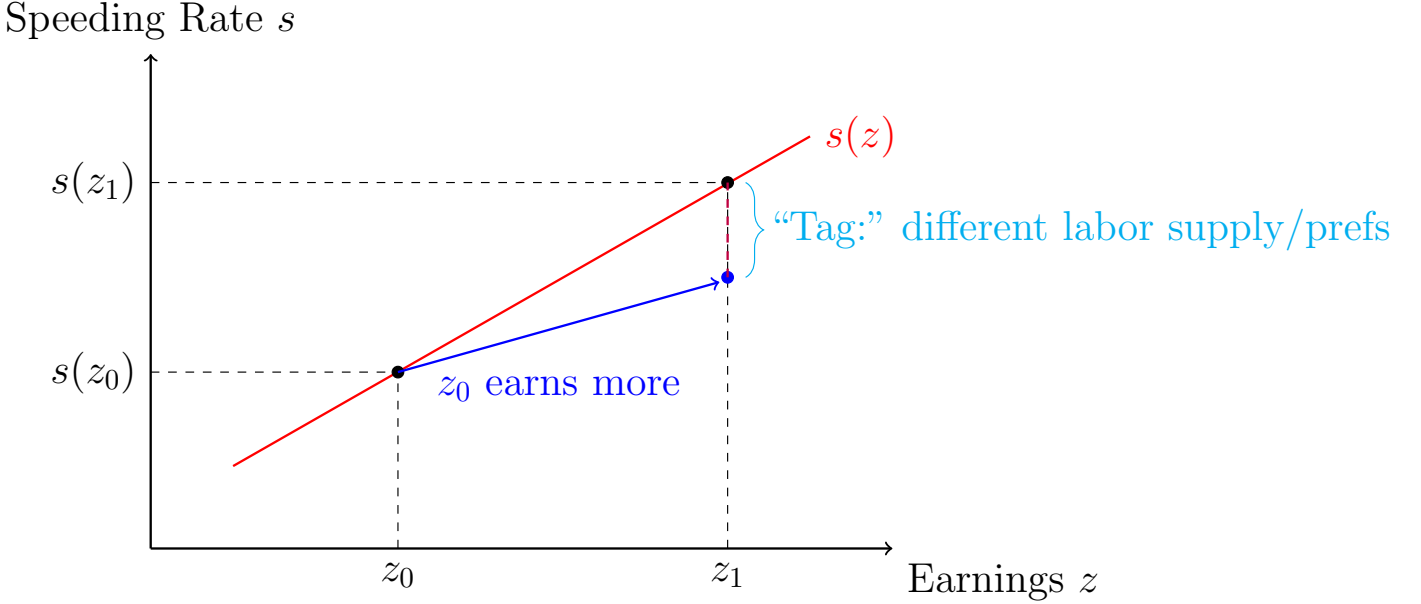
Causal earnings effects measure the change in speeding caused by a change in earnings *holding type fixed*; this is distinct from the *cross-sectional* slope $s'(z) := \frac{ds(z)}{dz}$, which measures the difference in observed speeding across individuals with different incomes. These earnings effects are crucial for evaluating the redistributive potential of income-based fines. The intuition is given in figure ??, which shows two adjacent earnings levels, z_0 and z_1 , and associated hypothetical speeding rates, $s(z_0)$ and $s(z_1)$. The key experiment we use to determine whether speeding behavior is (locally) informative about earnings ability is by inducing those earning z_0 to increase their labor supply and earn z_1 . As is shown in figure ??, despite having the same income z_1 , the two groups speed at different rates, reflecting differences in labor supply and/or preferences between the two

¹⁰Formally:

$$g(z) := \frac{\alpha(z)G'(V(z) - E(z))V'_c(z)}{\lambda}$$

where λ is the multiplier on the government budget constraint.

groups. Under assumption 1, z_1 -earners have a higher earnings ability θ than z_0 -earners; therefore, the difference in speeding rates reflects the fact that higher-ability types have stronger preferences for speeding. When speeding behavior is informative about type, the planner can use speeding fines to lower the efficiency costs of redistribution relative to using income taxes alone. We refer to the earnings ability tag from figure ?? as $s'_{het}(z) = s'(z) - s'_{inc}(z)$, following the notation in Ferey et al. (2024).



With this intuition in-hand, we now present the first result regarding the size of the fine gradient b , which is the key focus of our paper. Expressions for the optimal fixed component a and marginal income-tax rates $T'(\cdot)$ are shown in the Appendix.

Theorem 1. *Suppose assumption 1 holds, the income tax schedule $T(\cdot)$ is optimal, the fixed component $a^*(b)$ is optimal, and $s'_{\tau_s}(z) \perp z$. Then, at the optimum, the income-gradient of the fine schedule, b , satisfies the following:*

$$b^* = \frac{\text{Cov}[\bar{e}'(z)s'_{\tau_s}(z), z] + \text{Cov}\left[\hat{g}(z), \int_0^z (y - \bar{z})s'_{het}(y)dy\right]}{\bar{s}'_{\tau_s} \mathbb{V}[z]}$$

where $\hat{g}(z)$ are augmented social welfare weights.¹¹

Under conventional welfare weights that are monotonically declining in z , the optimal income gradient of the fine schedule depends essentially on how deterrence effects $\bar{e}(z) * s'_{\tau_s}(z)$ and earnings ability tags $s'(z) - s'_{inc}(z)$ co-vary with income. With respect to deterrence, we consider the special case where the externality $\bar{e}(z) = \alpha_e + \beta_e z$ is affine in z . This implies:

$$b^* = \beta_e + \frac{\text{Cov}\left[\hat{g}(z), \int_0^z (y - \bar{z}) s'_{het}(y) dy\right]}{\bar{s}'_{\tau_s} \mathbb{V}[z]}$$

In words, an optimally corrective fine gradient (ignoring redistribution) would simply set the fine gradient equal to the slope of marginal speeding costs, $b^* = \beta_e$.¹²

With regards to the redistributive term $\frac{\text{Cov}\left[\hat{g}(z), \int_0^z (y - \bar{z}) s'_{het}(y) dy\right]}{\bar{s}'_{\tau_s} \mathbb{V}[z]}$, we refer to the numerator as the “redistributive value:” it captures the value of using the fixed ability tags $s'_{het}(z)$ to move resources between individuals with different welfare weights $\hat{g}(z)$ through changes in the fine slope b . However, since speeding behavior is a *mutable* tag, the redistributive value is dissipated by a fiscal externality (*i.e.* a consumption distortion) term in the denominator, which depends on fine semi-elasticities $\bar{s}'_{\tau_s}$.

It can be shown that, when $s'_{het}(z) = \alpha_{het} + \beta_{het} * z$ is affine with $\beta_{het} > 0$, the redistributive term will be positive. In words, if the earnings ability tag $s'_{het}(z)$ is more informative at higher income levels, then fines that scale with income ($b^* > 0$) will have an instrumental redistributive value.

¹¹The augmented weight $\hat{g}(z)$ captures the full social value (including fiscal externalities) of a marginal transfer at a given income level z .

¹²The intercept term would be captured by the fixed component of the fine schedule: $a^* = \alpha_e$.

3.3 Incorporating Fairness

We now formalize the fairness rationales we consider and show how they enter our expression for the optimal fine gradient. As discussed in section 2, official argumentation for income-based fines in Finland cites a desire to equalize the “impact” of fines across incomes (Lahti, 2021). Does this “impact” refer to behavior (e.g. deterrence), or a financial impact? The criminal code details that day-fines should be calculated so as to be reasonable in accordance with ability to pay, which seems to point explicitly to the shape of the fine schedule (e.g. proportional punishment). Consistent with a deterrence-based story, recent legal theory (Kantorowicz-Reznichenko, 2013) emphasizes the potential of income-based fines to achieve deterrence across incomes.

We argue that these various motivations can be understood as either a desire for a particular fine schedule shape (e.g. proportional to income) or an induced distribution of behavior. In particular, if income-based fines are meant to equalize the marginal cost of speeding (“marginal deterrence”) across incomes, then proportional fines accomplish this goal when individuals have log-utility over consumption. In particular, we define $y = z - T(z)$ as net income and $u(c) = \ln(y - \tau(y) * s)$ as the sub-utility of income; suppose $\tau(y) = b * y$, *i.e.* is proportional to net income. Then, the marginal cost of speeding is given by:

$$u'_s(y - \tau(y) * s) = \frac{-b}{1 - b * s}$$

In words, when $\tau(y)$ is proportional to net-income, then the marginal cost of speeding is independent of income y . Indeed, income-based speeding fines are, for most of the income distribution, proportional to net-of-tax income by construction (equation 1).¹³

However, under this log-utility specification, proportional fines *also* generate equality in speeding rates across incomes. Specifically, under any monotonic and increasing sub-utility of speeding $v(s)$, it is easy to see that all individuals will speed at the same rate,

¹³Fines are also roughly proportional to pre-tax income; see figure 3.

regardless of their income. As we showed in figure 2, the income-based speeding fine system is associated with rough equality of speeding rates across incomes, in contrast to the inequality within the fixed-fine region of the schedule. Given this evidence, and motivated by theories of fairness that focus on the allocation/behavior dimension (e.g. specific egalitarianism),¹⁴ we separately evaluate two rationales for income-based fines: equality of behavior (which we call “equal compliance”) and proportional punishment.

To formalize these preferences within our optimal tax framework, we model each rationale in a reduced-form (sufficient-statistic) manner. In particular, suppose there is a prevailing fine schedule $\tau(z) = a^*(b) + b * z$, where $a^*(b)$ is optimal given a slope parameter b . Given this schedule, we define $\bar{e}_{\tau_s}(z)$ as the social value of a marginal speeding fine decrease for z -earners; similarly, we define $\bar{e}_s(z)$ as the social value of a marginal decrease in the speeding rate among z -earners. We call $\bar{e}_s(z)$ the “behavioral fairness” term (which captures local preferences over induced behavior) and $\bar{e}_{\tau_s}(z)$ the “fine fairness” term (which captures local preferences over the shape of the fine schedule).

Since it is proportional to behavioral responses $s'_{\tau_s}(z)$, the behavioral fairness term $\bar{e}_s(z)$ enters the expression for the optimal fine gradient in precisely the same manner as a standard externality. By contrast, $\bar{e}_{\tau_s}(z)$ produces effects proportional to the size of the fine change; as a result, it enters the optimal fine gradient in a manner similar to the redistributive value. To see this most clearly, suppose that both $\bar{e}_s(z) = \alpha_s + \beta_s * z$ and $\bar{e}_{\tau_s}(z) = \alpha_{\tau} - \beta_{\tau} * z$ are affine. Then, if $s'_{\tau_s}(z) \perp z$, our expression for the optimal fine gradient becomes:

$$b^* = \beta_e + \beta_s + \frac{-\beta_{\tau}\mathbb{V}[z] + \text{Cov}\left[\hat{g}(z), \int_0^z (y - \bar{z})s'_{het}(y)dy\right]}{\bar{s}'_{\tau_s}\mathbb{V}[z]}$$

Considering the fairness mechanisms in isolation, if the fairness-related desire to lower

¹⁴There is also empirical evidence consistent with specific egalitarianism; see Caspi et al. (2024).

speeding rates increases with income ($\beta_s > 0$) the optimal gradient b^* will be positive. Furthermore, if the desire to decrease fines falls with income ($\beta_\tau > 0$), then the optimal gradient will be positive since $\frac{\beta_\tau}{s_{\tau s}}$ is positive. In our survey, we test whether $\beta_s > 0$ by eliciting subjects' value for policies that lead to more equality in across-income speeding rates (relative to a distribution where high-income individuals speed more); similarly, we test whether $\beta_\tau > 0$ by varying the “steepness” of the fine schedule (*i.e.* how much larger the fine is at high incomes relative to low incomes).

3.4 General Expression for the Optimal Gradient

We now write the general expression for the optimal fine gradient that incorporates all the mechanisms above and assumes a constant fine elasticity ζ_τ (to replace the semi-elasticities in our previous equations). To reflect our various empirical approaches, we make the following assumptions:

Assumption 2. *Externality $\bar{e}(z)$, behavioral fairness $\bar{e}_s(z)$, and fine fairness $\bar{e}_\tau(z)$ are all composed of intercept and slope terms:*

$$\bar{e}(z) = \alpha_e + \beta_e * z$$

$$\bar{e}_s(z) = \alpha_s + \beta_s * z$$

$$\bar{e}_\tau(z) = \alpha_\tau - \beta_\tau * z$$

Furthermore, either $\alpha_\tau \leq 0$ or there exists some earnings level z_0 such that $\bar{e}_\tau(z_0) = 0$.

The last part of assumption 2 guarantees that β_τ determines the sign that fine-fairness contributes to the optimal fine gradient by ensuring that the intercept α_τ is not too large;¹⁵ therefore, knowledge of the sign of β_τ will be sufficient to test the fine-fairness

¹⁵Our claim that β_τ is determinative under this assumption is dependent on the joint distribution of fines, speeding rates, and earnings. Essentially, if $s(z)/\tau(z)$ is weakly increasing in z (as is the case when we consider average speeding rates/fines in Finland), then, under a constant fine elasticity, this claim will hold.

rationale.¹⁶

If income taxes $T(z)$ and the fixed-fine component $a^*(b)$ are optimal, then, under assumption 2, the optimal fine gradient satisfies:

$$b^* = \beta_e + \beta_s + \frac{-\beta_\tau \int_Z z * (z - \bar{z}^{rw}) dH(z) + \alpha_\tau (\bar{z} - \bar{z}^{rw}) + \text{Cov} \left[\hat{g}(z), \int_0^z (y - \bar{z}^{rw}) s'_{het}(y) dy \right]}{\zeta_\tau \int_Z z * (z - \bar{z}^{rw}) \frac{s(z)}{\tau(z)} dH(z)} \quad (3)$$

where $\bar{z}^{rw} := \frac{\mathbb{E}[\frac{s(z)}{\tau(z)} * z]}{\bar{s}/\bar{\tau}}$ is response-weighted average income. The key estimates we need to estimate in order to test our four hypothesized rationales are color-coded in equation 3 as follows:

- β_e : the gradient of marginal social speeding costs (estimated using linked administrative data).
- $s'_{het}(z)$: earnings ability tag, measured as $s'(z) - s'_{inc}(z)$, where $s'_{inc}(z)$ is the causal earnings effect on speeding. We estimate $s'_{inc}(z)$ using linked administrative data.
- β_s, β_τ : fairness preferences in relation to across-income compliance and fine-schedule steepness respectively (estimated using our survey).
- ζ_τ : fine elasticity, which we take from Kaila (2024).

We now discuss the administrative data sources that we use to estimate the marginal speeding cost gradient β_e and causal earnings effects on speeding $s'_{inc}(z)$.

4 Administrative Data

We now briefly describe the administrative data we use to assess externality- and redistribution-based rationales for income-dependent speeding fines.

¹⁶As is shown in equation 3, some assumption would need to be made about the intercept in order to calculate the optimal fine gradient predicted by the fine-fairness rationale. Since we are unable to reject that $\beta_\tau = 0$ using our survey evidence, we do not make an assumption on the intercept.

Our primary dataset is an annual panel of earnings, speeding-ticket receipt, and accident costs from 2006-2019. We construct this panel using the population register, income-tax register, crime report histories, and information from accident reports. We use taxable earned income as our earnings measure, though we also make use of information on transfers received and taxes paid from the tax returns. We construct a 0-1 indicator for speeding-ticket receipt in a given year as our measure of speeding behavior from the crime reports; using this data, we also construct an indicator for receipt of a non-speeding traffic ticket. We restrict this panel to individuals who are 18 years or older in a given year.

The accident report data is crucial for testing whether the marginal cost of speeding varies with income. To measure accident costs for a given individual i in year t , we count the number of injuries incurred by “externality-relevant” participants in all accidents involving i in year t . In our baseline results, we assign all participants *outside* of i ’s vehicle in an accident to the externality-relevant group and assume that the driver i internalizes the cost to everyone in their own vehicle. We consider an alternative definition where all others (except i) in the driver’s vehicle are also included as part of the externality-relevant group; qualitatively, this does not change our results.

We supplement these primary datasets with several additional registers. First, we make use of a vehicle register (2013-2023) to measure the mass of the cars that individuals own, as well as to construct an annual driving distance measure using odometer data from vehicle inspections (see Appendix section 10.2.4 for details). We also draw on the inheritance tax register (2013-2022), the universe of parent-child links, and the cause-of-death register (2013-2022) to construct inheritance-shock events around first parental deaths, which we use to measure income effects on speeding.

5 Externalities

We now discuss our approach for estimating the income-gradient of marginal speeding externalities, β_e , in Finland; in the absence of other concerns, value of β_e allows us to directly evaluate the externality-based rationale via equation 3. Our approach to measuring β_e has two key features. First, using linked data from accident reports, crime reports, and income tax returns, we exploit cross-sectional variation in the rate of receiving speeding tickets to measure the marginal cost of speeding with respect to injuries. Second, we leverage a “job-loss” design to calculate the monetary cost of injuries (both fatal and non-fatal) that result from accidents. We begin with the cross-sectional approach.

5.1 Cross-Sectional Approach

We measure the cost of speeding using accident-related injuries from our accident reports data. To measure marginal speeding costs, we use cross-sectional variation in speeding behavior conditional on income. In particular, we collapse our panel of earnings, speeding-ticket receipt, and externality-relevant injuries (section 4) into an average individual-level dataset¹⁷ with the following key variables:

- $\bar{z}_i$: average taxable earned income
- $\bar{s}_i$: average speeding rate, *i.e.* (# of years with speeding ticket/# of panel years)
- $\bar{I}_i^{ext}$: average number of injuries incurred by externality-relevant parties in accidents involving driver i

As discussed in section 4, we take as our baseline externality-relevant group individuals who are outside the focal driver’s vehicle (hereafter, “other-party” victims); our results are robust to including all other individuals besides the driver (hereafter, “other-person” victims). inside the driver’s vehicle (excluding the driver For individuals who are never

¹⁷We do not balance the panel before collapsing since missing status is endogenous (e.g. if an individual dies in an accident). We therefore weight our regressions by the number of years in which i appears in the population register from 2006-2018.

in an accident that causes an externality-relevant accident participant, $\bar{T}_i^{ext}$ will be equal to 0. Summary statistics for this dataset are available in Appendix table 27.

We distinguish between β_e^I , the gradient of marginal speeding costs with respect to injuries, and β_e^M , the money-metric gradient that we need to test the externality rationale. Since we measure the cost of each injury using a job-loss design, we first establish findings for the injury gradient β_e^I and explore mechanisms using this data, before turning to money-metric estimates. To estimate β_e^I , we use the following regression:

$$\bar{T}_i^{ext} = \rho_e^I * \bar{z}_i + \alpha_e^I * \bar{s}_i + \beta_e^I * \bar{z}_i * \bar{r}_i + \kappa * \bar{X}_i + \gamma_{a(i)} + \epsilon_i \quad (4)$$

where $\gamma_{a(i)}$ is a set of birth-cohort fixed-effects and $\bar{X}_i$ is a set of controls. The coefficient ρ_e^I measures the non-monetary cost-gradient for non-speeders, while α_e^I measures intercept of marginal social costs (corresponding to the constant in the optimal fine slope); our primary object of interest is β_e^I , which measures how the marginal effect of speeding varies with income.

We employ this cross-sectional approach for two primary reasons. First, due to the stochastic nature of ticket receipt, the average speeding rate $\bar{s}_i$ is a better representation of underlying behavior than ticket receipt $s_{i,t}$ in a given year, thereby mitigating attenuation concerns.¹⁸ Second, since accidents are rare, cross-sectional variation provides more powerful tests than behavioral variation induced by instruments such as price. Our approach relies on the following identification assumption:

$$\mathbb{E}[\bar{s}_i * \epsilon_i | \bar{z}_i, \bar{X}_i] = 0$$

In words, our measure of speeding behavior $\bar{s}_i$ must be uncorrelated with other determinants of accident costs conditional on income and covariates $\bar{X}_i$. However, we do allow

¹⁸Note that, because we implicitly take enforcement as given, we require behavior-driven variation in *observed* speeding behavior (*i.e.* ticket receipt) to measure the appropriate externality. Since we do not observe enforcement probabilities $p(z)$, it is difficult to make use of outside estimates for speeding externalities (van Benthem, 2015).

there to be across-income differences that affect how marginal speeding costs vary with income. For example, if income is positively correlated with vehicle mass, the marginal cost of speeding may be positive correlated with income because speeding with a heavier car can lead to more severe accidents. In other words, our goal is to measure a “second-best” gradient of marginal speeding externalities, allowing some unpriced margins to affect how the marginal cost of speeding varies with income.

In our baseline results, we include the following controls in $\overline{X}_i$:

- Average car ownership rates (for both an individual and their spouse), collected from the population register
- Average non-speeding traffic crime rates, constructed from crime reports
- Own accident-related injuries and involvement in non-injurious accidents

Car ownership serves as a baseline proxy of driving distance (and therefore accident exposure). The non-speeding traffic crime rate serves to control for non-speeding behavior that affects accident costs and may be correlated with speeding behavior. Given the cross-sectional nature of our main dataset, we also include controls for own accident-related injuries and involvement in non-injurious accidents to address reverse causality concerns.¹⁹ As an additional safeguard against reverse causality concerns, we drop from our sample any individuals involved in an accident from 2003-2005, prior to the 2006-2018 period we use to create our cross-sectional dataset.

We discuss the particulars of modifying our cross-sectional approach to measure the money-metric marginal cost gradient β_e^M in section 5.4.

¹⁹As we show with our job-loss design, accident-related injuries lead to earnings losses (and may have an impact on how likely someone is to subsequently speed). This can confound the relationship between income and marginal speeding costs.

5.2 Marginal Speeding Gradient (Injuries)

We now discuss our estimates of the marginal speeding cost gradient in terms of injuries (equation 4).

To motivate our main findings, figure 5 shows binscatters for expected other-party²⁰ injuries (per 1,000 individuals) separately for those who never receive a speeding ticket from 2006-2018 (“never-speeders”) and those who receive at least one ticket in that time-frame (“ever-speeders”). Despite a positive expected-cost gradient among non-speeders, the gradient among speeders is nearly flat, implying a negative marginal speeding gradient in the context of equation 4. As we show, while adding a rich set of controls attenuates the marginal speeding gradient towards zero, it never ceases to be negative.

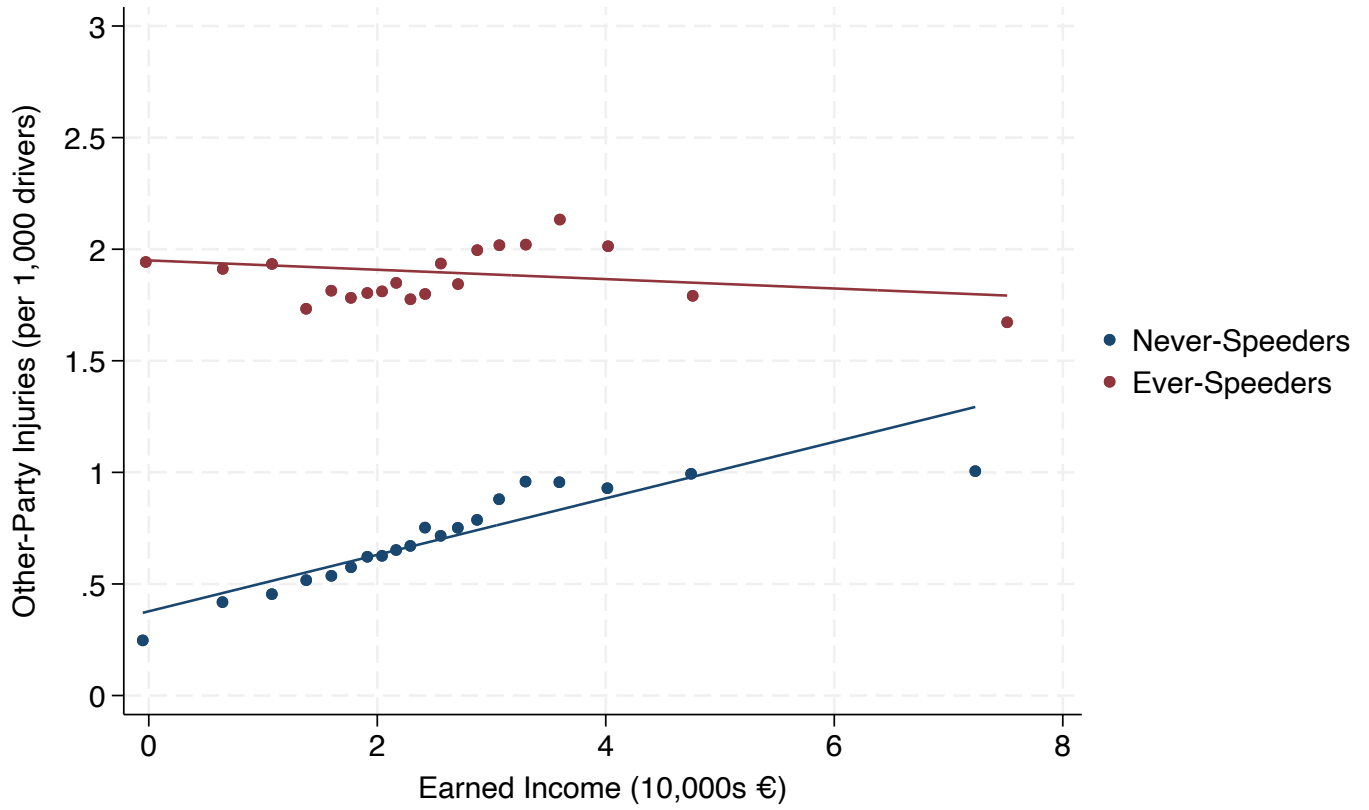
Appendix figure 31 replicates this finding for other-person victims (*i.e.* all participants except the focal driver i); the figure displays a more pronounced downward-sloping gradient among ever-speeders, which suggests that higher-income drivers have safer cars.

We now turn to our estimates of equation 4, which are displayed in table 1. To begin, in column 1, we include only the reverse-causality controls. Reassuringly, the baseline speeding effect on injuries (row 1) is positive, and the gradient (row 2) is negative, implying that the marginal effect of speeding on the likelihood of an injury decreases by 0.07 percentage points for every €10,000 increase in driver earnings. Column 2 conditions just on individuals who ever receive a speeding ticket; while the marginal speeding gradient decreases somewhat, it is clear that the negative gradient is not entirely driven by comparisons between never- and ever-speeders. In column 3, we include fixed effects municipality of residence,²¹ motivated by concerns that differential enforcement (rather than latent speeding behavior) might explain our results. However, compared to column

²⁰Recall that “other-party” means individuals outside of driver i ’s vehicle in an accident.

²¹Specifically, since this is a cross-sectional dataset, we include fixed-effects for the first and last municipality in which an individual appears during the 2006-2018 period.

Figure 5: Expected Injuries by Income and Speeding History



Binscatter relationships between average taxable earned income and average other-party injuries using the dataset described in section 5.1. Income and other-party injuries are residualized on birth-cohort fixed effects, and each individual is weighted by the number of years they appear in the population register from 2006-2018.

Table 1: Marginal Speeding Gradient (Injuries)

	(1)	(2)	(3)	(4)
	Other-Party Injuries (per 1,000 drivers)			
Speeding Rate (MSC Intercept)	8.484*** (0.163)	7.680*** (0.279)	8.374*** (0.164)	2.883*** (0.168)
Speeding X Earnings (MSC Gradient)	-0.741*** (0.0380)	-0.590*** (0.0605)	-0.742*** (0.0380)	-0.476*** (0.0391)
Earned Income (10,000s €)	0.122*** (0.00411)	0.0849*** (0.0113)	0.131*** (0.00422)	0.106*** (0.00608)
Constant	0.304*** (0.0104)	0.588*** (0.0455)	0.289*** (0.0105)	-0.0841*** (0.0135)
<i>N</i>	5033597	1588308	5033597	5033597
Cohort FE	X	X	X	X
Reverse-Causality Controls	X	X	X	X
Ever Speeding		X		
Municipality FE			X	
Car Ownership & Traffic Crime Controls				X

*Heteroskedasticity-robust standard errors in parentheses. Car ownership rates and non-speeding crime rates are also interacted with earnings when included. Each individual is weighted by the number of years they appear in the population register from 2006-2018. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

3, the speeding effect and marginal speeding gradient are nearly identical. Column 4 adds our car ownership and non-speeding traffic-crime controls (as well as interactions between these variables and earnings).²² The gradient attenuates by around 36% relative to column 1, but remains negative.

Table 2 explores whether two additional variables from the vehicle register can explain the results in Table 1: annual driving distance and vehicle mass. As described in section 4, we construct the driving distance measure using odometer data, using successive vehicle inspections for a given user-vehicle cell in order to allocate distance driven over time.²³ Due to the infrequency of inspections, this variable is often missing for a given

²²Coefficient estimates for these variables are available upon request.

²³See Appendix section 10.2.4 for details on the annual driving distance measure.

driver-year;²⁴ therefore, in column 1, we weight each individual by the number of years for which they have a distance measure.²⁵ While some of the result may be selection-driven, the distance control attenuates the marginal cost gradient relative to column 4 of Table 1 by more than 50%, though the gradient remains negative. In column 2 of Table 2, we instead control for vehicle mass, measured for each individual i as the average mass of vehicles for which i is the primary user from 2013-2018;²⁶ analogous to column 2, we weight each individual by the number of years for which that individual has non-missing vehicle mass information from 2013-2018.²⁷ The vehicle mass control leaves the marginal speeding gradient virtually unchanged relative to column 4 of Table 1. We view driving distance as a potential omitted variable that could be correlated with speeding rates conditional on income; we treat vehicle mass, on the other hand, as a mediating variable that may affect the marginal cost of speeding but is not itself a violation of our cross-sectional identification assumption. Therefore, we use column 1 of table 1 as our preferred specification, allowing across-income variation in vehicle mass to affect our second-best estimate of the marginal speeding gradient. Appendix table 29 reports the models in tables 1 and 2 using other-person victims as the externality-relevant group; echoing Appendix figure 31, we find somewhat more negative marginal speeding gradients.

To further explore potential mechanisms for our findings, we turn to information available from the accident reports. We caveat that, because such analyses are conditional on an accident, they cannot be used to fully explain the gradients in tables 1 and 2, which are measured using expected accident costs.²⁸ For this analysis, we take the indi-

²⁴Beyond the fact that inspections are relatively rare, we cannot assign distance between periods where a user switches vehicles. Among those in our cross-sectional sample, we observe, on-average, a bit more than 2 years (out of a possible 6 years) of driving distance information for any given individual from 2013-2018.

²⁵In practice, for each individual, we weight by the product of (# of years appearing in the population register from 2006-2018) and (# of years with distance information from 2013-2018).

²⁶We restrict the vehicle set to be typical road-driven vehicles: buses, cars, mopeds, motorcycles/motorbikes, trucks, and vans.

²⁷Among those in our cross-sectional sample, we observe, on-average, a bit more than 3 years of vehicle mass information for any given individual from 2013-2018.

²⁸In other words, some of the marginal speeding gradient is driven by across-income differences in the effect of speeding

Table 2: Marginal Speeding Gradient (Injuries): Distance and Vehicle Mass Controls

	Other-Party Injuries (per 1,000 drivers)	
	(1)	(2)
Speeding Rate (MSC Intercept)	1.370*** (0.217)	2.720*** (0.196)
Speeding X Earnings (MSC Gradient)	-0.219*** (0.0513)	-0.479*** (0.0453)
Earned Income (10,000s €)	0.138*** (0.0242)	0.180*** (0.0216)
Constant	-0.0373 (0.0783)	-0.435*** (0.0676)
<i>N</i>	2223285	2776588
Cohort FE	X	X
Reverse-Causality Controls	X	X
Car Ownership & Traffic Crime Controls	X	X
Annual Driving Distance	X	
Vehicle Mass		X

*Heteroskedasticity-robust standard errors in parentheses. Car ownership rates, non-speeding crime rates, driving distance, and vehicle mass are also interacted with earnings when included. Each individual is weighted using a composite weight. In column 1, individuals are weighted by the number of years they appear in the population register, multiplied by the number of years for which driving distance information is available from 2012-2018. In column 2, individuals are weighted by the number of years they appear in the population register, multiplied by the number of years for which vehicle mass information is available from 2013-2018. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

viduals from our cross-sectional sample and focus on the first accident from 2006-2018 in which they are involved as a driver.²⁹ Column 1 of Table 31 shows estimates of the conditional marginal speeding gradient in a baseline model where we include our standard controls (car ownership rate, non-speeding traffic-crime rate, and reverse causality controls), as well as a control for vehicle mass (a potential mechanism for the conditional gradient).³⁰ As in our unconditional results, we find a negative marginal speeding cost gradient (row 2), implying that damages conditional on an accident play a role in our unconditional findings. In unreported results, we examine a number of controls (accident municipality, speed limit, collision type,³¹ urban-vs-rural, and vehicle type) but find a persistently negative and significant gradient.³² However, when we control for the number of other-party participants (column 2), the gradient reduces by around 64% and becomes insignificant. Furthermore, the coefficient on earnings (row 3), *i.e.* the earnings gradient for non-speeders, attenuates to 0, suggesting that high-income drivers (regardless of their speeding behavior) tend to drive on busier roads (e.g. during rush hour), where other drivers' behavior plays a larger role in the likelihood and severity of an accident.

Taking stock, our results suggest that optimal externality-motivated fines may actually be *decreasing* in income, under the presumption that the cost of an external injury is unrelated to the driver's income. However, to motivate why such costs may not be constant, figure 6 uses accident-level data from the accident reports to plot the relationship between driver income (measured in the year prior to the accident) and "other-party" income (panel 6a) or "other-person" income (panel 6b). We see modest positive relationships in both figures, implying that higher-income drivers are involved in accidents with

on accident probability, as suggested by Appendix figure 38.

²⁹We also restrict to cases where these individuals are driving a typical road-driven motor vehicle (car, truck, bus, moped, motorbike, motorcycle, or van).

³⁰We also include interactions between earnings and these variables (excluding the reverse causality control). Though we use earnings in the year prior to the accident, there is an extant reverse-causality concern for our speeding rate measure, which is collapsed using 2006-2018 data. Since we're not concerned about own-injury effects on earnings, we use a lighter-touch reverse causality control: the total number of accidents in which each individual is involved from 2006-2018.

³¹(E.g. head-to-head collision, side-to-side collision, etc.

³²Among these, only collision-type attenuates the gradient (by around 42%).

Table 3: Conditional Marginal Speeding Gradient

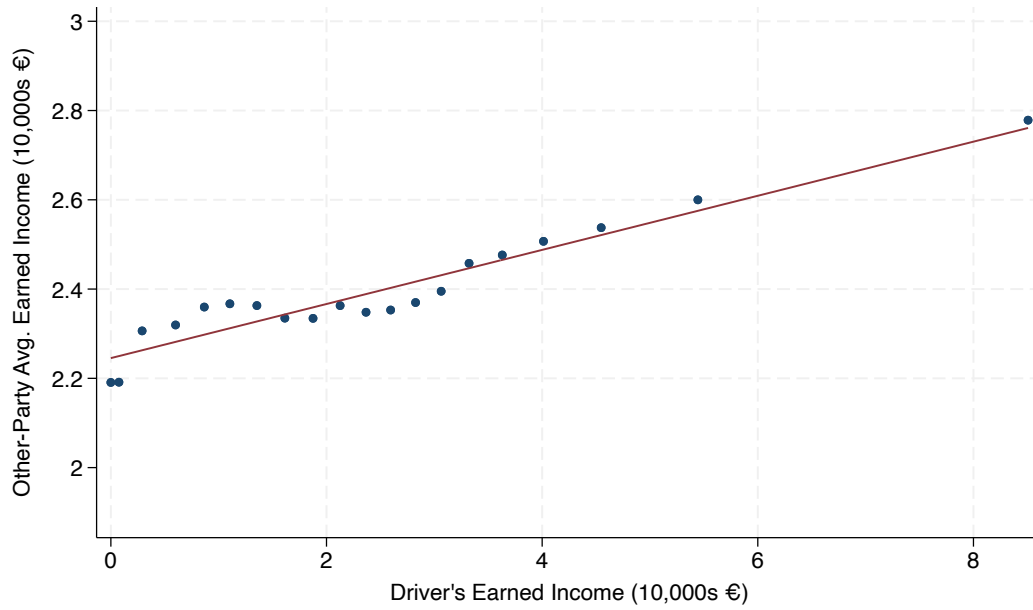
	(1)	(2)
	Other-Party Injuries (per 1,000 accidents)	
Speeding Rate (MSC Intercept)	18.64 (9.811)	2.581 (8.853)
Speeding X Earnings (MSC Gradient)	-10.85*** (2.626)	-3.872 (2.370)
Pre-Period Earnings (10,000s €)	5.235** (1.965)	-0.471 (1.773)
Other-Party Participants		153.7*** (0.574)
Constant	129.9*** (6.855)	-7.789 (6.207)
<i>N</i>	315026	315026
Age FE	X	X
Year FE	X	X
Reverse Causality Controls	X	X
Car Ownership & Other Traffic Crime Controls	X	X
Vehicle Mass Control	X	X

Description

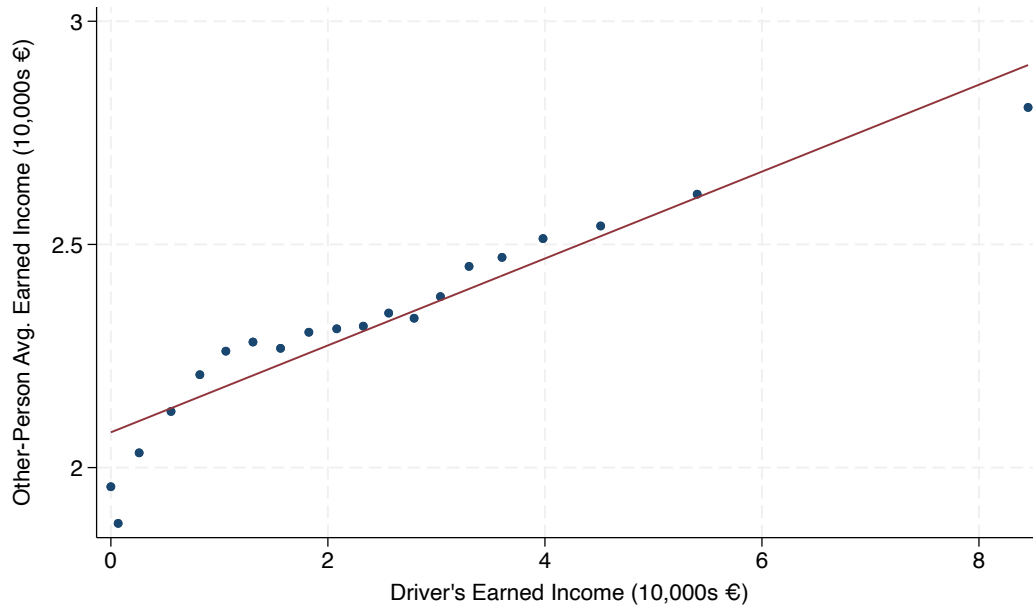
higher-income victims; to the extent that injury costs are correlated with income (e.g. due to lost labor time), this could weaken (or even flip) the negative marginal speeding cost gradient. More generally, victims' injury costs could be correlated with driver income for a number of reasons; we therefore turn to a job-loss design (von Wachter and Bender, 2006) that enables us to measure the cost of individual injuries. Importantly, this approach allows us to separately estimate fiscal externalities (the injury costs borne by the government) and net-income losses (the costs borne by victims); this separation allows us to apply income-dependent weights to net-income losses.

Figure 6: Binscatter Correlations Between Driver and Victim Income

(a) Other-Party Victims



(b) Other-Person Victims



Binscatter relationships between driver income and victim income. We use the sample of individuals described in 5.1 and restrict to the first accident in which an individual is involved as a driver from 2006-2018. We further restrict to drivers of a car, truck, bus, moped, motor bike, motorcycle, or van and exclude accidents involving trains, trams, tractors, snow mobiles, trailers, or other unidentified motor vehicles.

5.3 Money-Metric Costs of Injuries

Our approach to measuring the monetary costs of injuries relies on comparisons between individuals who are injured or killed in an accident and matched individuals from the general population. In particular, for each individual injured or killed in an accident that involves a motor vehicle (the treatment group), we assign two random control individuals from the population register with the same birth year, sex-at-birth, and education level³³ in the year prior to the accident. We then construct an event-time panel for each match-group around the accident event year, following individuals until 2023 (the last year in which we have detailed income data).

To illustrate our approach, we begin by estimating event-study models of the form:

$$Y_{i,t,k} = \beta_0 * \text{Treat}_{i,t} + \gamma_k + \sum_{k=-4, k \neq -1}^{2023-t} \beta_k * \text{Treat}_i * I(t+k \geq t) + \epsilon_{i,t,k} \quad (5)$$

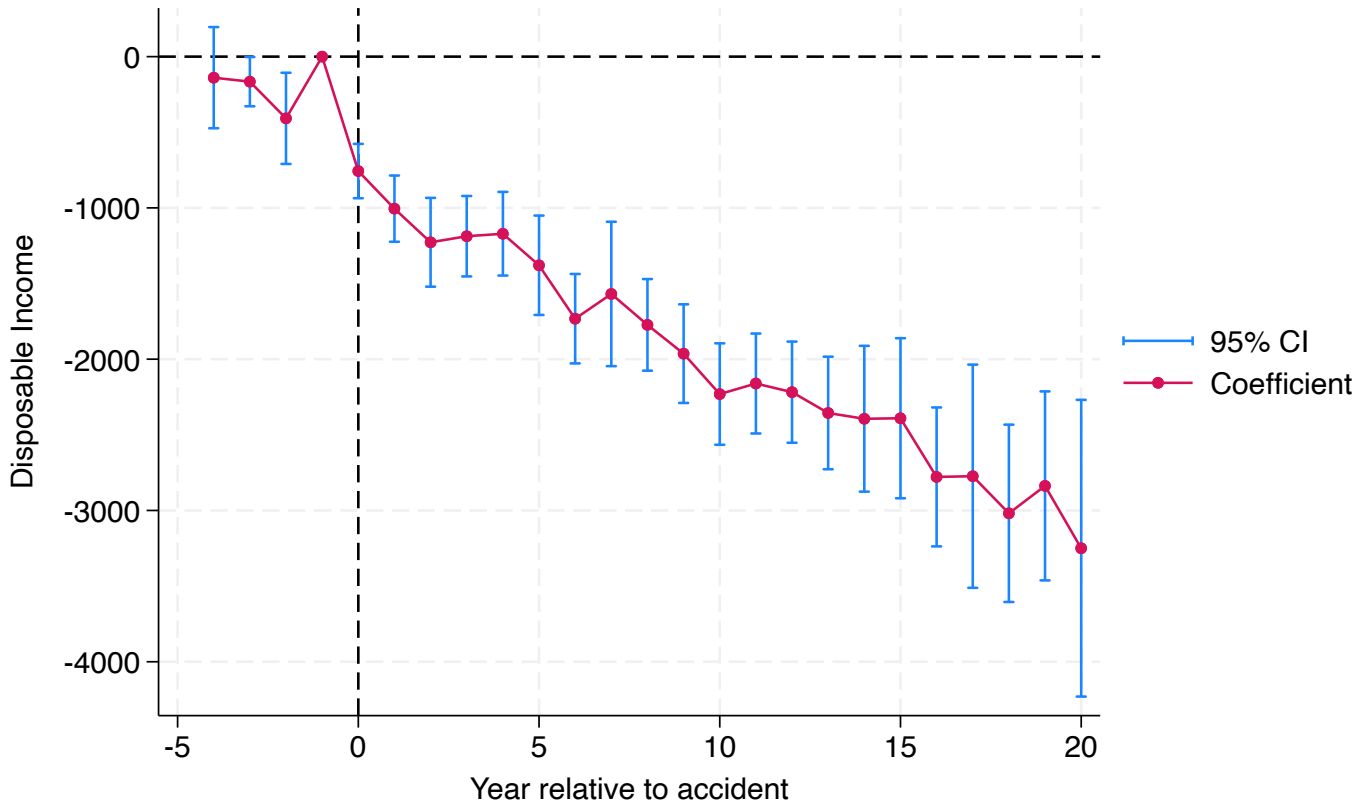
where $Y_{i,t,k}$ is an outcome for individual i in year $t+k$, $\text{Treat}_{i,t}$ is indicator for whether or not individual i is treated in year t , $I(t+k \geq t)$ is a post-period indicator, and γ_k is a set of event-time dummies.³⁴ Figure 7 shows event-study coefficients from equation 5, using disposable income (*i.e.* income net of taxes and transfers) as the outcome. We see no discernible pre-trends and an immediate income loss in the year of the accident, followed by a gradual divergence between the treatment and control groups. In Appendix figure 35, we restrict to non-fatal injuries; we see little-to-no decline in disposable income for several years following the accident, after which we see the same divergence pattern emerge as in figure 7.³⁵

³³We use four education categories (based on highest completed qualification): less than upper-secondary, upper secondary or post-secondary non-tertiary, short-cycle tertiary, and bachelor's or above.

³⁴Due to the nature of our data, in which we lose one accident year in each successive post-period, the event-time dummies are equivalent to including accident-year fixed-effects for the post-period.

³⁵Appendix figure 32 shows, for non-fatal injuries, that income before transfers and taxes declines immediately in the year of the accident, implying that the initial lack of decline in figure 35 is driven by insurance provided through the transfer-and-tax system.

Figure 7: Event Study Effects of Injury/Death on Disposable Income



Estimates of β_k from equation 5. The sample includes all accident-related injuries from 2003-2022, restricted to accidents that include a motor vehicle (cars, trucks, buses, vans, mopeds, motor bikes, and motorcycles). Injuries from accidents involving a snow mobile, trailer, tram, train, tractor, or other unidentified motor vehicles are excluded.

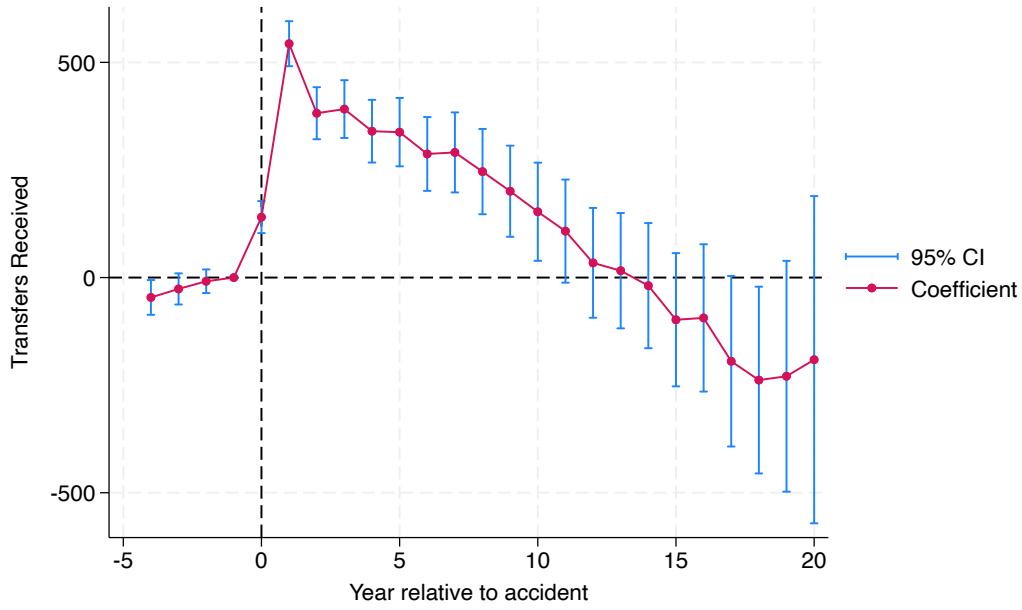
Our findings on net-income losses around accident-related injuries are consistent with Appendix figure 34, which shows event-study estimates of labor-force non-participation (measured as being employed for zero months during the year); we see immediate and persistent increases in non-participation in the year of the accident, regardless of whether or not we restrict to non-fatal injuries. In other words, accident-related injuries cause some share of treated individuals to drop out of the labor force; since the average injured individual is in their prime earnings-growth years (around 45 years old, see table 4), income gradually diverges between the treatment and control group.

While our findings on disposable income effects in Figure 7 are indicative of the consumption loss that victims experience, the government (through increased transfers and reduced tax payments) is smoothing the consumption losses of victims. In particular, Figure 8 shows event-study effects on transfers received (panel 8a) and taxes paid (panel 8b). One year after the accident, we see that the treatment group receives nearly 500 € in additional transfers relative to the control group, but this amount decays to zero after approximately 15 years. In panel 8b, we see that treated individuals pay less in taxes following the accident event; in line with the findings on income, taxes paid gradually decline over time relative to the control group.

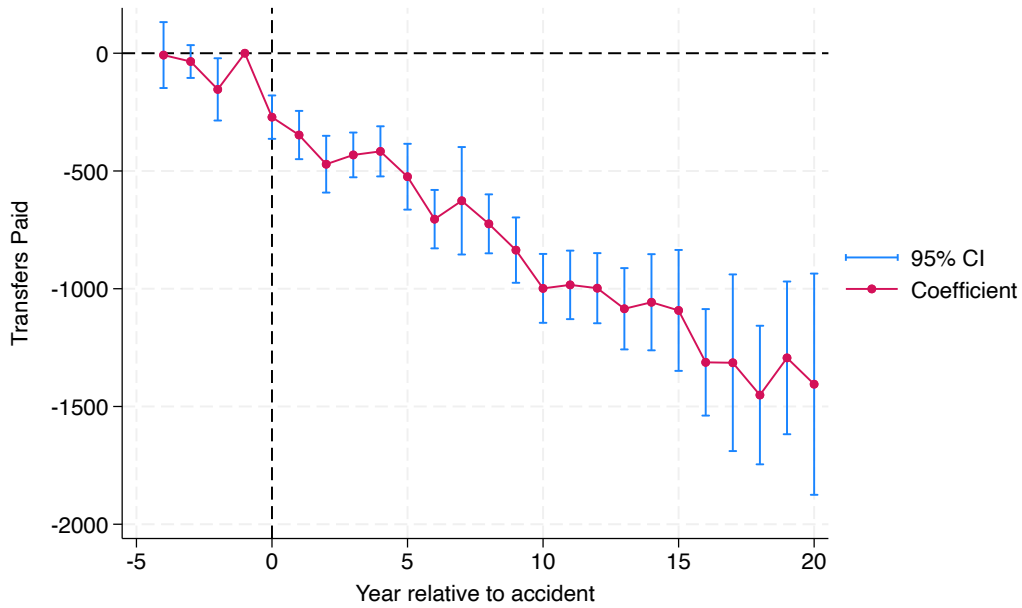
To compare these two cost margins, figure 9 plots event-study treatment effects from equation 5 separately for pre- and post-tax income; the red-shaded region corresponds to the losses in 7, while the blue-shaded region is the replacement income (provided through increased transfers and/or lower tax burdens) paid out by the government. In our main sample that combines fatal and non-fatal injuries, disposable income losses dominate (especially in later years); in Appendix figure 36, we show that, for non-fatal injuries, fiscal externalities play a larger role until well after the year of the accident.

How do we use these findings to calculate the cost of each accident-related injury?

Figure 8: Event Study Effects of Injury/Death on Transfers/Taxes



(a) Transfers Received



(b) Taxes Paid

Estimates of β_k from equation 5. The sample includes all accident-related injuries from 2003-2022, restricted to accidents that include a motor vehicle (cars, trucks, buses, vans, mopeds, motor bikes, and motorcycles). Injuries from accidents involving a snow mobile, trailer, tram, train, tractor, or other unidentified motor vehicles are excluded.

Figure 9: Event Study Effects of Injuries on Components of Social Costs



Description

We start by calculating net-present-value (NPV) fiscal externalities and disposable income losses for each injury. In particular, for each injury and post-period, we calculate difference-in-differences effects for each cost component relative to the two matched control individuals.³⁶ We then aggregate period-specific effects using a discount rate δ (equal to 3% in our baseline) to obtain NPV costs. The results of this aggregation are shown in table 4, separately for injuries and deaths. Under our baseline discount rate of 3%, non-fatal injuries generate around €10,600 in NPV costs to the government and €6,500 in NPV net-income losses to individuals with 13 years of post-accident data. For the small number of fatal injuries, we find cost *savings* for the government,³⁷ though these are swamped in magnitude by nearly €200,000 of NPV net-income losses.³⁸ Unsurprisingly, given that injuries cause individuals to exit the labor force (Appendix figure 34), we find in Appendix figure 37 that cumulative (un-discounted) fiscal externalities and disposable income losses are increasing in pre-period income.³⁹

column 1 of Appendix table 30 shows that an additional 1 € in pre-period income is associated with an additional 3.6 € in cumulative disposable income losses. The positive income-loss gradient is consistent with Appendix figure 34, which shows that a roughly constant share of individuals leave the labor force after an accident. The increasing profile of fiscal externalities is consistent with a social insurance system that replaces lost income in a proportional manner.

How should we combine the NPV fiscal externalities borne by the government and the NPV net-income losses borne by victims? In line with the optimal tax literature, we

³⁶For additional details, see Appendix section 10.2.1.

³⁷Savings can arise if the decrease in transfers received outweighs the decrease in taxes paid.

³⁸We do not project costs forward using our estimates; given our cross-sectional identification strategy described in sections 5.1 and 5.4, this is unproblematic as long as the year in which accidents occur is uncorrelated with speeding conditional on income. In unreported results, we residualize costs on accident year and re-estimate the money-metric gradient in section 5.4, which does little to change our results. Furthermore, we elect to use our estimates for the cost of fatal injuries, rather than relying on outside VSL estimates. Again, this will not be problematic as long as the bias in our approach is not systematically correlated with the speeding behavior of offenders conditional on income.

³⁹The result on fiscal externalities is unsurprising since higher-income victims lose more earnings, resulting in even more lost tax revenue for the government.

Table 4: Cumulative Social Costs of Accidents

	(1) Non-Fatal Injuries	(2) Fatal Injuries
# of Post-periods	12.98	12.91
Age at accident	44.43	50.92
Cumulative Fiscal Cost	€13,250	-€41,957
Cumulative Net Inc. Loss	€8,791	€244,961
$\delta = 0.03$		
Discounted Fiscal Cost	€10,564	-€35,261
Discounted Net Inc. Loss	€6,473	€199,547
$\delta = 0.05$		
Discounted Fiscal Cost	€9,226	-€31,817
Discounted Net Inc. Loss	€5,358	€176,700
$\delta = 0.07$		
Discounted Fiscal Cost	€8,149	-€28,973
Discounted Net Inc. Loss	€4,486	€158,157
N	78378	3489
Description		

apply welfare weights g_i to each individual's net-income losses, where, following Saez (2002b):

$$g_i = \frac{\kappa}{y_i^{-\nu}}$$

where y_i is i 's disposable income in the year prior to the accident and ν is a CRRA parameter that encodes the planner's distaste for inequality. In words, the planner weighs money-metric losses to each individual in proportion to the marginal utility from a CRRA utility function; the larger is ν , the more weight the planner puts on a given loss to low-income individuals relative to high-income individuals. We set κ such that $\bar{g} = 1$, meaning that the planner is indifferent between a €1 increase in revenue and distributing €1 equally among the population. The weights g are therefore measured relative to government revenue, implying that we can leave fiscal externalities unweighted. In our

baseline results, we set $\nu = 1$, though we evaluate higher and lower degrees of inequality aversion.

Robustness To claim our findings on the cost of accident-related injuries as causal, we rely on the standard parallel-trends assumption, *i.e.* that, in the absence of an injury/death, treated individuals would have progressed on the same trajectory as the control group. While we do not detect any significant pre-trends across various outcomes estimated using equation 5, one concern is that negative shocks *specific to the year of the accident* may simultaneously cause both an accident and earnings declines. For example, an individual who loses their job may become more risk-seeking and reckless and therefore prone to accidents. To assuage this concern, we estimate equation 5 separately for drivers and passengers; if individuals become reckless in the year of the accident due to a simultaneous negative shock, we would expect to see stronger effects for drivers (who are already on a downward trajectory). Appendix figure 33 shows the results for drivers (panel 33a) and passengers (panel 33b). Overall, the two trajectories are quite similar; if anything, the passenger effects are larger (at least in later years). Based on this evidence, simultaneous negative shocks seem to play little role in our estimates.

5.4 Marginal Speeding Gradient (Money-Metric)

To estimate the money-metric gradient of marginal speeding costs, β_e^M , we merge our estimates for the welfare-weighted NPV monetary costs of each injury from section 5.3 into our cross-sectional dataset (see section 5.1) on income and speeding rates, replacing average external accident-related injuries $\bar{I}_i^{ext}$ with its money-metric equivalent $\bar{M}_i^{ext}$.⁴⁰ We then estimate the money-metric gradient of marginal speeding costs, β_e^M , using the following regression:

⁴⁰In practice, the individual-specific $\bar{M}_i^{ext}$ are estimated. However, beyond adjusting standard errors for heteroskedasticity, we do not account for this estimation uncertainty when estimating equation 6; this introduces additional noise and inflates our standard errors. However, as we will see, across nearly all specifications, our standard errors are precise enough to rule out meaningfully positive estimates for β_e^M .

$$\overline{M}_i^{ext} = \rho_e^M * \bar{z}_i + \alpha_e^M * \bar{r}_i + \beta_e^M * \bar{z}_i * \bar{r}_i + \kappa * \overline{X}_i + \gamma_{a(i)} + \epsilon_i \quad (6)$$

Table 5 displays marginal speeding gradients using a 3% discount rate and no welfare-weighting (*i.e.*, $\nu = 0$), separately for other-person (columns 1 and 2) and other-party victims (columns 3 and 4). Columns 1 and 3 present estimates with only our reverse-causality controls; interestingly, in these specifications, we find a baseline speeding cost (MSC intercept α_e^M) of 169 € in the other-person case and 118€ in the other-party case. The former is nearly identical to the average fixed speeding fine in Finland (approximately €167), according to police ticket data, suggesting that the totality of our empirical approach (cross-sectional variation and job-loss design) does a reasonable job of capturing the marginal cost of speeding. Columns 2 and 4 of table 5, add the rest of our baseline controls (car ownership and non-speeding traffic-crime, as well as interactions between these variables and earnings), while columns 3 and 6 add annual driving distance. For the most part, the additional controls attenuate the intercept term (row 1) but do little to change the marginal speeding cost gradient (row 2). As discussed in section 5.2, the controls in columns 3 and 6 constitute our preferred specifications; for our preferred externality-relevant group (other-party victims), our estimates in column 6 indicate that fines should *decrease* by almost €19 for every €10,000 of income, well below the overall fine gradient we estimate in Finland (+12.4€, see section 2).

To see how different discount rates and welfare-weighting of net-income losses affect this baseline result, Table 6 displays a table of estimates for the marginal social cost gradient using our preferred specification. We vary the discount rate $\delta \in \{0.03, 0.05, 0.07\}$ we use to calculate NPV costs, as well as the planner's degree of inequality aversion ν for welfare-weighting net-income losses (see section 5.3). Regardless of the welfare-weighting scheme or the discount rate, we find consistently negative estimates and can rule out

Table 5: Money-Metric Speeding Gradient: 3% Discount Rate

	Other-Person Costs			Other-Party Costs		
	(1)	(2)	(3)	(4)	(5)	(6)
Speeding Rate (MSC Intercept)	168.8*** (25.49)	27.71 (26.57)	84.39* (34.35)	118.1*** (22.54)	63.63** (23.20)	75.43* (31.15)
Speeding X Earnings (MSC Gradient)	-27.19*** (5.978)	-9.371 (6.255)	-21.45* (8.425)	-17.54** (5.372)	-17.71** (5.585)	-18.60* (7.712)
Earned Income (10,000s €)	0.925 (0.681)	-1.276 (1.014)	0.160 (4.222)	2.100*** (0.617)	0.161 (0.922)	2.929 (3.743)
Constant	6.338*** (1.698)	3.315 (2.224)	-10.06 (13.08)	2.189 (1.507)	-0.680 (1.950)	-21.38 (11.44)
<i>N</i>	5033597	5033597	2223285	5033597	5033597	2223285
Cohort FE	X	X	X	X	X	X
Reverse Causality Controls	X	X	X	X	X	X
Car Ownership & Other Traffic Crime Controls		X	X		X	X
Annual Driving Distance			X			X

gradients larger than €1. We see that higher discount rates and stronger inequality aversion (higher ν) lead to more attenuation of the gradient. While the attenuation from the discount rate is mechanical, the role of inequality aversion is consistent with our findings in figure 6: since the victims of high-income drivers have larger injury costs (because they are higher-income), the negative injury gradient we measure in section 5.2 is “stretched” in money-metric terms, relative to a case where injury costs are constant across driver income. Therefore, when the planner down-weights the relative cost of injuries to high-income victims, the relative cost of injuries caused by high-income drivers decreases, causing the gradient to increase. We view a level of inequality aversion corresponding to $\nu = 4$ as a normatively appealing stopping point because, as shown by Appendix table 30, a value of $1 < \nu < 4$ effectively implies a constant weighted cost of injuries across victim incomes (in contrast to the upward-sloping relationship for unweighted injury costs).

Appendix Table 32 shows nearly identical quantitative results when we consider other-person victims. Therefore, to summarize, regardless of how we specify the externality-relevant group (other-party or other-person), we cannot identify a positive marginal speeding gradient across a range of discount rates and levels of inequality-aversion param-

Table 6: Marginal Speeding Cost Gradients by Discount Rate and Inequality Aversion: Other-Party Victims

	Discount Rate		
	3%	5%	7%
Welfare Weight			
$\nu = 0$	-18.6* (7.712)	-15.89* (6.694)	-13.67* (5.875)
$\nu = 0.25$	-16.07* (7.477)	-13.73* (6.504)	-11.84* (5.717)
$\nu = 1$	-7.617 (4.392)	-6.486 (3.839)	-5.568 (3.393)
$\nu = 4$	-7.55 (4.371)	-6.421 (3.821)	-5.507 (3.376)
N	5033597	5033597	5033597
Cohort FE	X	X	X
Reverse Causality Controls	X	X	X
Car Ownership & Other Traffic Crime Controls	X	X	X
Annual Driving Distance	X	X	X

Heteroskedasticity-robust standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

eters. As our preferred estimate that we use in section 8, we set the inequality-aversion parameter $\nu = 1$, the discount rate $\delta = 0.03$ (reflecting the yield on Finnish 10-year government bonds), and use other-party victims as the externality-relevant group; this yields a marginal speeding cost gradient of -7.6€ (se=4.4) under our preferred set of controls. We can therefore rule out gradients larger than €1 per €10,000 of income,⁴¹ below our estimate of the overall income-gradient of fine size in Finland (€12.4, see section 2) and well below the gradient within the day-fine region of the fine schedule (€88). We therefore reject externality-based motives as a rationale for the Finnish speeding fine system.

6 Redistribution

As elaborated in Section ??, we built our test of the redistributive rationale for income-based fines on the logic of “tagging:” fines serve a redistributive role if speeding behavior is a signal of earnings ability. This signal can be recovered by differencing observed across-income variation in speeding behavior with respect to causal earnings effects on speeding.⁴² Our goal is to measure causal earnings effects as an instrument for evaluating the tagging potential of speeding behavior.

We begin by providing a decomposition of the causal earnings effect into a labor-supply and income effect, before discussing our empirical approaches for estimating these objects.

6.1 Earnings Effects and Weak Separability

Recall from figure ?? that the earnings effect is identified from earnings changes induced by variation in *labor supply*; in general, such labor-supply induced effects carry additional information about latent wages that pure income effects (holding labor supply fixed) do

⁴¹Our standard errors are likely inflated since we do not account for the noise in our individual-specific injury-cost estimates.

⁴²By *causal*, we mean that, for an individual with a given income z , we can construct counterfactual speeding consumption at earnings levels $z + dz$. In other words, the variation in earnings dz should be exogenous to other determinants of consumption.

not capture.

To simplify notation, suppose individuals are indexed only by their implicit wage w ; given some tax-and-fine system, an individual with wage w chooses a speeding rate $s(z, l)$ as a function of pre-tax income $z = w * l$ and labor supply l . Under weak separability, this function would reduce to $s(z)$ (no dependence on labor supply except through pre-tax income); however, in the more general case, the derivative with respect to labor supply l (holding the wage w fixed) is given by:

$$\left. \frac{ds(z, l)}{dl} \right|_w = \frac{\partial s(z, l)}{\partial z} * w + \underbrace{\frac{\partial s(z, l)}{\partial l}}_{=0 \text{ under weak sep.}} \quad (7)$$

To obtain the earnings effect, we re-scale this expression by $\frac{1}{w}$ to obtain:

$$\left. \frac{ds(z, l)}{dz} \right|_w = s'_{inc}(z) = \underbrace{\frac{\partial s}{\partial z}}_{\text{Pre-tax income effect}} + \underbrace{\frac{\partial s}{\partial l} * \frac{1}{w}}_{\text{Re-scaled labor supply effect}}$$

In general, without a weak separability assumption, the earnings effect on speeding is composed of a pre-tax income effect and a wage-scaled (money-metric) labor supply effect. In what follows, we test for and identify the labor supply effect using within-individual variation in labor supply (holding earnings fixed); to estimate the income effect, we turn to an inheritance shock design.

6.2 Labor Supply Effects

Our approach for measuring the labor supply effect is motivated by a test for the weak separability assumption: weak separability would be violated if, for a given individual, variation in labor supply (holding earnings fixed) has an effect on speeding behavior. To demonstrate our approach, we estimate a single labor supply effect for the general population; in section 6.4, we estimate across-income heterogeneity, the key input for

determining the redistributive value of income-based speeding fines.

In particular, using our annual population panel from 2006-2019 (see section 4), we model ticket receipt $s_{i,t}$ for individual i in year t as:

$$s_{i,t} = \beta_1 * z_{i,t} + \beta_2 * \bar{w} * l_{i,t}^m + \gamma_i + \kappa_{a(i),t} + \epsilon_{i,t} \quad (8)$$

where $z_{i,t}$ is taxable earned income,⁴³ $\bar{w} * l_{i,t}^m$ is a measure of months employed⁴⁴ during the year (re-scaled into a money-metric using an average monthly wage),⁴⁵ γ_i is an individual fixed-effect, and $\kappa_{a(i),t}$ are year-by-cohort fixed effects. For demonstrating our approach, we use the 2012 average wage in order to facilitate comparability with our inheritance sample (see section 6.4). Under the null hypothesis of weak separability, $\beta_2 = 0$: labor supply has no independent effect on speeding behavior conditional on earnings. Our identification approach for the labor-supply effect β_2 relies on the assumption that, conditional on earnings and cohort-year effects, within-individual variation in labor supply is uncorrelated with other time-varying determinants of speeding behavior. We expect that, due to mis-measurement of true labor supply (proxied by employment months), our estimate of β_2 will be attenuated; we view this as unproblematic since it leads to underestimation of earnings effects, thereby biasing us in favor of finding a redistributive role for income-based fines.

Note that the coefficient β_1 ostensibly yields an estimate of the income effect (holding labor supply fixed). However, due to mis-measured labor supply, variation in earnings $z_{i,t}$ is likely measured using variation in wages *and* labor supply, leading to bias in the estimated income effect. We discuss the likely direction of this bias below.

⁴³This measure includes some taxable transfers like unemployment benefits and pensions. Since these transfers insure against within-person income risk rather than against lifetime income risk, we view this measure as corresponding to the static z in our model framework.

⁴⁴Employment months are measured in the register data based on the number of days an individual spent on a firm's payroll during the year.

⁴⁵Due to co-linearity, we cannot use an individual's imputed wage in this regression. We elect for a cross-sectional average wage since it is well-measured (as opposed to an individual's average wage) and less likely to be endogenous. When we estimate labor supply effects separately by income group, we calculate separate average wages for each group.

Table 7 displays our results. First, in column 1, we show the cross-sectional relationship between earnings and speeding rates: each €10,000 increase in earnings is associated with a 1.2 percentage point increase in speeding rates. Column 2 estimates equation 8. We find a significant wage-scaled labor supply effect: a labor supply increase that generates €10,000 of earnings causes a 0.29 percentage point increase in the likelihood of receiving a speeding ticket, leading us to reject weak separability in our setting. Overall, the labor supply effect explains around 25% of the cross-sectional gradient in column 1.

Table 7: Cross-Sectional Gradient & Labor Supply Effect

	Speeding Rate (pp)	
	(1)	(2)
Earned Income (€10,000s)	1.166*** (0.00609)	0.499*** (0.0106)
Re-scaled Labor Supply (€10,000s)		0.288*** (0.00768)
N	15207808	15172401
R^2	0.019	0.196
Pop. Sample	25%	25%
Cohort-Year FE	X	X
Individual FE		X

Standard errors (clustered at the individual level) in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Given our results, $\hat{\beta}_1$, our “naive” income effect estimate, is likely upward-biased. In particular, β_1 , the coefficient on earnings, is identified from wage changes (holding labor supply fixed). Therefore, if wage and labor supply changes are positively correlated, then $\beta_2 > 0$ implies that estimates of β_1 from equation 8 are upward-biased. Intuitively, some of the earnings variation used to identify β_1 will be driven by labor supply changes that are not perfectly captured by variation in $l_{i,t}^m$. Since the labor supply effects are

positive ($\beta_2 > 0$), there is upward bias. As a result, we turn to an alternative approach for estimate the income effect on speeding behavior. We take our fixed-effects approach at face value to estimate the labor supply effect, which likely leads to a conservative (lower-bound) estimate due to attenuation.

6.3 Income Effects

Our identification approach for estimating income effects builds on the method of Nekoei and Seim (2022), who exploit the differential timing of parental deaths to explore how inheritances contribute to inequality. However, the death of a parent may have concomitant effects on speeding that are unrelated to the wealth shock, *e.g.* due to grieving or caregiving. As we show, individuals who receive little-to-no inheritance do exhibit speeding trends around parental death. We therefore modify the approach in Nekoei and Seim (2022) and difference across inheritance sizes to address non-wealth effects in a triple-difference framework.

First Difference: Timing of Parental Deaths As in Nekoei and Seim (2022), the treatment event in our paper is the first parental death that a child experiences. We identify all such deaths from 2013-2022 (the period for which we have access to inheritance tax returns) using the cause-of-death register and parent-child links. We compare heirs whose first parental death occurs in year t (treated) to heirs whose first parental death occurs δ years after t (control). To address observable differences between parental-death cohorts, Nekoei and Seim (2022) also re-weight heirs to homogenize the distribution of education and age across cohorts. We follow a similar approach, re-weighting each cohort with respect to sex, age, and education. Since we are interested in population-level estimates, we normalize using the distribution of demographics in the 2012 population register.

In our baseline results, we set $\delta = 4$ and follow individuals for four years ($t = 0, 1, 2, 3$)

post-treatment; we examine robustness to using 5 post-periods below. This “fixed-control” method (one control cohort per treatment cohort) generates a clean comparison that is balanced across periods and avoids any contamination from using early treated observations as controls for late-treated observations. To create our event-time dataset, we construct, for each parental death year y , a control group using all individuals whose first parent dies in $y + 4$. We then stack observations across all death years y ; observations are therefore indexed by an individual i , a treated death year y , and an event time t (defined relative to y). We cluster all standard errors at the individual level and apply the weights described above.

Second Difference: Inheritance Size Measuring the income effect on speeding behavior using early- vs. late-treated comparison of heirs is potentially confounded by effects of parental death on speeding behavior. In particular, let β^{wealth} and α^{death} be the effects a given wealth shock and first parental death respectively on speeding behavior. A first-difference across parental death years measures the sum of both effects. However, assuming that the death effect α^{death} is common across inheritance size, a second comparison between those who receive positive inheritances $\Delta w > 0$ and those who receive none $\delta w = 0$ will remove the death effect:

$$\underbrace{\underbrace{(\beta^{\text{wealth}} + \alpha^{\text{death}})}_{\text{DID}|\Delta w > 0} - \underbrace{\alpha^{\text{death}}}_{\text{DID}|\Delta w = 0}}_{\text{Triple Difference}}$$

To motivate our construction of the $\Delta w > 0$ and $\Delta w = 0$ groups, 10 shows quintiles of cumulative net-of-tax inheritance flows around the year of first parental death.⁴⁶ We see that there are essentially no inheritance flows prior to the first parental death and

⁴⁶We start measuring the cumulative inheritance flows in $t - 4$. We allocate individuals to inheritance-flow quintiles within each death cohort d based on the cumulative inheritances received in $[d, d + 2]$ (*i.e.* using 3 years of post-death data). We do not use the full 4 years of post-death flows to define groups since each additional year we use forces us to drop a treatment cohort in our triple-difference specification. Figure 10 indicates that this decision does not affect the results since very little inheritance is received in year $d + 3$, three years after the death, for any inheritance-size groups.

the majority of flows occur within a year of parental death.⁴⁷ After $t + 1$, we see some additional flows, generated by a combination of above-average tax-processing times and correlated parental deaths. On average, cumulative inheritances are around €38,000 higher in the treatment group four years after parental death. However, the distribution is skewed: the bottom two quintiles receive around €2,300 on-average, while the top quintile receives close to €140,000. To construct the inheritance-size groups, we combine the bottom two quintiles to create the “bottom” group and the top three quintiles to create the “top” group. Relative to a true zero-inheritance group, we include the second quintile (which receives around €5,000 on-average) in the bottom group in order to improve power. While, in theory, this can lead to bias if income effects differ across inheritance size, that bias vanishes as $\frac{\Delta W^{\text{top}} - \Delta W^{\text{bottom}}}{\Delta W^{\text{top}}} \rightarrow 1$,⁴⁸ given that the bottom group receives around €2,300 and the top group around €63,000, this bias should be negligible.

Estimating Income Effects We begin by showing how speeding behavior evolves around parental deaths for each of our inheritance-size groups. In particular, figure 11 plots (re-weighted) event study estimates from the following regression:

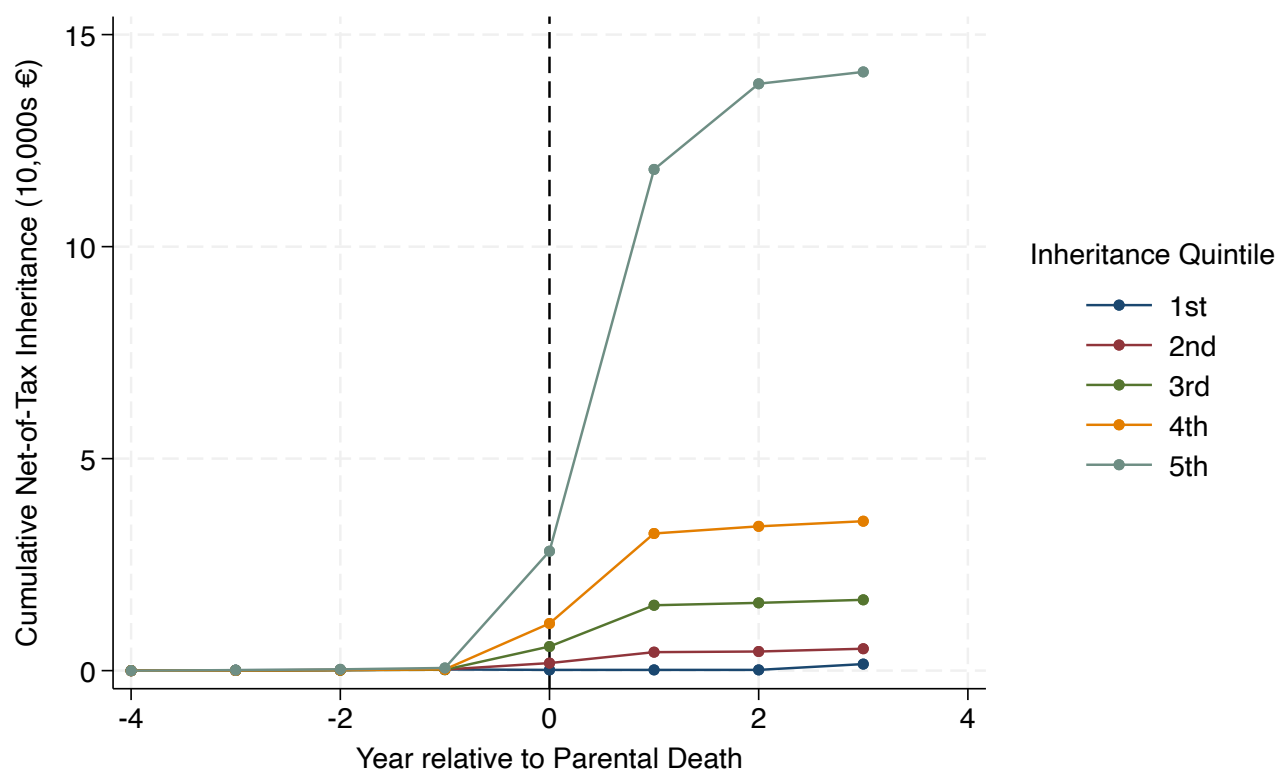
$$s_{i,y,t} = \gamma_t + \rho * I_{YOD(i)=y} + \sum_{k=-3}^4 \beta^k * I_{YOD(i)=y} * I_{t=k} + \epsilon_{i,y,t} \quad (9)$$

where $s_{i,y,t}$ is an indicator for whether i received a speeding ticket in year $y + t$, γ_t captures event-time fixed effects, $I_{YOD(i)=y}$ is an indicator for whether i ’s first parental death occurred in year y , and β^k is the treatment coefficient for event year k . We use $t - 4$ as the omitted year in order to clarify the existence of anticipation effects, which appear for both inheritance-size groups in year $t - 1$ of figure 11. Following this anticipation effect, the two groups diverge, with the top group exhibiting an overall increase in speeding rates. Appendix figure 22 shows event studies by inheritance-size group where we con-

⁴⁷The estate inventory must be filed with the tax authority within 3 months of the parent’s death, after which the tax authority must assess the inventory and make a tax decision (which usually takes 6-12 months). Therefore, most inheritance decisions are made in the year following the parental death.

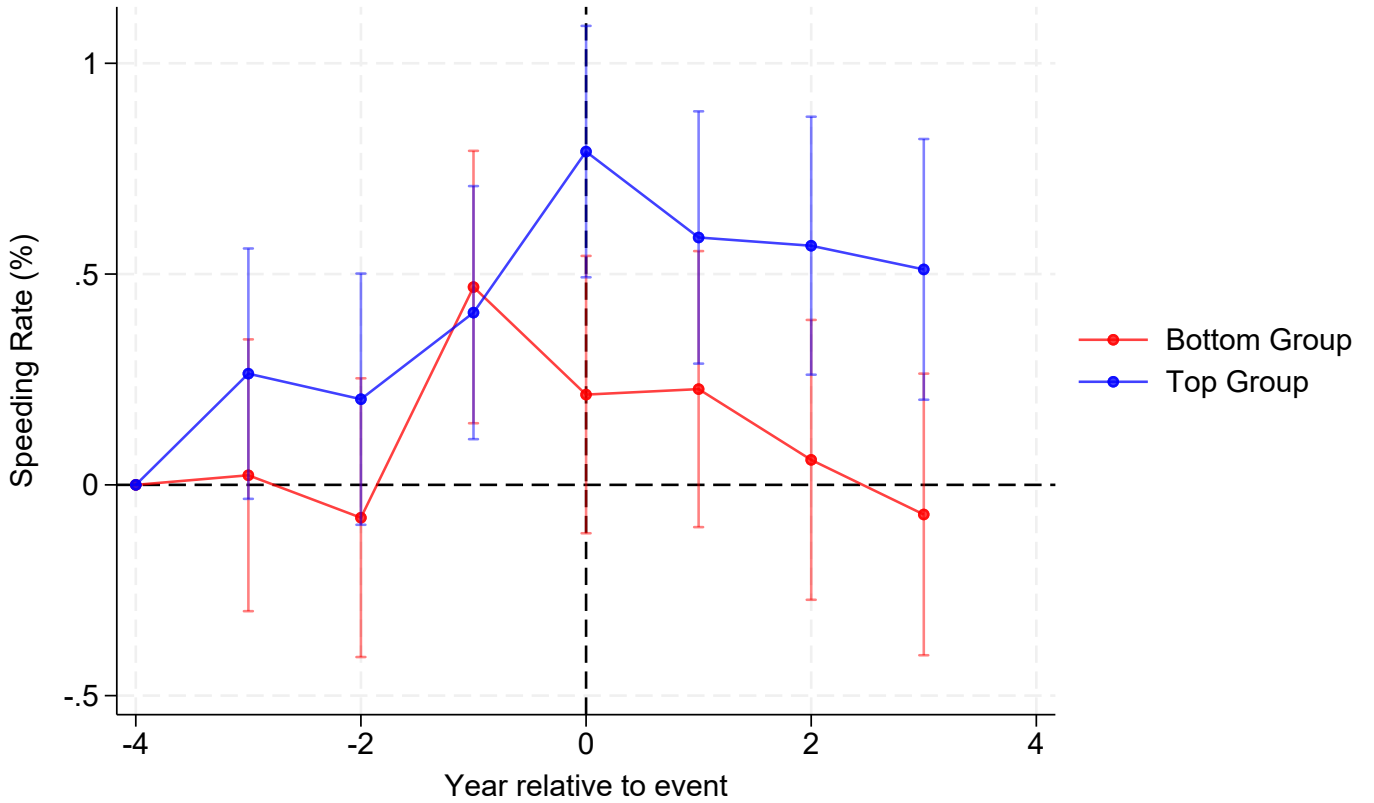
⁴⁸See Appendix section 10.2.2 for an expression of the bias.

Figure 10: Cumulative Inheritances Around Parental Death



Caption

Figure 11: Event Study of Speeding Rates around Parental Death, by Inheritance Group

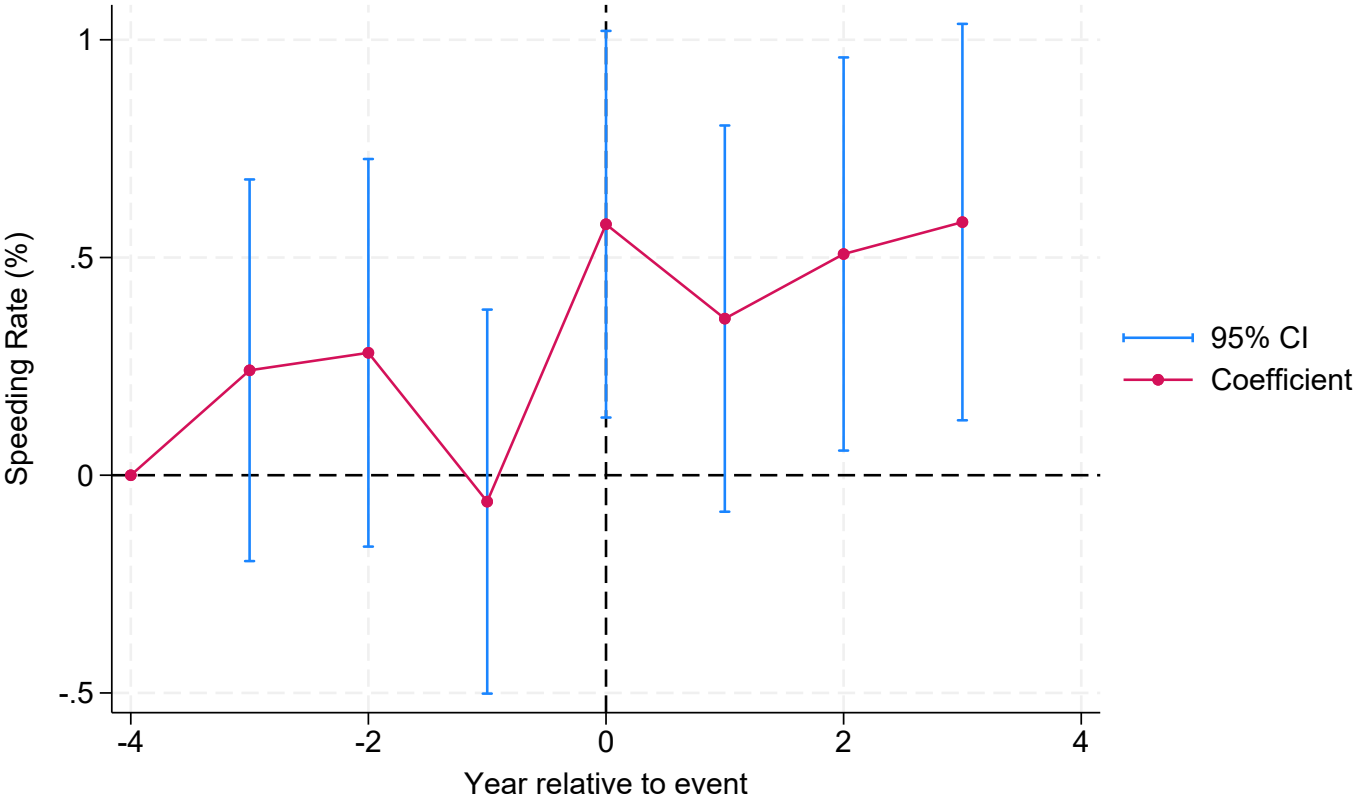


Caption

struct the bottom inheritance group using only the bottom quintile of inheritance flows (rather than the bottom two quintiles); the results are qualitatively similar, though the series for the bottom group is much more noisy, which motivates our decision to use the bottom two quintiles as the bottom group in our baseline results.

Given that the common anticipation effect in $t - 1$ is present in the bottom inheritance group, which receives little-to-no-inheritance, this spike is unlikely to be driven by anticipated wealth shocks. To bolster this claim, Appendix figures 20 and 21 show that there are no anticipatory effects on earnings or car ownership. We conclude that this common anticipation effect reflects non-wealth effects (e.g. travel for care-giving purposes) that we want to difference out of our income-effect estimates. To this effect, figure 12 plots the difference in event-study coefficients, $\beta_{\text{top}}^t - \beta_{\text{bottom}}^t$, from figure 11 for each t . While the

Figure 12: Event Study of Speeding Rates around Parental Death: Triple Difference



Caption

series is somewhat noisy, we see a clear level difference between pre- and post-treatment periods and the absence of statistically significant pre-trends.

The estimates in figure 12 correspond to the reduced-form in the triple-difference IV approach we use to measure income effects. To measure the denominator (first-stage) of our income-effect estimates, we use the cumulative inheritances received in the post-treatment period. We view this approach as conservative: implicitly, it assumes that the extent to which the inheritance wealth shock will be “spent” on speeding fines is confined to our post-treatment window. If consumption responses are positive and persistent, our estimate of the income effect would be biased downwards.⁴⁹ While we are

⁴⁹Furthermore, we use the assessed value of the estate as the denominator in our income effect. Since we have no information on the asset mix of inheritances and it is costly to liquidate some asset types, using the assessed value is, again, likely to be conservative.

limited in our ability to explore longer treatment horizons, we do find that our income effect is somewhat larger with a 5-year horizon, as opposed to the 4-year horizon in our baseline results. However, the 4-year horizon yields more power for estimating income-effect heterogeneity by earnings level, which is important for evaluating the redistributive potential of income-based fines.

To demonstrate our approach, we estimate an average income effect using a DDD-IV design, before estimating heterogeneity in this effect in section 6.4. In particular, we estimate the following system:

$$\begin{aligned} \Delta W_{i,y,t} = & \theta_0 + \theta_1 * I_{t \geq 0} + \theta_2 * I_{YOD(i)=y} + \theta_3 * \text{Top}(i) + \theta_4 * I_{t \geq 0} * I_{YOD(i)=y} + \theta_5 * I_{t \geq 0} * \text{Top}(i) \\ & + \theta_6 * I_{YOD(i)=y} * \text{Top}(i) + \theta * I_{t \geq 0} * I_{YOD(i)=y} * \text{Top}(i) + \epsilon_{i,y,t} \end{aligned} \quad (10)$$

$$\begin{aligned} s_{i,y,t} = & \beta_0 + \beta_1 * I_{t \geq 0} + \beta_2 * I_{YOD(i)=y} + \beta_3 * \text{Top}(i) + \beta_4 * I_{t \geq 0} * I_{YOD(i)=y} + \beta_5 * I_{t \geq 0} * \text{Top}(i) \\ & + \beta_6 * I_{YOD(i)=y} * \text{Top}(i) + \beta * \Delta W_{i,y,t} + \nu_{i,y,t} \end{aligned} \quad (11)$$

where $\Delta W_{i,y,t}$ is the wealth shock (flow) received by individual i in year $y + t$, $\text{Top}(i)$ is an indicator for whether i is in the top inheritance group (quintiles 3-5), and $I_{t \geq 0}$ is a post-event indicator. We estimate $\hat{\beta}$ (equation 11) via two-stage least-squares. Table 8 shows the results, with the reduced-form (equation 11) in column 1, the first stage (equation 10) in column 2, and the DDD-IV estimate in column 3. We recover an income effect of 0.26 pp for every €10,000 of income received through inheritances, similar in magnitude to the average labor supply effect we estimate in section 6.2. Thus, the income effect explains around 22% of the cross-sectional relationship between income and speeding (column 1 of table 7). Therefore, as a rough approximation, the average earnings effect (labor supply effect + income effect) explains nearly one-half of the cross-sectional

gradient.

Table 8: DDD-IV Estimates of the Income Effect

	(1)	(2)	(3)
	Reduced-Form	First-Stage	IV
DDD Coefficient	0.391*** (0.115)	1.511*** (0.0344)	
Inherit. Shock (10,000s €)			0.259*** (0.0765)
Constant	7.316*** (0.0540)	0.00701*** (0.000927)	7.314*** (0.0541)
<i>N</i>	4912424	4912424	4912424

Overall, we view the income effect we measure in column 3 of table 8 to be somewhat conservative for several reasons: (1) we assume that heirs “spend” the full inheritance on speeding tickets within our 4-year treatment horizon, (2) some inherited assets are less liquid than cash, and (3) inheritances are, to some extent, anticipated. As with our attenuated labor supply effects (section 6.2), weaker income effects likely bias us in favor of finding that speeding behavior is a tag for income. We now probe the robustness of these baseline results before measuring across-earnings heterogeneity in our earnings effects and calculating the redistributive value of speeding fines.

Robustness

Labor Supply and Earnings Responses Our inheritance-shock approach is meant to measure the effect of income on speeding, holding labor supply fixed. However, if labor supply responds to inheritance receipt, our income-effect estimates may be contaminated by the labor supply effect on speeding. Similarly, to the extent that earnings

respond to inheritance receipt, we may be mis-measuring the denominator in our income effect, since earnings changes will have an additional income effect on consumption. Appendix figures 24 and 23 show triple-difference event studies of months employed and earnings respectively around the parental death event. We see a decrease in earnings following inheritance receipt; for months employed, we observe anticipation effects and a subsequent decline following parental death. However, the magnitude of these effects are quantitatively negligible: in any given period, earnings decrease by no more than €400 and employment months increase by no more than 0.07 months.⁵⁰ Given the size of the inheritance shocks we use to estimate income effects, we view these sources of contamination as negligible for our estimates.

Effect Horizon We elect to maintain a 4-year effect horizon in order to increase our power for estimating across-income heterogeneity in income effects (section 6.4). However, we examine robustness to including an additional post-treatment year, the results for which can be found in Appendix table 23.⁵¹ Column 3 indicates that speeding rates increase by 0.3 pp for every €10,000 shock, which is about 15% higher than our baseline estimate in column 3 of table 8. Appendix figure 28 suggests that this increase arises due to a small extant speeding response in event year $t = 5$; since inheritance shocks in this period are essentially zero, the total speeding effect ds increases relative to the total shock Δw . These results support the view that our baseline estimates are somewhat conservative.

Comparability of Death Effects A key assumption we use to identify income effects is that the death effect is common across inheritance-size groups, which allows us to attribute any differences in speeding responses to differences in the size of the inheritance

⁵⁰If anything, the direction of these effects is likely to bias our income effect downwards since (1) the net income shock (after earnings reductions) is smaller than the inheritance shock and (2) labor supply appears to decrease following parental death, which would have a negative effect on speeding if labor supply is positively associated with speeding (column 3 of table 7).

⁵¹For this analysis, we use individuals whose first parent dies in $[y + 1, y + 5]$ as a control group for individuals whose first parent dies in y in order to increase power.

shock.

One reason this assumption may fail is if certain causes-of-death (e.g. car crashes) are systematically correlated with inheritance size and/or the year of parental death. Indeed, using cause-of-death data, we show in Appendix figure 25 that later parental death cohorts are less likely to die in a transport-related accident or an accident involving a motor-vehicle;⁵² Appendix table 19 similarly shows that parental estates in the top inheritance-size group are more likely to die from a transport-related or motor-vehicle accident.⁵³ However, these causes-of-death are quite rare in our sample, constituting no more than 0.6% of any given death cohort's cause-of-death. Therefore, when we re-estimate the income effect after dropping all transport-related accidents (Appendix table 20), our estimate is unchanged relative to our baseline.

Another potential reason that death effects aren't comparable across inheritance groups is that these groups have different speeding preferences. Indeed, column 1 of Appendix table 21 shows that the bottom inheritance group has a speeding rate that is 0.342 percentage points larger than the top group at event time $t = -4$. While this difference is small relative to baseline speeding rates (slightly above 7%), we consider an alternative to our baseline assumption that death effects are constant in levels: we assume that death affects are a constant proportion of baseline speeding rates. In particular, we measure speeding effects for each inheritance-size group relative to its baseline speeding rate; we then difference the relative effects across inheritance-size groups and calculate the implied level effect from this difference.⁵⁴ Appendix figure 26 compares these adjusted effects to the baseline event study effects from figure 12. The effects are almost perfectly aligned, suggesting that preference differences across inheritance groups are unlikely to drive our

⁵²See Appendix section 10.2.3 for details on how these categories are defined. Motor-vehicle accidents are a subset of transport-related accidents and correspond as closely as possible to deaths caused by a road-collision with a motor vehicle.

⁵³While wealthier individuals likely drive safer cars, they also drive more and are therefore more exposed to accident risk.

⁵⁴The level effect is obtained by multiplying the differenced relative effect by the baseline speeding rate for the top group; this is the group for which we can measure income effects.

results.

Finally, death effects may not be comparable if the amount of effort involved in disbursing the estate varies across inheritance-size groups in a way that systematically affects the likelihood of speeding. For example, larger inheritances may be more complex and therefore require more travel to-and-from the parent’s residence. To evaluate the role of inheritance complexity, we examine two features of the estate: (1) the number of (potential) child heirs and (2) the number of tax revisions we observe for an heir’s inheritance. Appendix figure 27 shows inheritance size is negatively correlated with the number of potential child heirs (panel 27a) and positively correlated with the number of tax revisions (panel 27b). To quantify the role these differences play in our income-effect estimates, Appendix table 22 shows the income-effect estimates from our triple-difference approach when we add “placebo” controls for the number of child heirs (column 1) or the number of tax revisions (column 2);⁵⁵ our income effect estimates are virtually unchanged, even increasing slightly after controlling for the number of child heirs. We therefore do not view estate complexity as a meaningful confound in our estimates.

6.4 Optimal Redistributive Fine Gradient

We now use our estimates of the income and labor-supply effects to identify the optimal redistribution-driven gradient of speeding fines. It is easy to show from our results in section 3 that, under a constant fine-elasticity ζ_τ , the optimal redistribution-driven gradient b_{redist}^* is given by:

$$b_{redist}^* := \frac{\text{Cov}[\hat{g}(z), \int_0^z (y - \bar{z}^{rw}) s'_{het}(y) dy]}{\int_Z z * s'_{\tau_s}(z) (z - \bar{z}^{rw}) dH(z)} \quad (12)$$

The numerator captures the redistributive value of fines and depends on the covariance

⁵⁵In particular, this set of controls consists of all possible interactions between the number of heirs/revisions, the treatment indicator $I_{YOD(i)=y}$, and the post-indicator $I_{t \geq 0}$. In other words, we add a set of controls that substitutes the top-inheritance group indicator for the number of heirs/revisions, allowing us to test whether these estate characteristics are the true underlying treatment dimension that explains our results.

between welfare weights and the earnings ability tag $s'_{het}(z) = s'(z) - s'_{inc}(z)$. The denominator captures the consumption distortion induced by increasing the fine gradient, which depends on how responsive individuals are to fine size (the elasticity ζ). We take the fine-elasticity estimated in Kaila (2024), which exploits a regression-discontinuity-design around the income-based fine cutoff to measure extensive margin effects of larger speeding fines on future recidivism.⁵⁶ We also explore variation around this estimate, though, if $s(z)/\tau(z)$ is increasing in income, it can be shown that the sign of the denominator is always negative if the elasticity is constant.

Therefore, the key determinant for the sign of b^*_{redist} is how $s'_{het}(z)$ evolves with z : if $s'_{het}(z)$ is increasing in z , the numerator of our expression for b^*_{redist} will be negative under conventional welfare weights that decrease with z . Thus, if the earnings ability signal $s'_{het}(z)$ is larger at higher incomes, it is optimal to have speeding fines scale with income.

To obtain measures of $s'_{het}(z) = s'(z) - s'_{inc}(z)$ across earnings levels, we estimate heterogeneity in our labor supply and income effects. To make these two sets of estimates internally consistent, we estimate the labor supply effects using individuals from the top inheritance group in our inheritance-shock sample; this is the group for which we are able to recover income effects using our DDD-IV approach. Appendix table 25 estimates the overall cross-sectional gradient and average labor-supply effect within this sample using our within-individual approach and applying the population demographic weights used in our inheritance-shock design;⁵⁷ the results are extremely similar to those in table 7 for the general population, suggesting that selection into the inheritance-shock sample is largely driven by observable characteristics. To measure heterogeneity, we assign individuals in this sample into 8 equal-sized income groups based on 2012 age-adjusted earnings.⁵⁸ We then estimate the re-scaled labor supply effect using equation 8 separately for each

⁵⁶Recall that Kaila (2024) shows individuals do not bunch at fine discontinuities (see section 2).

⁵⁷For this estimation, we construct a panel of top-inheritance-group individuals starting in 2006 and follow them until one year before their first parental death event.

⁵⁸We use 2012 as our benchmark year since it is one year prior to the first wave of parental deaths in our sample.

group⁵⁹ and estimate income effects separately for below- and above-median earnings groups.⁶⁰ We combine the labor supply effects and income effects to obtain $s'_{inc}(z_j)$ for each income group $j \in \{1, 2, \dots, 8\}$. While we are more limited in our ability to estimate income-effect heterogeneity with the inheritance-shock approach, we find only a small difference (0.076 pp, see Appendix table 26) between the below- and above-median groups, despite a large earnings difference;⁶¹ this suggests that heterogeneity may be minimal for the income effect.⁶²

After calculating earnings effect heterogeneity, we use the cross-sectional profile of speeding rates in 2012 $s(z)$ (see Appendix figure 29) to calculate across-income variation $s'(z)$,⁶³ from which we subtract our earnings effects $s'_{inc}(z)$ to obtain the earnings ability tag $s'_{het}(z)$. Figure 13 summarizes our results.⁶⁴ Cross-sectional variation $s'(z)$ (blue dashed line) initially increases with earnings at low incomes, before declining, while the earnings effects (red-dashed line) are relatively flat across incomes. As a result, the ability tag profile (solid purple line) takes on the shape of the cross-sectional profile.

Recall that our first test of whether the fine-gradient should be positive is whether the purple line in figure 13 is increasing with earnings.⁶⁵ While the covariance with earnings is indeed negative, the true metric indicated by equation 12 is how *welfare weights* $g(z)$ (*i.e.* the social value of a marginal transfer to a given income level) co-vary with the ability tag $s'_{het}(z)$. For our construction of welfare weights, we take the same approach

⁵⁹For these estimates, we scale the labor-supply variable using the average 2012 wage calculated separately for each group.

⁶⁰To increase power, we augment the control group for each parental death cohort y to include parental death cohorts $[y + 1, y + 4]$, as opposed to our baseline approach which only uses the $y + 4$ cohort as a control. Column 3 of Appendix table 24 shows that expanding the inheritance control group has little impact on our main income effect estimates from column 3 of table 8.

⁶¹In particular, the above-median group earns 3x more than the below-median group.

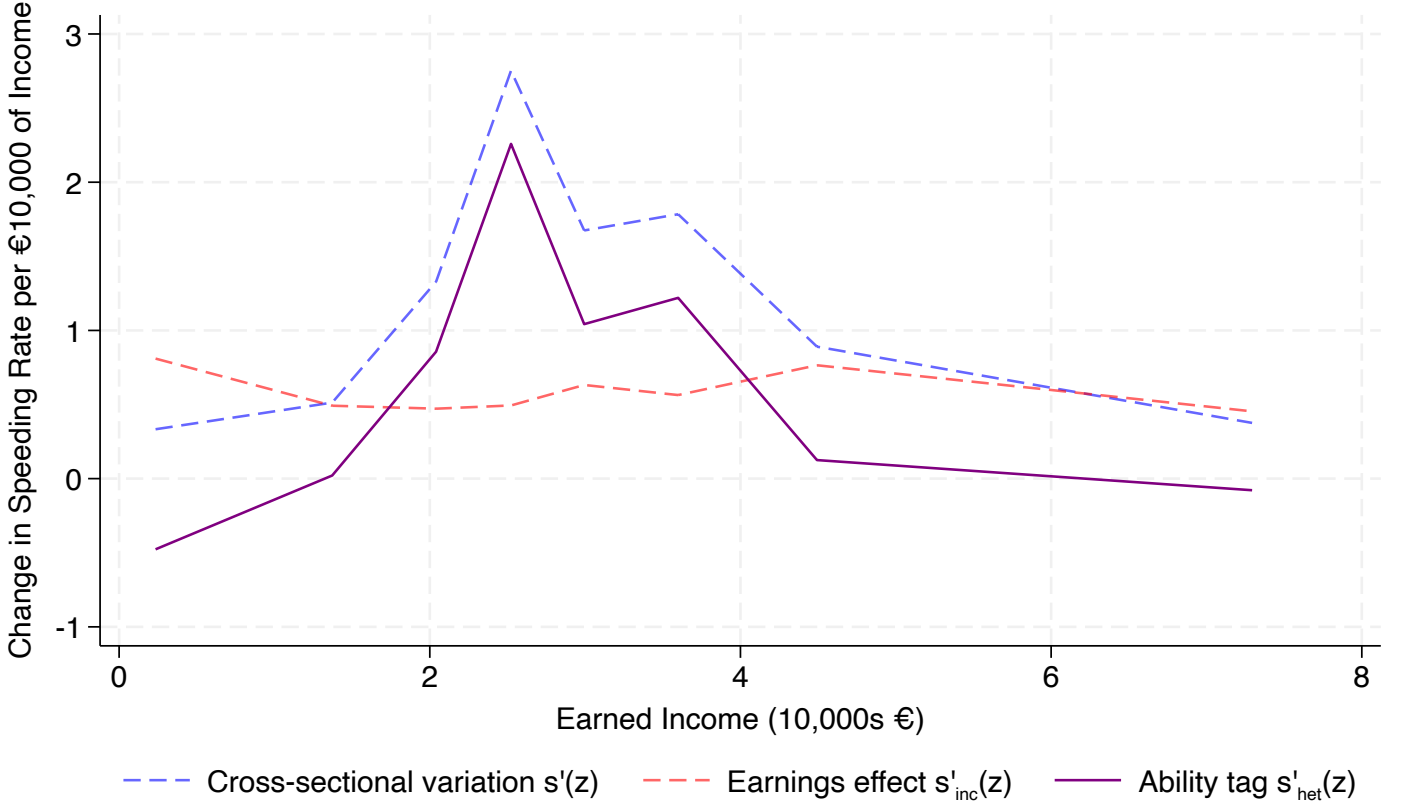
⁶²Interacted income effects from the DDD-IV approach (available upon request) indicate that income effects follow an inverted u-shape; however, the interactions with earnings are highly imprecise. Furthermore, the predicted differences across incomes are small, with most of the income distribution falling in an effect-range of $[0.2, 0.25]$.

⁶³In particular, for all income groups $j < 8$, we calculate $s'(z_j)$ as $\frac{s(z_{j+1}) - s(z_j)}{z_{j+1} - z_j}$; for the top income group, we estimate use a regression to estimate the cross-sectional relationship between earnings and speeding rates within this group, as in column 1 of table 7.

⁶⁴See Appendix table 26 for the estimates underlying this graph.

⁶⁵With a more general non-linear speeding-fine system, the purple profile suggests that the fine schedule $\tau(z)$, which is the marginal cost of speeding for a z -earner, should follow an inverted u-shape (Ferey et al., 2024).

Figure 13: Relationship between Ability Tag and Earnings

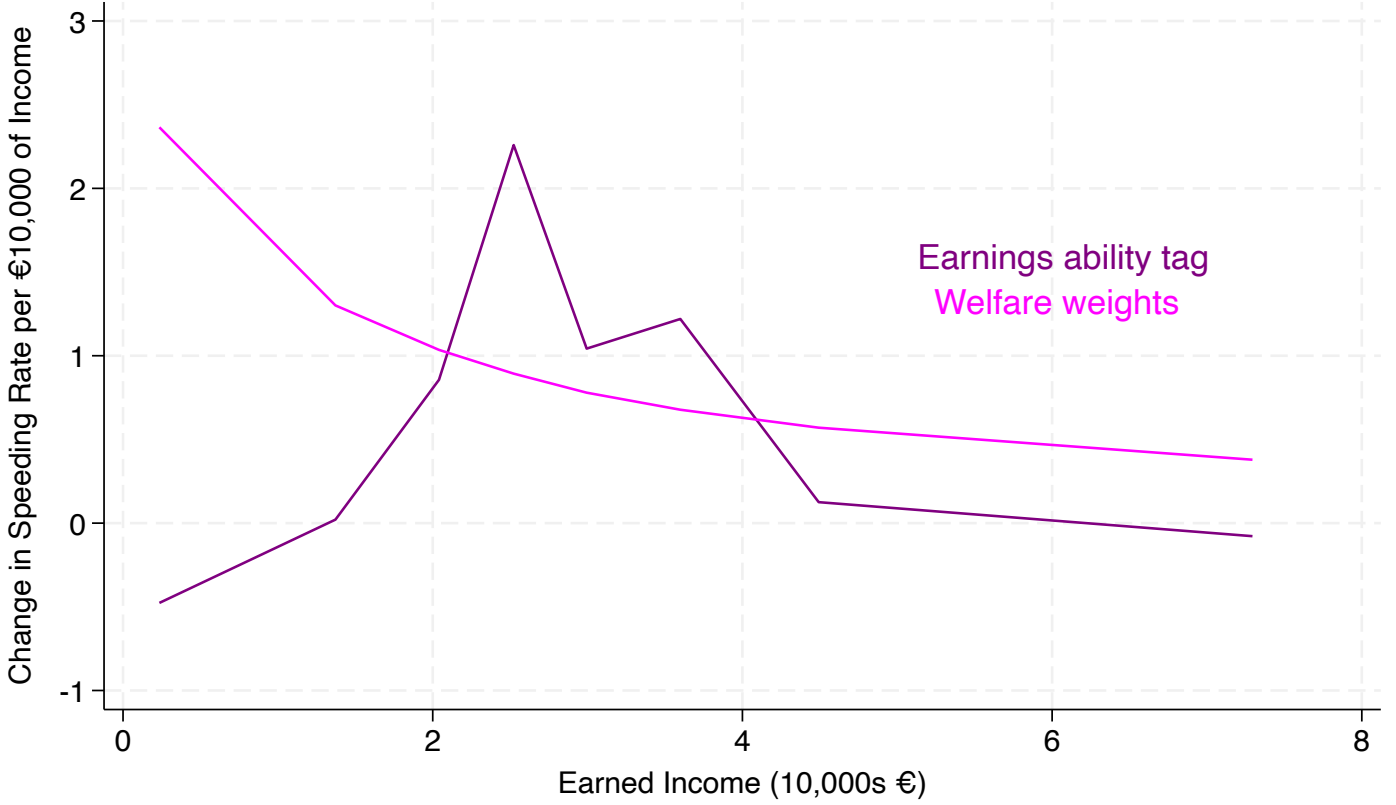


Description

detailed in section 5.3, using welfare weights $g(z) \propto y^{-\nu}$, where y is net income Saez (2002b). Figure 14 plots the earnings ability tag $s'_{het}(z)$ alongside welfare weights derived from $\nu = 1$ (*i.e.* assuming log utility). The two series move in opposite directions at low incomes, before co-moving for the rest of the earnings distribution. Under this parameterization of welfare weights ($\nu = 1$) and using the fine elasticity $|\zeta| = 0.157$ from Kaila (2024), we find that the optimal redistributive gradient consistent with equation 12 is equal to -12€ per €10,000 of earnings. In other words, given our estimates and the current distribution of income X speeding rates X speeding fine levels,⁶⁶ fines should *decrease* by €12 for every €10,000 of income at the optimum; this contrasts with our estimate of the overall speeding-fine gradient (+12.4 €) and the gradient for income-dependent speeding fines (+88 €) from section 2.

⁶⁶For the distribution of these moments that we use (from 2012), see Appendix figure 26.

Figure 14: Covariance between Welfare Weights and Ability Tag



To assess the robustness of our results, Appendix figure 30 shows a heatmap of the optimal redistributive speeding-fine gradient as a function of the fine elasticity $|\zeta| \in \{0.1, 0.157, 0.2\}$ and the degree of inequality aversion $\nu \in \{0.25, 1, 4\}$. Across all considered values, we find a negative gradient, with the strongest (*i.e.* most negative) gradients arising from stronger redistributive preferences and lower elasticities. While the latter is mechanical (higher elasticities increase the distortion in the denominator of equation 12), the relationship with the degree of inequality aversion ν reflects the fact that, as ν increases, the planner wants to move closer to maximizing revenue collection at higher incomes for the purpose of redistribution; the negative relationship between the ability tag and earnings in figure 14 implies that this is most efficiently done by increasing income tax rates at high incomes and lowering fines.

In summary, we find that the earnings ability tag latent in speeding behavior is *decreasing*

in earnings overall, implying that speeding fines are less efficient means of redistribution at high incomes. As a result, given our estimates, optimally redistributive speeding fines would *decrease* with income under any conventional parameterization of welfare weights. We therefore reject the redistributive rationale for income-based speeding fines in Finland.

7 Fairness Preferences

We now discuss our survey, which we use to evaluate the two fairness-driven rationales for income-dependent speeding fines outlined in our theory (section ??). In particular, we posit that income-based fines may be valuable as instruments for achieving “equal compliance” (*i.e.* equal speeding rates across incomes) or proportional punishment (which may reflect a desire to equalize marginal deterrence, see section 3). We build our survey to quantify each of these forces, as well as any residual value that respondents may place on income-dependent fines. The key question in our survey asks respondents to hypothetically compare a given fixed-fine policy to a (randomly varied) income-dependent policy; we begin by showing how a simple version of such a comparison can be used to test whether either of the fairness rationales can rationalize income-based fines.

7.1 Conceptual Approach

Suppose there is a prevailing fine schedule $\tau(z) = a^*(b) + b * z$, where $a^*(b)$ is optimal given a slope parameter b . Suppose that assumption 2 holds, *i.e.* that the behavioral- and fine-fairness terms $\bar{e}_s(z)$ and $\bar{e}_\tau(z)$ are affine. Then, the key objects we need to identify in order to determine how each fairness rationale enters the optimal fine gradient b^* are β_s and β_τ , the slope terms for behavioral- and fine-fairness respectively (see equation 3). The key result informing our survey design is that these slope parameters can be identified from comparisons between simple two-tiered fine policies $\{\tau_{below}(z), \tau_{above}(z)\}$ that assign a fine $\tau_{above}(z)$ for above-median earners and a fine $\tau_{below}(z)$ for below-median

earners. To see this, consider a given two-tiered speeding fine policy with associated speeding behavior $\{s_{below}(z), s_{above}(z)\}$. Then, randomly varied $\{\{-d\tau, d\tau\}, \{ds, -ds\}\}$ reforms, *i.e.* those with equal-and-opposite fine/behavior changes, can be used to identify β_s and β_τ . The intuition is provided in table 9; because the two income groups are equal-sized, the equal-and-opposite nature of the reform eliminates each of the α intercept terms, implying that total fairness effects depend only on the β slope terms.

Table 9: Fairness Effects of Increasing Fine Slope

	$z < z_{med}$	$z > z_{med}$	Total
Fine Changes	$-d\tau$	$d\tau$	0
Fine-Fairness Effects ⁶⁷	$0.5(\alpha_\tau - \beta_\tau \bar{z}_{below})d\tau$	$-0.5(\alpha_\tau - \beta_\tau \bar{z}_{above})d\tau$	$\beta_\tau * d\tau * 0.5 * \Delta \bar{z}$
Behavioral Responses	ds	$-ds$	0
Behavioral-Fairness Effects ⁶⁸	$-0.5(\alpha_s + \beta_s \bar{z}_{below})ds$	$0.5(\alpha_s + \beta_s \bar{z}_{above})ds$	$\beta_s * ds * 0.5 * \Delta \bar{z}$
Residual	$\beta_0 - \beta_0 * I(b > 0)$		

In table 9, we also introduce a residual preference for income-dependence that “turns on” when the fine schedule exhibits *some* income-dependence;⁶⁹ this term captures preferences for income-dependence that are unrelated to induced behavior or the steepness of the fine schedule.⁷⁰ To identify this residual preference (which is constant in comparisons between two income-dependent fine policies), in addition to the behavioral- and fine-fairness effects, we therefore ask survey respondents to make comparisons between hypothetical fixed- and income-based fine policies, where we vary the induced behavior and fine schedule of the income-dependent policy across respondents.⁷¹

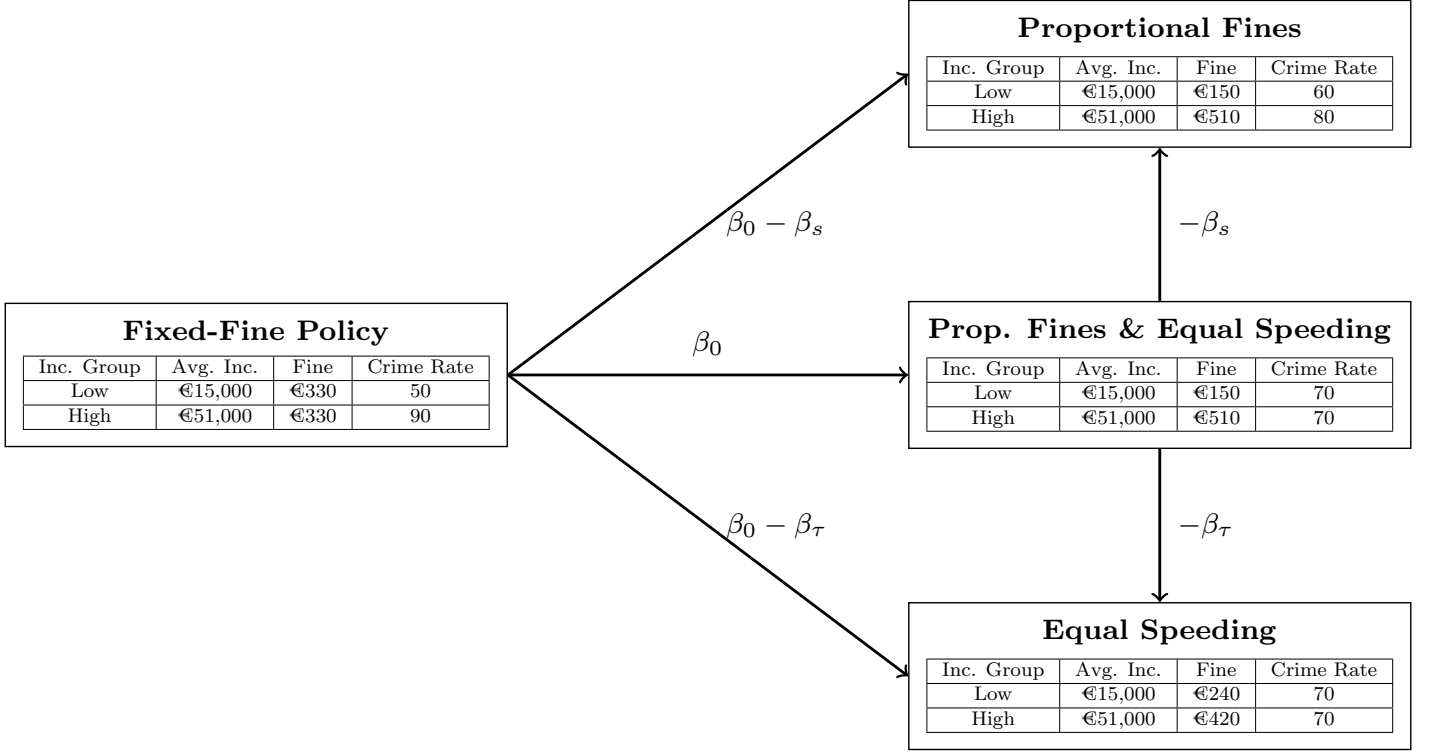
Figure 15 shows our identification approach, where β_s and β_τ are normalized as the effects of a 10-unit crime rate change and €90 fine change respectively. The policy in the left

⁶⁹Note that, conditional on having a positive fine slope $b > 0$, the fine- and behavioral-fairness terms inform the desired “steepness” of the slope, while the warm-glow term only contributes to extensive-margin choice between $b = 0$ and $b > 0$.

⁷⁰These preferences are also orthogonal to externality- or redistribution-based rationales, which we’ve addressed elsewhere.

⁷¹We discuss the reason for an across-respondent design in section 7.2.

Figure 15: Survey Identification



column has a fixed fine and inequality in crime rates across incomes, whereas the middle policy in the right column levies fines proportional to income and induces equal speeding rates across incomes. To mirror our pre-registered regression, β_0 measures the value of an income-based policy with *proportional fines and equal speeding rates*. If equal compliance (*i.e.* equal speeding rates) is the instrumental objective of income-based fines, then respondents should value the income-based policy in the first row (“Proportional Fines”) less than the income-based policy in the second row (“Proportional Fines and Equal Speeding”), implying $\beta_s > 0$. Similarly, if proportional punishment is the instrumental objective, then respondents should value the income-based policy in the second row more than the income-based policy in the third row, implying $\beta_\tau > 0$. Any residual value not attributable to these effects is ascribed to β_0 .

Beyond simplicity, an advantage using equal-and-opposite reforms is that they do not change the overall speeding rate since (recall that the income groups are equal-sized); therefore, unless respondents believe that the cost of speeding depends on the income

of the speeder, safety-related costs should not be a confound between the two policies.⁷² However, three other confounds might arise. First, the comparison across income-dependent policies with different behavior distributions is confounded by the behavior *change* induced by the income-based policy.⁷³ We therefore also randomize the size of the equal-and-opposite behavior change induced by a given income-based policy; implicitly, this provides another way to test the behavioral fairness rationale since, for a given income-dependent policy, increasing the behavior change implies more inequality in the fixed fine policy. If respondents value increases in behavioral equality, they should place a higher value on income-dependent policies that induces larger behavior changes.

Second, in comparison to a fixed-fine policy, fine policies that lower fines for low-income individuals may be desirable due to liquidity concerns, a rationale that is arguably separate from the fairness concerns we’re interested in.⁷⁴ To deal with this concern, we vary the fixed-fine level, which, by construction, also equals the average fine under the income-dependent policy:⁷⁵ if liquidity drives demand for income-dependent fines, then demand should decrease as we lower the fixed-fine level.

Finally, the comparison between fixed and income-dependent policies may be confounded by redistributive preferences: respondents may favor income-dependent fines simply because they implicitly transfer money to infra-marginal low-income individuals (at the expense of high-income individuals). To first-order, increasing the speeding fine by $d\tau$ for an individual with a speeding rate s_i is equivalent to taking $d\tau * s_i$ € away from individual i ; our approach to addressing redistribution builds on this first-order approach. In particular, we ask respondents to compare a baseline fixed transfer to an income-

⁷²See section 7.2 for how we make this aspect salient to respondents. We also conduct robustness checks based on free-response data, see section 7.6.

⁷³For example, in figure 15, the equal-and-opposite behavior change (relative to the fixed-fine policy) induced by the income-based policy in the first row is 10; for the income-based policy in the second row, the change is 20.

⁷⁴E.g., the formalization of the marginal deterrence rationale outlined in section 3 is a static model with no liquidity concerns.

⁷⁵By average, we mean an unweighted average between the statutory fine levels, *i.e.* not a behavior-weighted average.

dependent transfer, where the difference between the two transfers captures the implicit (first-order) redistribution implied by income-dependent fines.⁷⁶ We then difference out respondents' valuations for the income-dependent transfer from their valuations of the income-dependent fine policy, yielding a net-of-redistribution value for income-dependent fines.

7.2 Survey Design

We now discuss the key elements of our survey design that facilitate the hypothetical policy comparisons discussed in section 7.1, beginning with the comparison between fine policies. This comparison consists of three steps: (1) an explanation + comprehension question, in which we ensure that respondents understand what we mean by a speeding-crime rate;⁷⁷ (2) a qualitative choice between fixed and income-dependent fine policies; and (3) a slider that respondents use to indicate how much government revenue they would be willing to forgo in order to implement their preferred fine policy. In the preamble to the qualitative fine-policy comparison (shown in Appendix section 10.2.6), respondents are told that their own hypothetical fine is unaffected and to make a choice based on what would be better for Finland overall; this removes own-fine concerns from consideration and is intended to put respondents in the frame of a planner making trade-offs between social objectives.⁷⁸

After this preamble, respondents are shown two fine policies, as in figure 16. Note that, at the bottom of each policy table, we indicate the total speeding tickets under each policy. To control for safety concerns, this number is held fixed, a fact that we emphasize using the following instructions that follow the policy tables:

Please note that the systems differ in how much speeding is done in different

⁷⁶For details on how we construct these transfers, see Appendix section 10.2.5.

⁷⁷We force respondents to correctly answer this question before proceeding.

⁷⁸The instructions also emphasize that income groups are equal-sized, which is meant to control the relative weight that subjects place on the two groups. It also ensures that inferences about the total speeding rate are consistent with what we tell respondents about total speeding behavior.

Figure 16: Example Fine-Policy Comparison

Fixed fines		
<u>Income Group</u>	<u>Speeding Fine</u>	<u>Fines per 1000 people</u>
Low (<i>Average Income: 15 000 €</i>)	330 €	50
High (<i>Average Income: 51 000 €</i>)	330 €	90
<u>Total speeding tickets per year:</u> 315 000		

Income-based fines		
<u>Income Group</u>	<u>Speeding Fine</u>	<u>Fines per 1000 people</u>
Low (<i>Average Income: 15 000 €</i>)	150 €	70
High (<i>Average Income: 51 000 €</i>)	510 €	70
<u>Total speeding tickets per year:</u> 315 000		

income groups and how large the fines are. However, the overall speeding rate is the same in both systems. The systems are otherwise completely similar.

After making a choice between the two policies, we tell respondents that the government could hypothetically save on administrative costs by implementing their dis-preferred system. Respondents then use a slider (see Appendix figure 40) to indicate the minimum annual savings that would compel them to switch to their dis-preferred policy. We display the savings both in both total and per-capita terms; the slider is capped at €275 of annual savings per Finnish adult, which corresponds to a total savings of €1.238 billion.

We elicit valuations for income-dependent transfers in the same manner, starting with a qualitative comparison (see Appendix figure 41) followed by a slider to elicit willingness-to-pay in terms of government revenue. As in the fine policy comparison, we tell respondents that they would not receive a transfer personally and to choose the better policy for Finland overall.⁷⁹ We randomize whether or not the transfer questions appear before the fine policy questions and include an attention check (described in section 7.3) between the two sets of policy questions. We collect only one set of responses (*i.e.* one fine-policy comparison and one transfer comparison) per respondent for two reasons: (1) the survey as-is takes the median respondent around 13 minutes to complete and (2) we wish to avoid showing respondents mutually inconsistent fine policies.⁸⁰

In addition to these questions, we ask two questions that are meant to determine whether individuals prefer income-based fines (1) without priming and (2) with minimal information (*i.e.* outside the context of a policy comparison). We deal with (1) using a vignette question in which we describe a hypothetical individual who receives a speeding ticket and ask respondents to determine an appropriate punishment (see section 7.6 for more

⁷⁹One difference between the two comparisons is that the transfers are motivated as a one-time occurrence, not as a policy. The one-time story is more natural to motivate in the transfer scenario; because the government savings being considered in the transfer question are also one-time savings (as opposed to annual savings in the fine-policy comparison), the distinction does not matter for comparing valuations between the two policy types.

⁸⁰*E.g.*, we might show, over the course of two questions, income-dependent policies with the same fines but different speeding rates.

details). We address (2) using a straightforward question: *“Do you think the size of speeding tickets should be determined based on an individual’s income?”* Both of these questions are elicited at the beginning of the survey. At the end of the survey, we ask respondents to provide background characteristics and describe beliefs about speeding behavior that we detail in subsequent sections.

Finally, after each slider question, we ask respondents to describe the factors that influenced their qualitative choice;⁸¹ we use these responses to evaluate the persistence of confounds in our elicitation (see section 7.6).

7.3 Survey Recruitment

We collected 3,038 survey responses from a participant pool of Finnish residents maintained by Norstat; this pool regularly participates in surveys distributed by Norstat. To create representative survey samples, Norstat monitors the demographics of respondents as surveys are completed and dynamically adjusts its contact rate for different demographic groups. As a result, our survey is approximately representative on income, age, and gender based on demographic information we collect from respondents at the end of the survey (see Appendix table 34). With respect to rates of speeding ticket receipt and car-ownership, our sample is also broadly consistent with statistics from administrative data.⁸²

Consistent with our pre-registration, we drop respondents that fail an attention check⁸³

⁸¹For a respondent that chose income-based fines over fixed fines, we ask *“Please use the text box below to describe the factors that influenced your decision to choose Income-based fines over Fixed fines.”*

⁸²In our administrative data on speeding ticket receipt, available through 2019, around 7% of individuals receive a ticket in a given year. In our survey, we find that the share of respondents receiving a ticket in the last 2 years is slightly above 7%, though it is well known that enforcement rates have declined following the implementation of a new system in 2020. For car ownership rates, a 2016 household survey (see here) found that around 74% of households have at least one car; for our survey respondents, this rate is around 80%.

⁸³We use the following attention check, taken from Rees-Jones and Taubinsky (2019):

Sometimes, respondents to online surveys do not read all the instructions carefully and proceed too quickly. With this question, we want to make sure that respondents to this survey pay attention to the instructions provided. We now ask you to simply press the “Next” button at the bottom of the page. Do not select any of the options listed

(104 respondents) or who choose the maximum slider value in *both* the fine and transfer policy questions (210 respondents) in order to construct our main analysis sample.⁸⁴ The summary statistics of our main estimation sample are shown Appendix table 34; differences between the analysis sample and the full set of respondents are negligible.

7.4 Estimation

Our main dependent variable is net-of-redistribution WTP for income-dependent fines, WTP_i^{net} for each respondent i . To calculate this variable, we take $WTP_i^{inc-fine}$, respondent i 's WTP for the income-based fine policy, and subtract $WTP_i^{inc-transfer}$, respondent i 's WTP for the income-based transfer.⁸⁵ Our main regression assesses how this variable responds to the randomized features of the fine-policy comparison, which are displayed in table 10; details on how we choose the average speeding rate, fine levels, and below-/above-median incomes are discussed in Appendix section 10.2.7. Two points are worth emphasizing about table 10. First, starting from an induced speeding behavior distribution with $s_{below}^{inc} = s_{above}^{inc}$ (equal speeding rates), we consider induced distributions with either $s_{above}^{inc} < s_{below}^{inc}$ ((low-income individuals speed less often)) or $s_{above}^{inc} > s_{below}^{inc}$ (low-income individuals speed *more* often) in order to test whether respondents' preferences over inequality depend on which group speeds more. Second, given the average earnings levels of the below- and above-median income groups, the income-based fine schedules we consider can be grouped into three discrete categories:⁸⁶ proportional to income, below-proportional, and above-proportional.⁸⁷ In our regressions, we collapse income-based fine schedules into these categories and test whether individuals are responsive to this categorization. Similar to our treatment of induced behavior, we use below-proportional fines as

below. You can now move forward by pressing "Next".

Respondents are provided three answers, "Never," "Sometimes," and "always;" any respondent who fills in one of these answers fails the attention check.

⁸⁴For respondents that choose the maximum slider value for both the fine and transfer question, we cannot infer the sign of net-of-redistribution WTP for income-dependent fines.

⁸⁵For individuals that prefer fixed fines or fixed transfers, the WTP measures are negative.

⁸⁶This holds regardless of the fixed-fine level.

⁸⁷Formally, proportionality means that $\tau_{above}^{inc}/\tau_{below}^{inc} = \bar{z}_{above}/\bar{z}_{below}$. Below-proportional fines are more equal than the income ratio, while above-proportional fines are more unequal than the income ratio.

Table 10: Survey Randomization

<u>Characteristic</u>	<u>Values</u>
Behavior	
Behavior Change	$s_{below}^{inc} - s_{below}^{fixed} = \Delta s \in \{20, 30\}$
Income-Dependent Speeding Distribution	$(s_{below}^{inc}, s_{above}^{inc}) \in \{(60, 80), (70, 70), (80, 60)\}$
Fines	
Fixed-fine level	$\tau^{fixed} \in \{200, 330\}$
Income-Based Fines ($\tau^{fixed} = 200$)	$(\tau_{below}^{inc}, \tau_{above}^{inc}) \in \{(145, 255), (90, 310), (35, 365)\}$
Income-Based Fines ($\tau^{fixed} = 330$)	$(\tau_{below}^{inc}, \tau_{above}^{inc}) \in \{(240, 420), (150, 510), (60, 600)\}$

our baseline and measure responses to policies with proportional and above-proportional fine schedules.

Echoing our notation elsewhere in the paper, we use b to refer to the fine gradient of the income-based policy and designate b_{prop} as the proportional fine gradient. Letting $I(\cdot)$ be an indicator function, our main regression is given by:

$$\begin{aligned}
 WTP_i^{net} = & \beta_0 - \beta_s^{equal} * I(s_{below}^{inc} < s_{above}^{inc}) + \beta_s^{below} * I(s_{below}^{inc} > s_{above}^{inc}) + \beta_s^{fixed} * \Delta s_i \\
 & - \beta_\tau^{prop} * I(b < b_{prop}) + \beta_\tau^{steep} * I(b > b_{prop}) + \rho X_i + \epsilon_i
 \end{aligned} \tag{13}$$

where X_i can include an indicator for the fixed-fine level (200 € or 330 €) and question-order effects. The parameters β_s^{equal} and β_s^{below} measure the fairness effects of changing the behavior *distribution* induced by the income-dependent policy (holding behavior *change* constant). The coefficient β_s^{prop} measures the value of changing from a policy with $s_{below}^{inc} < s_{above}^{inc}$ to a policy with $s_{below}^{inc} = s_{above}^{inc}$ and therefore corresponds to the β_s parameter in figure 15; meanwhile, β_s^{below} can be used to test whether preferences over inequality depend on which group speeds more often. Similarly, β_τ^{prop} and β_τ^{steep} measure the fairness effects of fine-schedule changes, where the former corresponds to β_τ from

figure 15. We normalize the sign of β_s^{equal} and β_τ^{prop} so that positive coefficients are consistent with the equal-compliance and proportional-punishment rationales respectively: in other words, they measure the effects of achieving equal speeding rates (proportional fines), starting from a lower speeding rate (higher fine) for the low-income group. This normalization allows us to interpret behavior coefficients in the same units: all β_s coefficients measure the effect of increasing the low-income speeding rate (and simultaneously lowering the high-income speeding rate) by 0.01. Similarly, both β_τ coefficients measure the effect of increasing the high-income group's fine (and lowering the low-income group's fine) by €72.5 on-average.⁸⁸ We primarily examine β_s^{above} and β_τ^{flat} to inform our tests of the equal-compliance and proportional-punishment rationales, though, in practice, this distinction does not matter for our results.

The β_s^{fixed} parameter measures the effect of increasing behavior change between the two fine policies. As discussed in section 7.1, this parameter provides an alternative test of the equal-compliance rationale: conditional on a given behavior distribution under income-based fines, β_s^{fixed} measures the effect of increasing speeding-rate inequality under the fixed-fine policy. If respondents dislike speeding-rate inequality in general, then $\beta_s^{fixed} > 0$, *i.e.* respondents place lower value on fixed-fine policies with more inequality. As in figure 15, the constant term β_0 measures WTP for a policy with proportional fines and equal speeding rates.

7.5 Results

7.5.1 Extensive-Margin

We begin by discussing respondents' average extensive-margin preferences. Table 11 shows the share of our main sample that (1) strictly prefers the income-dependent fine policy (row 1), (2) strictly prefers the income-dependent transfer (row 2), and (3) has positive

⁸⁸Given the randomized values in table 35, the implied average fine change between adjacent fine-schedule categories is equal to €72.5, *i.e.* the average of 90 and 55, when we include both fixed-fine levels in our regression. This value will change when we condition on a fixed-fine level.

net-of-redistribution WTP (row 3). More than 3/4 of our sample chooses the income-dependent fine policy and indicates a positive WTP for it; for the income-dependent transfer, that share is slightly lower, but still more 2/3. On net, a bit more than 1/2 of our sample prefers the income-dependent fine policy after accounting for redistributive preferences expressed in the transfer comparison.

Table 11: Preferences for Income-Dependent Fines

$\Pr(WTP_i^{\text{inc-fine}} > 0)$	0.768*** (0.008)
$\Pr(WTP_i^{\text{inc-transfer}} > 0)$	0.678*** (0.009)
$\Pr(WTP_i^{\text{net}} > 0)$	0.558*** (0.009)
$\Pr(WTP_i^{\text{fine}} > 0) - \Pr(WTP_i^{\text{net}} > 0)$	0.209*** (0.01)
N	2727

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 12 estimates equation 13 using extensive-margin outcomes. Column 1 examines preferences for the income-dependent fine policy prior to accounting for redistribution; we see a very high probability (80%) of choosing the proportional-and-equal omitted policy in row 1. In the second row, we see that the the probability of choosing income-based policies with equal speeding rates is 2.9pp *lower* than that of income-based policies with $s_{\text{below}}^{\text{inc}} < s_{\text{above}}^{\text{inc}}$, although the results are quite imprecise. In rows 3 and 4, we see that further increasing the speeding rate for the low-income group leads to successive noisy decreases in the probability of choosing the income-dependent policy. In the bottom two rows, we see that the response to steeper fine schedules is quite muted. In column 2, we examine results using the net-of-redistribution probability and find similar effects on the behavioral margin and some weak evidence that respondents are less likely to choose

steeper fine schedules (fifth row). Because the behavior effects are roughly constant, in Appendix table 33, we combine these coefficients⁸⁹ and find a significant average effect of -2.5 percentage points per 0.01-unit increase in the low-income speeding rate; we caution, however, that this is not a pre-registered analysis. These results suggest that, on the extensive-margin, the equal-compliance rationale cannot account for income-based fine policies. We now turn to our WTP results to evaluate this claim further.

Table 12: Extensive Margin Regressions

	(1) $Pr(WTP^{\text{inc-fine}} > 0)$ (pp)	(2) $Pr(WTP^{\text{net}} > 0)$ (pp)
Proportional Fines & Equal Speeding	79.69*** (3.125)	58.26*** (3.674)
Behavior		
$\Delta s_{\text{below}}^{\text{inc}} = 0.01$ ($s_{\text{below}}^{\text{inc}} < s_{\text{above}}^{\text{inc}}$)	-2.922 (1.996)	-2.625 (2.346)
$\Delta s_{\text{below}}^{\text{inc}} = 0.01$ ($s_{\text{below}}^{\text{inc}} = s_{\text{above}}^{\text{inc}}$)	-1.888 (1.998)	-2.130 (2.348)
$\Delta s_{\text{below}}^{\text{fixed}} = 0.01$	-2.018 (1.619)	-2.831 (1.903)
Fine Schedule		
$\Delta \tau_{\text{below}} = \text{€}72.5$ (Below-Proportional)	-0.537 (1.976)	-3.791 (2.323)
$\Delta \tau_{\text{below}} = \text{€}72.5$ (Proportional)	-1.395 (2.007)	0.929 (2.359)
N	2727	2727

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

⁸⁹In particular, our independent variable becomes $\underbrace{s_{\text{below}}^{\text{inc}} - s_{\text{below}}^{\text{fixed}}}_{\text{Behavior change}} + \underbrace{s_{\text{below}}^{\text{inc}} - 0.06}_{\text{Behavior distribution}}$, which captures both dimensions (behavior change and induced distribution) that we use to increase the speeding rate of the low-income group relative to the high-income group.

7.5.2 WTP

Table 13 displays average WTP measures (in terms of government revenue) for our main sample. WTP for the income-dependent fine policy is €75 per-capita on average, while average WTP for the income-dependent transfer policy is €43 per-capita. As a result, average net WTP for the income-dependent fine policy is €32 per-capita, corresponding to a total WTP of €144 million.⁹⁰ This indicates a strong fairness-driven support for income-dependent fines on-average and provides reassurance that our elicitation is not purely noise.⁹¹ However, we can also see from table 13 that median net WTP is about 25% as large as the average, though we can reject that the median is equal to 0 and the total implied WTP (€36 million) is still substantial.

Table 13: Mean WTP

$\overline{WTP}^{\text{inc-fine}}$	75.25
$\overline{WTP}^{\text{inc-transfer}}$	43.27
$\overline{WTP}^{\text{net}}$	31.98*** (2.94)
Median WTP_i^{net}	8*** (1.80)
N	2727

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 14 presents estimates of equation 13; columns 1-3 present OLS estimates, while

⁹⁰In our survey, “per-capita” means for every Finnish adult, of which there are around 4.5 million.

⁹¹In particular, respondents can rank policies using our WTP measure.

column 4 examines a quantile (median) regression. Starting with the OLS results, column 1 uses $WTP^{\text{inc-fine}}$, *i.e.* WTP for the income-dependent fine policy before account for transfers, as the outcome, while column 2 uses our main outcome, net WTP. While we see a large difference across these columns in the WTP for the minimal income-based policy (row 1), we see relatively little difference in the coefficients on randomized behavior or fine-schedule characteristics, suggesting that our differencing procedure does not meaningfully affect our tests for fairness rationales. Therefore, focusing on column 2, we find that net WTP is very weakly responsive to randomized behavior or fine-schedule characteristics. Furthermore, while the estimates are noisy, the sign on the effects (which are uniformly negative) runs contrary to the predictions of the equal-compliance and proportional-punishment rationales: starting from unequal speeding rates and below-proportional fines, respondents do not prefer to either increase the speeding rate for the low-income group or increase the fine for the high-income group. Column 3 considers an alternative net WTP measure that assigns a higher WTP value to those who select a maximum slider value to examine whether these observations drive our results.⁹² However, the results using this alternative measure are nearly unchanged relative to column 2. Column 4 estimates a quantile (median) regression; while WTP for the minimal income-based policy is smaller than the estimate from our OLS results, our qualitative results are largely unchanged.

In summary, respondents place a large premium on income-based policies but are on-average insensitive to (1) the distribution of behavior induced by income-dependence, (2) the steepness of the fine schedule, and (3) the behavior change caused by the income-based policy. To probe the heterogeneity in these baseline results, table 15 presents OLS estimates of equation 13 separately for different randomized characteristics: the fixed-fine level, induced behavior change, and question order. Beginning with the fixed-fine level, columns 1 and 2 show estimates of equation 13 for $\tau^{\text{fixed}} = 200$ and $\tau^{\text{fixed}} = 330$ re-

⁹²In our baseline net WTP measure, we assign the maximum slider value, €275, as the WTP for those choosing the maximum value. In our alternative measure, we assign a value of €325.

Table 14: Baseline Regressions

	(1)	(2)	(3)	(4)
	WTP Inc-Dep Fine (per capita)	Net WTP (per capita)	Alternative Net WTP (per capita)	Net WTP (per capita)
Proportional Fines & Equal Speeding	87.56*** (9.557)	45.74*** (11.36)	47.09*** (11.99)	16* (6.337)
Behavior				
$\Delta s_{below}^{inc} = 0.01$ ($s_{below}^{inc} < s_{above}^{inc}$)	-7.483 (6.104)	-3.473 (7.256)	-2.906 (7.659)	-2 (4.048)
$\Delta s_{below}^{inc} = 0.01$ ($s_{below}^{inc} = s_{above}^{inc}$)	-7.899 (6.109)	-7.090 (7.262)	-6.567 (7.666)	-4 (4.051)
$\Delta s_{below}^{fixed} = 0.01$	-4.823 (4.951)	-8.496 (5.886)	-8.618 (6.213)	-6 (3.283)
Fine Schedule				
$\Delta \tau_{below} = \text{€}72.5$ (Below-Proportional)	6.191 (6.044)	-5.022 (7.185)	-5.961 (7.584)	-4 (4.008)
$\Delta \tau_{below} = \text{€}72.5$ (Proportional)	-8.429 (6.137)	-4.641 (7.295)	-5.081 (7.700)	2 (4.069)
N	2727	2727	2727	2727

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

spectively. Focusing on the effect of behavior and fine-schedule characteristics, results are quite noisy when $\tau^{fixed} = 330$; however, we see relatively precise responses to randomized behavior characteristics when $\tau^{fixed} = 200$. Furthermore, these responses are exactly the *opposite* of those predicted by the equal-compliance rationale: when $\tau^{fixed} = 200$, respondents *dislike* policies that increase speeding rates for low-income individuals relative to high-income individuals. We also see that respondents dislike fines that are steeper than proportional when $\tau^{fixed} = 200$; however, this may be due to the fact that the fine is extremely small (€35) for the low-income-group under such a policy.⁹³ Despite the effects that emerge conditional on $\tau^{fixed} = 200$, only one of the differences between the coefficients in columns 1 and 2 are significant at the 5% level. In columns 3 and 4, we examine results conditional on different levels of behavior change; we see no discernible differences across the two columns. In columns 5 and 6, we examine the role of question order (whether respondents answer the transfer or fine comparison first) to test

⁹³Note that we do not see any response to above-proportional fine schedules with $\tau^{fixed} = 330$, when the low-income fine is €60.

how salience and/or learning effects might affect our estimates. Within this comparison, we again find no significant differences, though there is some evidence that respondents *dislike* proportional fines when the transfer question is answered first (fifth row).

Overall, across the 28 coefficients (other than the constant terms) in table 15, only 3 are significant at the 5% level. Furthermore, only one of the 17 differences between coefficients is significant at the 5% in table 15, implying that our baseline findings in table 15 are not particularly sensitive to the features we randomize. Overall, our findings indicate that respondents highly value income-based policies regardless of the behavior distribution or fine-schedule they entail.

Table 15: Heterogeneity by Randomized Characteristics

	Outcome: Net WTP					
	Fixed Fine		Behavior Change Δs		Question Order	
	(1) €200	(2) €330	(3) 20 tickets per 1,000	(4) 30 tickets per 1,000	(5) Transfer Q 1st	(6) Fine Q 1st
Proportional Fines & Equal Speeding	59.92*** (15.81)	29.57 (16.26)	43.05*** (9.951)	23.46* (9.220)	44.54** (15.07)	46.70** (16.83)
Behavior						
$\Delta s_{below}^{inc} = 0.01$ ($s_{below}^{inc} < s_{above}^{inc}$)	-16.12 (10.04)	8.938 (10.43)	-1.682 (10.41)	-4.707 (10.15)	5.135 (9.520)	-10.59 (10.87)
$\Delta s_{below}^{inc} = 0.01$ ($s_{below}^{inc} = s_{above}^{inc}$)	-4.643 (9.916)	-8.067 (10.59)	-16.02 (10.49)	1.623 (10.09)	-17.13 (9.649)	1.208 (10.75)
$\Delta s_{below}^{fixed} = 0.01$	-18.96* (8.055)	3.472 (8.566)			-5.181 (7.758)	-11.21 (8.770)
Fine Schedule						
$\Delta \tau_{below}$ (Below-Proportional)	1.513 (9.863)	-10.20 (10.41)	-1.168 (10.26)	-9.167 (10.07)	-19.85* (9.517)	7.014 (10.68)
$\Delta \tau_{below}$ (Proportional)	-22.85* (9.937)	14.31 (10.64)	-6.743 (10.47)	-2.981 (10.19)	2.912 (9.494)	-11.72 (11.00)
N	1358	1369	1356	1371	1313	1414

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

In Appendix table 36, we explore heterogeneity with respect to background character-

istics (including income, age, and education) and beliefs we collect from respondents at the end of our survey. For the latter, we ask respondents (1) how often they think drivers are aware of the speeding fine they would pay and (2) whether they believe speeding behavior is responsive to fine size.⁹⁴ Overall, we find remarkably little heterogeneity across sub-groups: WTP for the baseline proportional-and-equal policy is consistently large and there are no significant differences within any given sub-group comparison.

To summarize our results thus far, we consistently identify large and significant WTP for a baseline proportional-and-equal fine policy. However, we fail, across a large swath of heterogeneity checks, to find significant responses to (1) the across-income distribution of speeding rates induced by the income-dependent fine policy, (2) the steepness of the income-dependent fine schedule itself, and (3) the behavior change induced by the income-dependent policy. Are these estimates sufficiently small and precise to able to reject a positive fairness-driven fine gradient? To make progress on this question, we translate our estimates into an implied fine gradient using the identification results in table 9 and equation 3. To increase our power, we assume that the effect of increasing the speeding rate of the low-income-group is constant across baseline levels of inequality, which allows us to combine the three behavioral effects from our baseline estimates (column 2, table 14). Similarly, we assume that each €1 increase in the high-income fine (accompanied by a €1 in the low-income fine) has the same effect, motivated by the uniformly negative effects in column 2 of table 14; in our regression, we therefore also include an indicator for the fixed-fine level in our regression estimate to ensure that policy comparisons occur within each fixed-fine level.⁹⁵ Additional details on our approach are available in Appendix section X. In effect, our process amounts to a change in units of the behavior and fine-schedule characteristics; we then regress net WTP on these adjusted variables. Our results are shown in table 16; column 1 shows results from an OLS regression, while column 2 uses a quantile (median) regression. We note four main findings.

⁹⁴See Appendix section 10.2.8 for additional detail on these elicitations.

⁹⁵If we did not do this, we would confound changes in the fine schedule with comparisons across different fixed-fine levels.

First, consistent with our baseline findings in table 14 and our extensive-margin results from table 12, the implied slopes are negative; this arises because respondents expressed a slight distaste for fine policies with steeper schedules and higher speeding rates among the low-income group. As is well-known, hypothetical WTP elicitation may be subject to inflation bias; while our differencing approach deals with some of this, some residual likely remains.⁹⁶ Second, the implied slopes are extremely large; this follows because respondents are willing to sacrifice several euros *per-capita* for policies that ultimately affect a small share of the population. Third, after combining the various behavioral effects, we have enough power to reject behavior-related motives for income-based fines, as in our extensive-margin regression (see Appendix table 33). However, we caution that the combination of these coefficients was not pre-registered. Our fourth finding is that, even after assuming a constant fine-schedule effect, standard errors are large enough that we cannot reject schedule-motivated fairness rationales. In particular, while the estimated coefficient is negative, the size of the standard errors are such we cannot rule out the degree of income-dependence found in the day-fine region of the Finnish speeding schedule (€88, see figure 3). This uncertainty is emphasized in column 2, which estimates a quantile regression and finds a (median) fine fairness effect equal to 0, with large standard errors.

To assess both rationales together, we calculate a total implied gradient (last row of table 16) as the sum of the behavioral- and fine-fairness slopes; in our OLS results, the estimated day-fine gradient of €88 lies right at the edge of our 95% confidence interval: [-1287.1, 85.7]. Therefore, considering the impact of both rationales on the optimal fine gradient, we can reject the degree of income-dependence in the day-fine region of the Finnish speeding-fine schedule.

In summary, responses to increases in the low-income speeding rate, relative to the high-

⁹⁶E.g. if a respondent likes the income-dependent transfers half as much as the income-dependent fine policy, then inflation can persist if the base elicitation (how much they like the fine policy) is inflated.

Table 16: Optimal Fairness-Driven Fine Gradients

	Optimal Fine Gradient b^* (per €10,000 of income)	
	(1)	(2)
Behavioral Fairness	-331.9* (168.0)	-222.2* (98.11)
Fine Fairness	-268.8 (307.5)	0 (179.6)
Residual β_0 (per-capita)	52.41*** (12.61)	20** (7.363)
N	2727	2727
Fixed-Fine Control	X	X
Estimation	OLS	Quantile
Total Gradient	-600.7 (350.2)	-222.2 (204.5)

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

income rate, are precise enough when considered in aggregate to allow us to reject the equal-compliance rationale for income-based fines. Our estimates of the response to increases in the steepness of the fine schedule are more imprecise; we therefore cannot reject this rationale in isolation, though we do not find that respondents place higher value on fine policies with steeper fine schedules. Considering both rationales jointly (table 16), our results are precise enough to reject the degree of income-dependence in the day-fine region of the Finnish speeding-fine schedule. Furthermore, in comparison to residual WTP for income-based fine policies, WTP for behavior or schedule-based characteristics is small. Therefore, our read is that the survey evidence overall is inconsistent with either the equal-compliance or proportional-punishment motives.

We now discuss some additional analyses to probe the robustness of our results.

7.6 Robustness

Safety Concerns While we emphasize to respondents that total speeding is unresponsive to which fine policy is implemented, respondents might therefore believe that the incidence of accident costs is different between the two policies (or that particular groups' speeding behavior is more dangerous). Therefore, to address this potential confound, we asked respondents, after making their slider decision, to provide free-text reasoning for why they made their particular fine-policy choice.⁹⁷ To examine the prevalence of safety-related concerns, we tag all responses that include one of the following strings: {safe, danger, accident}; we do not address the tone in which these strings are mentioned.⁹⁸ In total, only 113 responses (4%) include one of these strings; furthermore, average net WTP among this group is €28, not meaningfully different from all other responses (€32).⁹⁹ This evidence suggests that safety-related concerns do not drive our results.¹⁰⁰

8 Discussion

We now conduct two simple counterfactual exercises in order to compare magnitudes of the effects we've identified. In particular, we estimate the welfare consequences of setting different fine gradients b using equation 3. First we consider decreasing the total fine gradient (measured by combining all speeding crime) from its observed level (€12.4 per €10,000 of earnings) to the optimum implied by redistribution- and externality-based rationales (-€19.6 under our preferred estimates from sections 5.4 and 6.4). This exercise clarifies the consequences of eliminating income-dependence altogether and using standard economic rationales to generate policy prescriptions. In our second counterfactual,

⁹⁷*E.g.* someone who chooses the income-based fine policy sees the following prompt: “Please use the text box below to describe the factors that influenced your decision to choose Income-based fines over Fixed fines.”

⁹⁸*E.g.*, someone might write “Accidents are the same in both policies.”

⁹⁹In unreported results, we re-estimate equation 13 after dropping this group and find little difference in our coefficients.

¹⁰⁰For an expanded notion of safety (which may also include fairness-motivated rationales), we also include responses with the string “deter;” this expands the number of safety-related responses to 230, nearly 10% of our sample. Again, this group has similar net WTP to the rest of the sample, and re-estimating equation 13 with or without these responses makes little difference.

we show the consequences of proportionality by increasing the fine gradient from €12.4 to €88, the latter being the gradient we estimate in the day-fine region of the Finnish fine schedule. Throughout, we assume that the fixed component of the fine schedule $a^*(b)$ and the income tax schedule $T(\cdot)$ optimally adjust; together, these imply that our counterfactuals amount to rotations of the fine schedule around response-weighted average earnings $\bar{z}^{rw}$.¹⁰¹

Throughout, we assume that our estimated behavioral responses (and moments of the income distribution) are constant over the range of changes we consider.¹⁰² As a result, these exercises are merely suggestive.¹⁰³ We show the effects in figure 17; we separate the effects into revenue, externality (*i.e.* reallocation of optimal deterrence), and redistribution components. The first bar shows the effects of optimally setting the gradient to optimally redistribute and correct externalities; overall the effects are quite small. Most of the positive gains are on the redistribution side, with the revenue losses being quite modest. As a result, this reform implies a marginal value of public funds (MVPF) around 5.5. The proportionality reform, on the other hand, is quite distortionary, generating over €5.5 million in losses, which is around 8% of total spending on speeding fines per year in Finland.

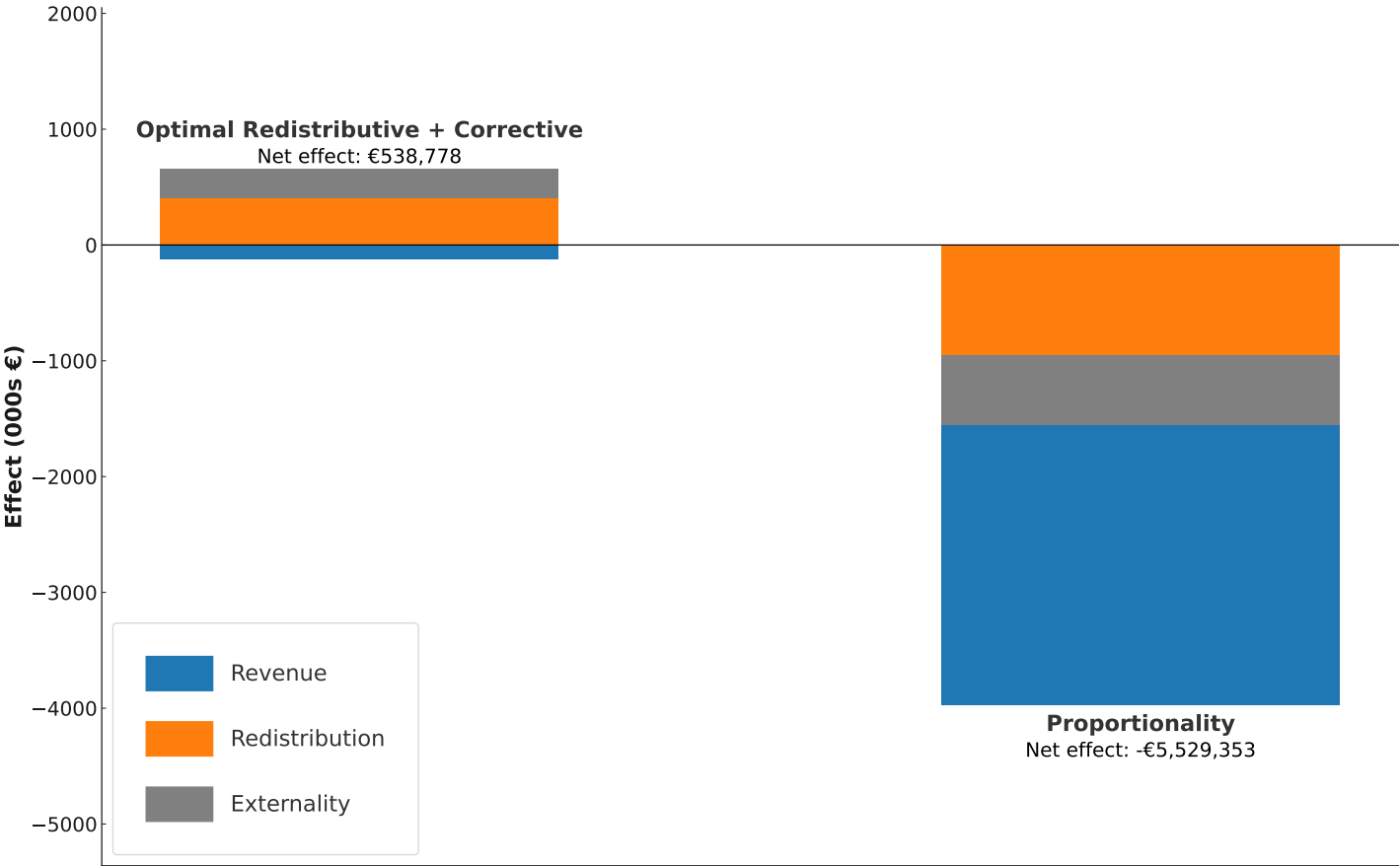
How do these effects compare to respondents' WTP for income-based fine policies? On average, respondents are willing to forgo €144 million in government revenue in order to implement income-based fine policies, and our results indicate that this WTP is relatively insensitive to fine-schedule steepness or induced behavior. Therefore, we assume this value would be forgone by moving to the optimal redistributive + corrective gradient and would not augment by moving to proportionality. As a result, the fairness losses of setting optimal redistributive + corrective fines would be orders of magnitude larger than

¹⁰¹In particular, equation 15 for the optimal fixed component implies that, at any level of b , the fine for those earning $\bar{z}^{rw}$ is held constant.

¹⁰²For details on the moments we use, which are common with those used in our section 6.4, see table 26.

¹⁰³A proper account would allow fine semi-elasticities to change as the gradient changes, for example.

Figure 17: Counterfactual Fine Gradient Policies



losses on other dimensions; this is true even if we use the median WTP in our survey (€36 million). On the other hand, our survey results show no compelling fairness-driven reason to increase the steepness of the fine schedule; as a result, the proportionality counterfactual generates sizable large losses and no fairness gains.

Overall, if we take the WTP of respondents in our survey at face value, the optimal fine gradient should include *some* modest level of income-dependence, but need not include the extremes found in the income-based region of the Finnish speeding-fine schedule, given the sizable welfare losses associated with such a system.¹⁰⁴

9 Conclusion

In this paper, we evaluate rationales for a controversial policy: income-based fines. Using the Finnish speeding fine system as our laboratory, we first consider the Beckerian motive: does the marginal social cost of speeding scale with the income of the offender? Drawing on linked data from accident reports, income tax returns, and crime histories, we exploit cross-sectional variation in speeding histories conditional on income, together with a job-loss design, to recover the income-gradient of marginal social costs. We find that this gradient is *negative*, leading us to reject this rationale.

We then evaluate a standard motivation in the public finance literature: in a Mirrleesian redistributive framework, fines can be exploited to lower the distortionary costs of redistribution if speeding is an informative signal for latent earnings ability. As clarified by recent work in optimal redistributive commodity taxation, this signal can be identified by differencing across-income variation in speeding rates with respect to causal earnings effects on speeding. To identify these causal earnings effects, we leverage two complementary strategies: within-individual earnings variation and differences in the timing

¹⁰⁴In our survey, the most minimal income-based policy we pose to respondents includes a gradient of around €30, somewhat higher than the overall gradient of €12.4 but well below the proportional gradient of €88.

and size of inheritance shocks. We find that the estimated earnings ability signal is *negatively* correlated with income, implying that Mirrleesian redistributive rationales can only rationalize a *lower* speeding fine for the rich.

Finally, we consider two fairness rationales alluded to in both statutory motivations for the policy and the law literature: equal compliance across incomes and proportional punishment. To evaluate these motives, we use data from an original survey in which Finnish residents make trade-offs between fixed- and income-based fine policies, where the latter randomly varies across respondents with respect to both schedule “steepness” of the schedule and the induced across-income distribution of speeding rates. We find that income-based fines are popular: net of redistribution, respondents are willing to forgo €144 million in government revenue in order to implement an income-based speeding fine policy. However, these valuations are insensitive to both the distribution of behavior induced by fines and the steepness of the fine schedule itself. Therefore, our evidence is inconsistent with the equal-compliance and proportional-punishment rationales.

Overall, our evidence suggests that oft-cited rationales for income-based fines are insufficient to justify their popularity. Furthermore, we explicitly reject two natural economic objectives for such policies, externality mitigation and redistribution. Therefore, accounting for these policies requires enriching either our normative criteria (e.g. to include procedural fairness or deontological concerns) or our economic models (e.g. by adding liquidity concerns and/or income-correlated mis-perceptions).

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10 Appendix

10.1 Expressions for Fines and Taxes

Given $\tau(z) = a + b * z$, the optimal marginal income tax rate satisfies the following at each earnings level z^0 :

$$T'(z^0) + b * s(z^0) + [\tau(z^0) - \bar{e}'(z^0)]s'_{inc}(z^0) = \frac{\int_{z \geq z^0} (\hat{g}(z) - 1) dH(z)}{h(z^0)z'_{T_z}(z^0)} \quad (14)$$

where $z'_{T_z}(z^0)$ is the (semi-)elasticity of labor supply with respect to the marginal income tax rate and $h(\cdot)$ is the earnings density function. Under an optimal income-tax schedule, if assumption 1 holds, then the fixed component of fines satisfies the following for any given b :

$$a^*(b) = \frac{\mathbb{E}[s'_{\tau_s}(z) * \bar{e}'(z)] + \text{Cov}[s'(z) - s'_{inc}(z), \hat{g}(z)] - b * \mathbb{E}[s'_{\tau_s}(z) * z]}{\bar{s}'_{\tau_s}} \quad (15)$$

10.2 Empirical Appendix

10.2.1 Procedure for Calculating Money-Metric Cost of Injuries and Deaths from Accidents

We proceed as follows

1. For each driver j involved in an accident a_t in year t , we identify the set of externality-relevant accident victims $\{v_1(j, a_t), v_2(j, a_t), \dots, v_n(j, a_t)\}$.
2. For each victim $v_m(j, a_t)$ we randomly select two “control” individuals, $c_1(v_m)$ and $c_2(v_m)$, from the FOLK population register who have the same birth year, sex, and education level as v_m in year $t - 1$.
3. We compute disposable income differences for each victim-year $\{\Delta \hat{y}_{v_m, t-1}, \Delta \hat{y}_{v_m, t}, \dots, \Delta \hat{y}_{v_m, 2023}\}$ until 2023, where $\Delta \hat{y}_{v_m, t+r} := y_{v_m, t+r} - [(y_{c_1(v_m), t+r} + y_{c_2(v_m), t+r})/2]$ for all $r \in \{-1, 0, 1, \dots, 2023 - t\}$.
4. Similarly, we compute differences in net transfers received for each victim-year $\{\Delta \hat{T}_{v_m, t-1}, \Delta \hat{T}_{v_m, t}, \dots, \Delta \hat{T}_{v_m, 2023}\}$ until 2023, where $\Delta \hat{T}_{v_m, t+r} := T_{v_m, t+r} - [(T_{c_1(v_m), t+r} + T_{c_2(v_m), t+r})/2]$ for all $r \in \{-1, 0, 1, \dots, 2023 - t\}$.
5. We calculate victim-specific DID net-income losses in each event year $r \geq 0$ as $\Delta \hat{y}_{v_m, t+r}^{DID} := \Delta \hat{y}_{v_m, t+r} - \Delta \hat{y}_{v_m, t-1}$. We analogously calculate victim-specific DID for net transfer received, $\Delta \hat{T}_{v_m, t+r}^{DID}$.
6. We calculate a welfare-weighted, NPV cost for each victim v_m as

$$M_{v_m} := \sum_{k=0}^{2023-t} \delta^k \left[\underbrace{\Delta \hat{T}_{v_m, t+k}^{DID}}_{\text{Transfers Paid by Govt.}} - \underbrace{g(y_{v_m, t-1}) * \Delta \hat{y}_{v_m, t+k}^{DID}}_{\text{Welfare-weighted income gain}} \right]$$

under some exponential discount rate δ and $g(\cdot)$ denotes a set of social marginal welfare weights

7. The total welfare-weighted external income loss imposed by driver j in accident a_t is

$$M_j^{ext}(a_t) := \sum_{m=1}^n M_{v_m}$$

10.2.2 Bias in Income Effects Estimated Using Inheritance Shocks

Let α be the death effect (assumed homogeneous), η_q be the income effect for quintile q , and ΔW_q be the size of the total inheritance shock received by quintile q . Then, for a given q , the total effect of parental death is $\alpha + \eta_q * \Delta W_q$. Differencing effects across two quintiles q and q' yields $\eta_q * \Delta W_q - \eta_{q'} * \Delta W_{q'} = (\eta_q - \eta_{q'}) * \Delta W_q + \eta_{q'}(\Delta W_q - \Delta W_{q'})$. Dividing by $\Delta W_q - \Delta W_{q'}$ (*i.e.* the first-stage) yields $\eta_{q'} + (\eta_q - \eta_{q'}) * \frac{1}{1 - \Delta W_{q'}/\Delta W_q}$, which equals η_q if $\Delta W_{q'} = 0$, $\eta_{q'} = \eta_q$, or $\eta_{q'} = 0$. Furthermore, as $\Delta W_{q'}/\Delta W_q \rightarrow 0$, this expression converges to η_q .

10.2.3 Defining Transport-Related and Motor-Vehicle Accidents

Transport Related Accidents: This corresponds to Chapter V deaths in the cause-of-death code.

Motor-vehicle accidents: Within Chapter V, any death that occurs when a [pedestrian, cyclist, motorcycle rider, occupant of 3-wheeled motor vehicle, car occupant, occupant of a pick-up truck or van, occupant of a heavy transport vehicle, bus occupant] is killed in a collision with a [two- or three-wheeled motor vehicle, car, pick-up truck, van, heavy transport vehicle, bus]. We ignore collisions with pedestrians, cyclists, an animal, trains, non-motor vehicle, or stationary objects and any non-collision transport-related deaths. Chapter code 2: [0,7], chapter code 3: [2,4].

10.2.4 Imputing Driving Distance Using Odometer Data

We start with a quarterly panel of vehicles from the vehicle registration data and keep only quarters in which we observe an odometer reading for a vehicle. We then group our data into vehicle-by-primary-user cells. Each of these cells constitutes a panel where we observe odometer readings over time for each vehicle-user. Between any two successive odometer readings for a vehicle-user cell, we calculate distance driven between inspections. We then uniformly assign this distance across quarters between the two inspection dates, assuming that inspections occur at the beginning of each quarter.¹⁰⁵ Aggregating across vehicles and quarters, we obtain a user-by-year panel of driving distance. Driving distance is non-missing for a given user-year (i, t) in our panel if and only if at least one quarter in year t falls between two inspections for a given vehicle registered to user i . Note that, if a user switches from an old vehicle h to a new vehicle h' , we will not be able to calculate distance driven between the last inspection date for the old vehicle h and the first inspection date for the new vehicle h' .

10.2.5 Construction of Survey Transfer Policies

Let $\{\bar{s}_{below}^{inc}, \bar{s}_{above}^{inc}\}$ be speeding rates under the income-dependent fine policy and $d\tau = \tau_{above}^{inc} - \tau_{above}^{fixed}$ be the fine difference for the high-income group. We ask individuals to compare a fixed transfer policy that gives 25 € to all Finnish adults with an income-dependent transfer that allocates $25 + d\tau * s_{below}^{inc}$ to the low-income group and $25 - d\tau * s_{above}^{inc}$ to the high-income group. We use the speeding rates under the income-dependent policy, rather than those under the fixed-fine policy, since this choice guarantees that revenue effects are the same for income-dependent transfer and fine policies. To see this, note that the revenue effect of the income-dependent fine policy is given by:

$$\Delta R_{fine} = 0.5[s_{below}^{inc} * \tau_{below}^{inc} + s_{above}^{inc} * \tau_{above}^{inc}] - 0.5 * \tau^{fixed} * [s_{below}^{fixed} + s_{above}^{fixed}]$$

¹⁰⁵ Assuming the inspections occur at the end of the quarter makes no difference for our results.

Since total speeding is the same under both policies, we can replace $s_{below}^{fixed} + s_{above}^{fixed}$ with $s_{below}^{inc} + s_{above}^{inc}$ and it immediately follows that:

$$\Delta R_{fine} = 0.5 * d\tau * [s_{above}^{inc} - s_{below}^{inc}]$$

Thus, increasing the low-income transfer by $d\tau * s_{below}^{inc}$ (*i.e.* spending an additional $d\tau * s_{below}^{inc}$ per low-income person) and lowering the transfer by $d\tau * s_{above}^{inc}$ for the high-income group yields an equivalent revenue effect.

In general, our method for constructing the income-dependent transfer will tend to inflate the role of redistribution when (1) “altruism” weights α on each group are positive and (2) $\alpha_{below} > \alpha_{above}$. In particular, for baseline speeding rates $\{s_{below}, s_{above}\}$, the welfare effect of transfers $\{d\tau * s_{below}, -d\tau * s_{above}\}$ is given by:

$$dW = d\tau * \alpha_{below} * (s_{below} - \frac{\alpha_{above}}{\alpha_{below}} * s_{above})$$

With $\alpha_{above} > 0$ and $\alpha_{above} < \alpha_{below}$, this value will be maximized when s_{below} is large and s_{above} is small. Since the income-dependent policies we consider increase the speeding rate for the low-income and decrease it for the high-income group, dW will be larger if we use the speeding rates under the income-dependent policy to construct the income-dependent transfer, making our estimates for the value of income-dependent fines more conservative. This result also holds if $\alpha_{above} < 0$ with $\alpha_{below} > |\alpha_{above}| > 0$.

10.2.6 Instructions for Fine-Policy Comparisons (Preamble)

Respondents see the following text above the two fine policies they are asked to compare:

This question compares two hypothetical speeding ticket systems that could be introduced in Finland:

1. *Fixed fines - the fines are the same for everyone.*

2. *Income-based fines - the size of the fine depends on whether a person belongs to a low or high income group.*

We ask you to choose which system would be better for Finland overall; suppose that the fine YOU would have to pay would not be determined by the systems described above.

The tables below present these two systems for two different income groups with different income levels. Each income group covers half of the adult population in Finland. The low-income group includes the half of the population with the lowest income. The high-income group includes the half with the highest income.

10.2.7 Choice of Fine Policy Parameters

Average Incomes: We set the above- and below-median incomes (€15,000 and €51,000 respectively) using the 2023 distribution of earned income; we lower the income of the low-income group relative to these calculations in order to make proportional fines more apparent in the income-based policy.

$\tau^{\text{fixed}} = \mathbf{330}$: We calculate the higher fixed-fine of €330 as the average of the proportional fines €150 and €510; these proportional fines, which are 1% of average income for each group, are reflective of the degree of income-dependence we estimate on-average within the day-fine region of the Finnish fine schedule (see figure 3).

$\tau^{\text{fixed}} = \mathbf{200}$: The fixed-fine of €200 is close to the average speeding fine received in Finland (around 215 €). We calculate proportional income-based fines relative to this average fine level using a slightly lower proportionality rate of 0.6%, which guarantees that the fine difference between the fixed and income-based policies is the same in relative terms for both levels of the average fine. In particular, $(200-90)/200 \approx (330-150)/330$.

$d\tau$: We choose the size of the fine perturbations $d\tau$ (€90 for $\tau^{fixed} = 330$ and €55 for $\tau^{fixed} = 200$) around the proportional policy to be roughly equivalent relative to the level of the fixed fine.

Speeding Rate: We choose the average speeding rate to reflect the average in Finland in 2019, our last complete year of data.

10.2.8 Survey Belief Elicitations

Fine Awareness We ask respondents to indicate how often (from never to always) they believe “drivers are aware of the fine they would receive if they were caught speeding.” In Appendix table 36, we separate respondents into those who answer “rarely” or “never” (column 3) and those who answer at least as often as “sometimes” (column 4); see Appendix figure 42 for the distribution of responses.

Fine Responsiveness We ask respondents to indicate how much they agree with the following statement:

“Drivers will speed less if speeding fines are increased.”

In Appendix table 36, we group respondents into {“strongly disagree,” “disagree,” “neither agree nor disagree”} (column 5) and {“agree,” “strongly agree”} (column 6); the distribution of responses is shown in Appendix figure 43.

10.2.9 Counterfactual Analysis

When the fixed component of the fine schedule $a^*(b)$ is optimal and the income tax $T(\cdot)$ is optimal, then a small increase in the fine slope b yields a welfare effect (ignoring fairness considerations) given by:

$$dW = (b - \beta^M) * db * \int_Z s'_{\tau_s}(z) * z * (z - \bar{z}^{rw}) dH(z) \\ - db * \text{Cov}[\hat{g}(z), \int_{z_{min}}^z (y - \bar{z}^{rw})(s'(y) - s'_{inc}(y))]]$$

where β^M is the income-gradient of the marginal social cost of speeding (assumed constant). To calculate the revenue, efficiency, and redistributive consequences of a change in the slope from b to b^{alt} , we integrate the expression above.

10.3 Supplemental Tables

10.3.1 Setting

Table 17: Income-Gradient of Excess Speed

	Excess Speed				
	(1)	(2)	(3)	(4)	(5)
Earned Income (€10,000s)	-0.342*** (0.00358)	-0.0175*** (0.00371)	-0.0436*** (0.00366)	-0.00708** (0.00224)	-0.00959** (0.00348)
Constant	15.66*** (0.0164)	14.46*** (0.0166)	14.56*** (0.0163)	11.23*** (0.0103)	14.43*** (0.0155)
<i>N</i>	753881	753877	753876	603762	753876
Age-FE		X			
Age-Year FE			X	X	X
Camera Only				X	
Municipality FE					X

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 18: Triple-Difference Income Effect Estimates with First Inheritance Quintile as Differencing Group

	(1) Speeding Rate (pp)	(2) Inherit. Shock (10,000s €)	(3) Speeding Rate (pp)
DDD Coefficient	0.499*** (0.137)	1.118*** (0.0246)	
Inherit. Shock (10,000s €)			0.446*** (0.123)
Constant	7.622*** (0.0782)	0.00670*** (0.000831)	7.619*** (0.0785)
<i>N</i>	4912424	4912424	4912424
Control Group	Inherit=0	Inherit=0	Inherit=0
Model	Reduced-Form	First-Stage	DDD-IV

Standard errors (clustered at the individual level) in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 19: Cause-of-Death by Inheritance-Size Group

	(1) Motor-Vehicle Acc. (%)	(2) Transport-Related Acc. (%)
I(Top Group)	0.112*** (0.0105)	0.218*** (0.0174)
Constant	0.112*** (0.00748)	0.356*** (0.0124)
<i>N</i>	614053	614053

Standard errors (clustered at the individual level) in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 20: DDD-IV Income Effect Estimates, Excluding Transport-Related Causes-of-Death

	(1) Speeding Rate (pp)	(2) Inherit. Shock (10,000s €)	(3) Speeding Rate (pp)
DDD Coefficient	0.311*** (0.0818)	1.424*** (0.0118)	
Inherit. Shock (10,000s €)			0.219*** (0.0575)
Constant	7.361*** (0.0348)	0.00648*** (0.000552)	7.359*** (0.0349)
<i>N</i>	10491513	10491513	10491513
Control Group Model	No Transport Acc. Reduced-Form	No Transport Acc. First-Stage	No Transport Acc. DDD-IV

Standard errors (clustered at the individual level) in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 21: Speeding Rate Differences Across Inheritance-Size Groups

	Speeding Rate (pp) (1)
I(Bottom Group)	0.342*** (0.0851)
Constant	7.259*** (0.0575)
<i>N</i>	614053

Description

10.3.2 Preference Heterogeneity

10.3.3 Externalities

10.3.4 Fairness Preferences

10.4 Supplemental Figures

10.4.1 Setting

10.4.2 Preference Heterogeneity

10.4.3 Externalities

10.4.4 Fairness Preferences

Table 22: DDD-IV Income Effect Estimates with Controls

	Number of Heirs (1)	Number of Revisions (2)
Inherit. Shock (10,000s €)	0.279*** (0.0831)	0.258** (0.0876)
Constant	6.944*** (0.0944)	7.313*** (0.0541)
<i>N</i>	4912424	4912424
DDD-IV	X	X

Table 23: DDD-IV Income Effect Estimates: 5-year Effect Horizon

	(1) Reduced-Form	(2) First-Stage	(3) IV
DDD Coefficient	0.364*** (0.108)	1.219*** (0.0272)	
Inherit. Shock (10,000s €)			0.299*** (0.0892)
Constant	7.358*** (0.0401)	0.00471*** (0.000674)	7.356*** (0.0402)
<i>N</i>	10179538	10179538	10179538

Table 24: DDD-IV Income Effect Estimates: Expanded Control Group

	(1) Speeding Rate (pp)	(2) Inherit. Shock (10,000s €)	(3) Speeding Rate (pp)
DDD Coefficient	0.379*** (0.103)	1.512*** (0.0344)	
Inherit. Shock (10,000s €)			0.251*** (0.0684)
Constant	7.228*** (0.0400)	0.00744*** (0.000784)	7.226*** (0.0400)
<i>N</i>	10521577	10521577	10521577
Specification	Reduced-Form	First-Stage	DDD-IV

Standard errors (clustered at the individual level) in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 25: FE Estimates of s'_{inc} : Inheritance Sample

	(1)	(2)	(3)
	Speeding Rate (pp)		
Taxable Earned Income (10,000s €)	1.165*** (0.0133)	0.591*** (0.0248)	0.474*** (0.0267)
Re-scaled Labor Supply (10,000s €)			0.281*** (0.0227)
Constant	3.285*** (0.0389)	4.846*** (0.0677)	4.426*** (0.0751)
N	9593075	9593075	9593075
R^2	0.014	0.206	0.206
Top Inherit. Group	X	X	X
Cohort-Year FE	X	X	X
Individual FE			X

Standard errors (clustered at the individual level) in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 26: Redistributive Value: Sufficient Statistics

	Earnings Group							
	1	2	3	4	5	6	7	8
Panel (a): Earnings Effects & Cross-Sectional Variation								
<i>Inherit. Income Effect</i>	0.211 (0.0866)	0.211 (0.0866)	0.211 (0.0866)	0.211 (0.0866)	0.287 (0.105)	0.287 (0.105)	0.287 (0.105)	0.287 (0.105)
<i>FE Labor Supply Effect</i>	0.599 (0.0917)	0.281 (0.0619)	0.261 (0.0778)	0.283 (0.0852)	0.345 (0.0936)	0.277 (0.0840)	0.478 (0.0845)	0.167 (0.0576)
$s'_{inc}(z)$	0.810	0.492	0.472	0.494	0.632	0.564	0.765	0.454
$s'(z)$	0.333	0.513	1.328	2.753	1.675	1.784	0.890	0.376
Panel (b): Earnings Ability Tag								
$s'_{het}(z) := s'(z) - s'_{inc}(z)$	-0.477	0.021	0.856	2.259	1.043	1.220	0.125	-0.078
Panel (c): Additional Moments (2012)								
Avg. Earnings (€)	2350	13720	20405	25231	29943	35992	44927	72940
Avg. Disposable Income (€)	8143	14803	18600	21561	24707	28423	33764	50834
Speeding Rate (%)	4.368	4.747	5.090	5.731	7.028	8.041	9.635	12.129
Observed Fine Level (€)	173	188	196	202	208	215	226	261
$\bar{z}^{rw}$	€35,124							

Description

Table 27: Summary Statistics for Collapsed Accident Panel

	Mean (SD)
Earned Income (10,000s €)	2.576 (1.788)
Car Ownership Rate	0.508 (0.446)
Speeding Rate	0.0573 (0.104)
Ever Receive Speeding Ticket (%)	35.12 (47.73)
Other Traffic Crime Rate	0.0238 (0.0657)
Ever Receive Other Traffic Ticket (%)	18.17 (38.56)
Ever in Accident (%)	7.965 (27.08)
Ever in Injurious Accident (%)	1.765 (13.17)
Ever Injured in Accident (%)	0.798 (8.898)
<i>N</i>	5033598
Description	

Table 28: Correlations Between Year of Accident, Injuries, and Earnings

	(1)	(2)
	oth_p_did_tot_d5_wz	min_accident_year
min_accident_year	-14.57*** (2.585)	
earn_inc_p_t_10k		0.0328*** (0.00324)
_cons	29460.8*** (5197.6)	2010.9*** (0.0110)
<i>N</i>	363950	363950
Cohort FE	X	X

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Description

Table 29: Marginal Speeding Gradient (Other-Person Injuries)

(1)	(2)	(3)	(4)	(5)	(6)
Other-Person Injuries (per 1,000 drivers)					
10.70*** (0.198)	9.202*** (0.330)	10.58*** (0.199)	2.911*** (0.207)	1.498*** (0.254)	2.848*** (0.233)
-1.103*** (0.0445)	-0.837*** (0.0695)	-1.105*** (0.0445)	-0.463*** (0.0463)	-0.253*** (0.0593)	-0.503*** (0.0524)
0.145*** (0.00479)	0.107*** (0.0131)	0.157*** (0.00491)	0.148*** (0.00698)	0.176*** (0.0289)	0.227*** (0.0250)
0.338*** (0.0128)	0.680*** (0.0546)	0.316*** (0.0129)	-0.171*** (0.0163)	-0.124 (0.100)	-0.624*** (0.0854)
5033597	1588308	5033597	5033597	2223285	2776588
X	X	X	X	X	X
X	X	X	X	X	X
	X				
		X			
			X	X	X
				X	
					X

*Heteroskedasticity-robust standard errors in parentheses. Car ownership rates, non-speeding crime rates, driving distance, and vehicle mass are also interacted with earnings when included. Each individual is weighted by the number of years they appear in the population register. In column 5, this weight is multiplied by the number of years for which driving distance information is available from 2012-2018. In column 6, this weight is multiplied by the number of years for which vehicle mass information is available from 2013-2018. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

Table 30: Income Loss Gradients, Separately by ν

	(1)	(2)	(3)	(4)
	did_net_income_t	did_net_income_t_ww_025	did_net_income_t_ww_1	did_net_income_t_ww_4
net_income_p_t_pre	3.707*** (0.591)	2.497*** (0.230)	1.826** (0.601)	-1.176 (1.363)
_cons	-46385.8*** (9882.7)	-30074.9*** (3743.1)	-39834.9* (16552.4)	45111.5 (49712.7)
<i>N</i>	100761	99803	99803	99803
<i>R</i> ²	0.027	0.028	0.007	0.004
Coeff. RRA	0	0.25	1	4
Age-Year FE	X	X	X	X

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Description

Table 31: Conditional Injury Gradients, Separately for Fatal- and Non-Fatal Injuries

	(1)	(2)
	Non-Fatal	Fatal
Pre-Period Earnings (10,000s €)	3.269*** (0.422)	0.129 (0.0687)
Constant	134.8*** (1.278)	4.267*** (0.208)
<i>N</i>	379382	379382
<i>R</i> ²	0.002	0.001
Age FE	X	X
Year FE	X	X

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Description

Table 32: Marginal Speeding Cost Gradients by Discount Rate and Inequality Aversion: Other-Person Victims

	Discount Rate		
	3%	5%	7%
Welfare Weight			
$\nu = 0$	-21.45* (8.425)	-18.45* (7.302)	-15.99* (6.398)
$\nu = 0.25$	-18.06* (8.146)	-15.55* (7.079)	-13.5* (6.215)
$\nu = 1$	-9.027 (4.809)	-7.74 (4.193)	-6.703 (3.696)
$\nu = 4$	-9.064 (4.783)	-7.761 (4.169)	-6.713 (3.674)
N	5033597	5033597	5033597
Cohort FE	X	X	X
Reverse Causality Controls	X	X	X
Car Ownership & Other Traffic Crime Controls	X	X	X
Annual Driving Distance	X	X	X

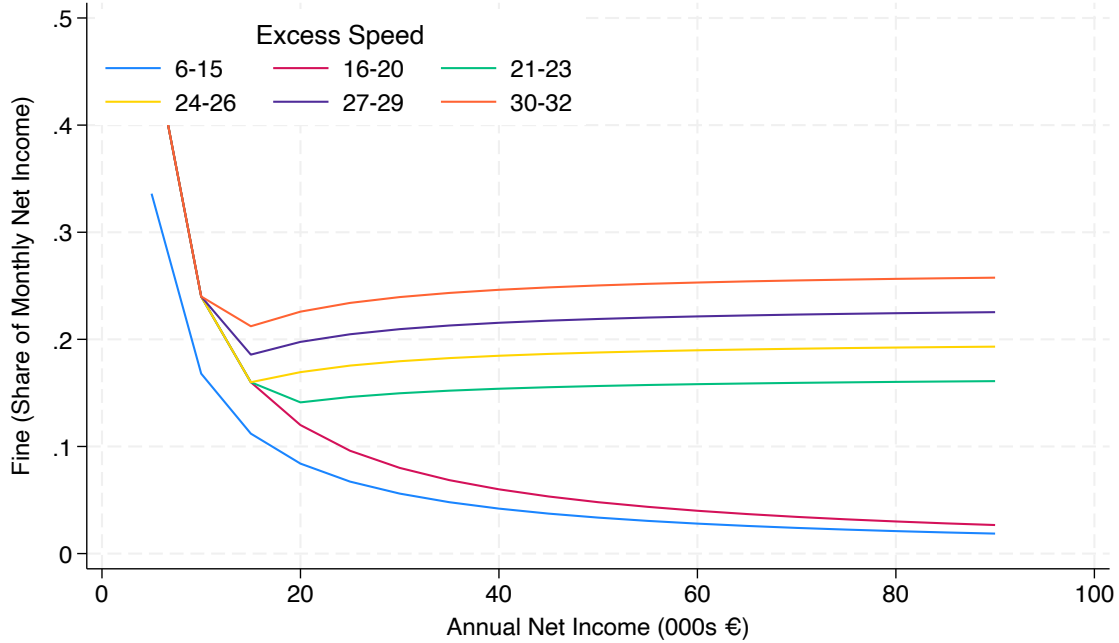
Heteroskedasticity-robust standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 33: Extensive Margin Regressions

	(1) $Pr(WTP^{\text{net}} > 0)$ (pp)
Proportional Fines & Equal Speeding	62.93*** (3.811)
Behavior	
$\Delta s_{\text{below}} = 0.01$	-2.503* (0.978)
Fine Schedule	
Below-Proportional Fines ($\Delta = \text{€}72.5$)	-3.805 (2.322)
Above-Proportional Fines ($\Delta = \text{€}72.5$)	0.937 (2.357)
N	2727

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ **Figure 18:** Hypothetical Fine Schedule (100 km/h limit)

Net-of-tax income consists of earned and capital income, as well as any transfers received, after the payment of taxes. We assume no dependents for the formula in equation 1.

Table 34: Survey Respondent Summary Statistics

	(1)	(2)
	All Responses	Main Sample
Income Group (%)		
0 € - 14,999 € (1st quintile)	20.3	19.6
15,000 € - 24,999 € (2nd quintile)	16.3	16.4
25,000 € - 34,999 € (3rd quintile)	20.9	21.0
35,000 € - 49,999 € (4th quintile)	22.3	22.7
50,000 € or above (5th quintile)	20.2	20.2
Education completed (%)		
Primary Education or lower	7.0	6.2
Secondary Education	48.1	47.7
Higher Education	44.9	46.1
Age group (%)		
18-34 (1st quartile)	27.1	26.3
35-50 (2nd quartile)	26.0	25.9
51-65 (3rd quartile)	21.3	21.3
Over 65 (4th quartile)	25.6	26.4
Gender (%)		
Man	49.9	49.3
Woman	49.3	49.9
Other/decline to answer	0.8	0.8
Driving frequency (%)		
Never	13.0	12.3
Less than once per month	10.6	10.4
Monthly	6.8	6.8
Weekly	69.5	70.5
I(Speeding ticket in last 2 years) (%)	7.27	7.00
HH Car Ownership Rate (%)	79.3	79.9
Median completion time (mins.)	12.9	13.05
<i>N</i>	3038	2727

Table 35: Values for Survey Random Variables

Feature	Randomization
τ_{fixed}	Unconditional
$\Delta \bar{r}_{fixed}$	Conditional ($\Delta \bar{r}_{fixed} > 0$)
$\bar{r}_{fixed}$	Unconditional
$d\tau^{above}$	Conditional ($d\tau^{above} > 0$)
$-d\tau^{below}$	Conditional ($-d\tau^{below} < 0$)
$-d\bar{r}^{above}$	Conditional ($-d\bar{r}^{above} < 0$)
$d\bar{r}^{below}$	None ($d\bar{r}^{below} = d\bar{r}^{above}$)

Table 36: Heterogeneity by Background Characteristics

	Outcome: Net WTP									
	Income		Fine Awareness Beliefs		Fine Responsiveness Beliefs		Age		Education	
	(1) <35,000 €	(2) ≥35,000 €	(3) Unaware	(4) Aware	(5) Unresponsive	(6) Responsive	(7) ≤ 50	(8) > 50	(9) Secondary Educ.	(10) Higher Educ.
Proportional Fines & Equal Speeding	52.19*** (14.90)	38.59* (17.51)	45.76* (18.08)	44.71** (14.60)	34.32 (17.81)	53.96*** (14.68)	32.46* (14.83)	61.11*** (17.50)	38.65* (16.63)	52.54** (17.05)
Behavior										
$\Delta s_{below}^{inc} = 0.01$ ($s_{below}^{inc} < s_{above}^{inc}$)	1.240 (9.572)	-10.72 (11.11)	6.412 (11.61)	-10.27 (9.313)	-2.134 (11.37)	-5.123 (9.376)	-14.33 (9.405)	9.088 (11.14)	-7.467 (10.43)	1.205 (10.82)
$\Delta s_{below}^{inc} = 0.01$ ($s_{below}^{inc} = s_{above}^{inc}$)	-11.07 (9.434)	-2.505 (11.35)	-10.46 (11.79)	-4.758 (9.216)	-16.16 (11.26)	0.907 (9.463)	-1.558 (10.03)	-13.72 (10.82)	-6.580 (10.48)	-8.769 (11.02)
$\Delta s_{below}^{fixed} = 0.01$	-5.707 (7.712)	-12.60 (9.092)	-9.041 (9.411)	-7.361 (7.553)	-14.56 (9.147)	-3.927 (7.650)	-3.668 (7.817)	-13.91 (8.924)	-0.723 (8.370)	-15.84 (8.912)
Fine Schedule										
$\Delta \tau_{below} = \text{€}72.5$ (Below-Proportional)	-6.203 (9.487)	-1.836 (11.00)	-2.291 (11.46)	-7.086 (9.223)	-15.94 (11.07)	4.450 (9.409)	-9.195 (9.350)	0.299 (10.97)	-12.59 (10.60)	-1.197 (10.67)
$\Delta \tau_{below} = \text{€}72.5$ (Proportional)	-2.974 (9.617)	-8.436 (11.20)	9.145 (11.77)	-13.78 (9.293)	21.97 (11.33)	-24.52** (9.494)	-7.855 (9.559)	-0.887 (10.85)	4.913 (10.37)	-19.78 (10.91)
N	1555	1172	1110	1617	1117	1610	1425	1302	1300	1257

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 19: Binscatter Relationships Between Income and Speeding Crime Characteristics

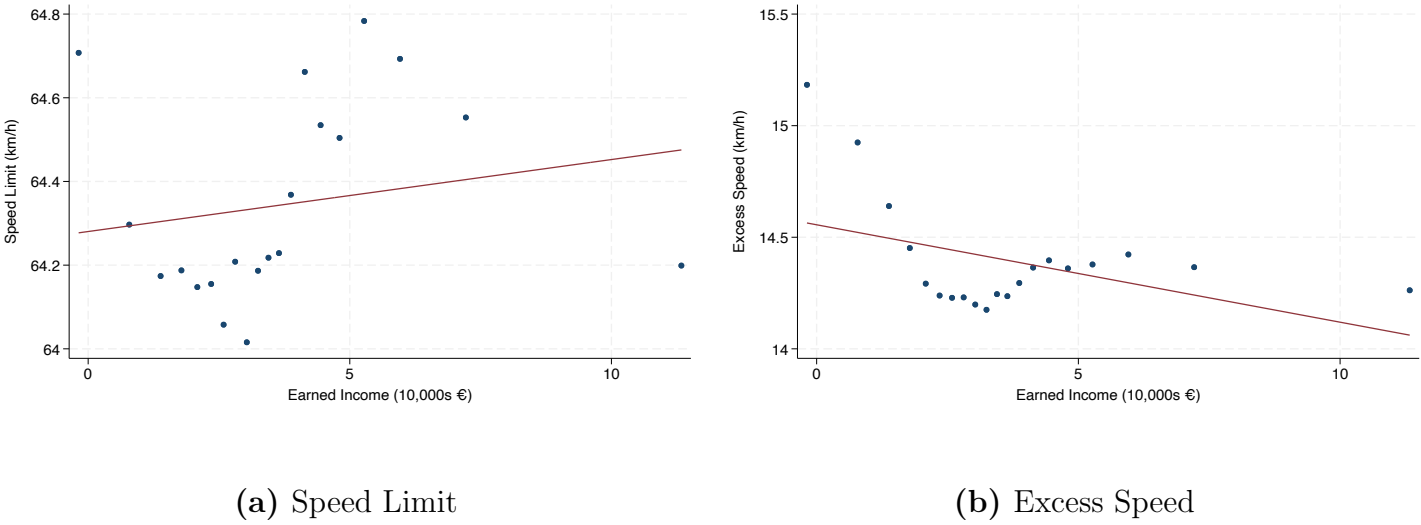
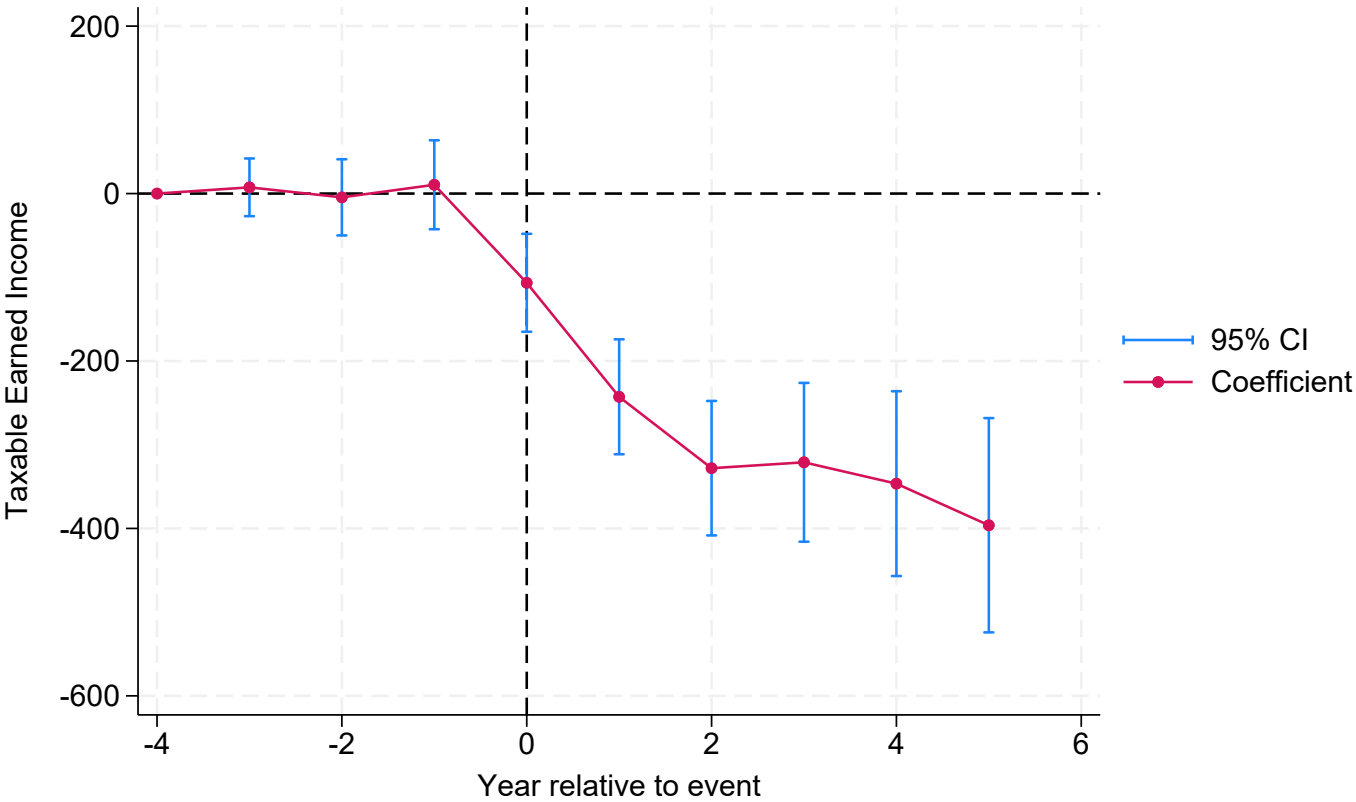
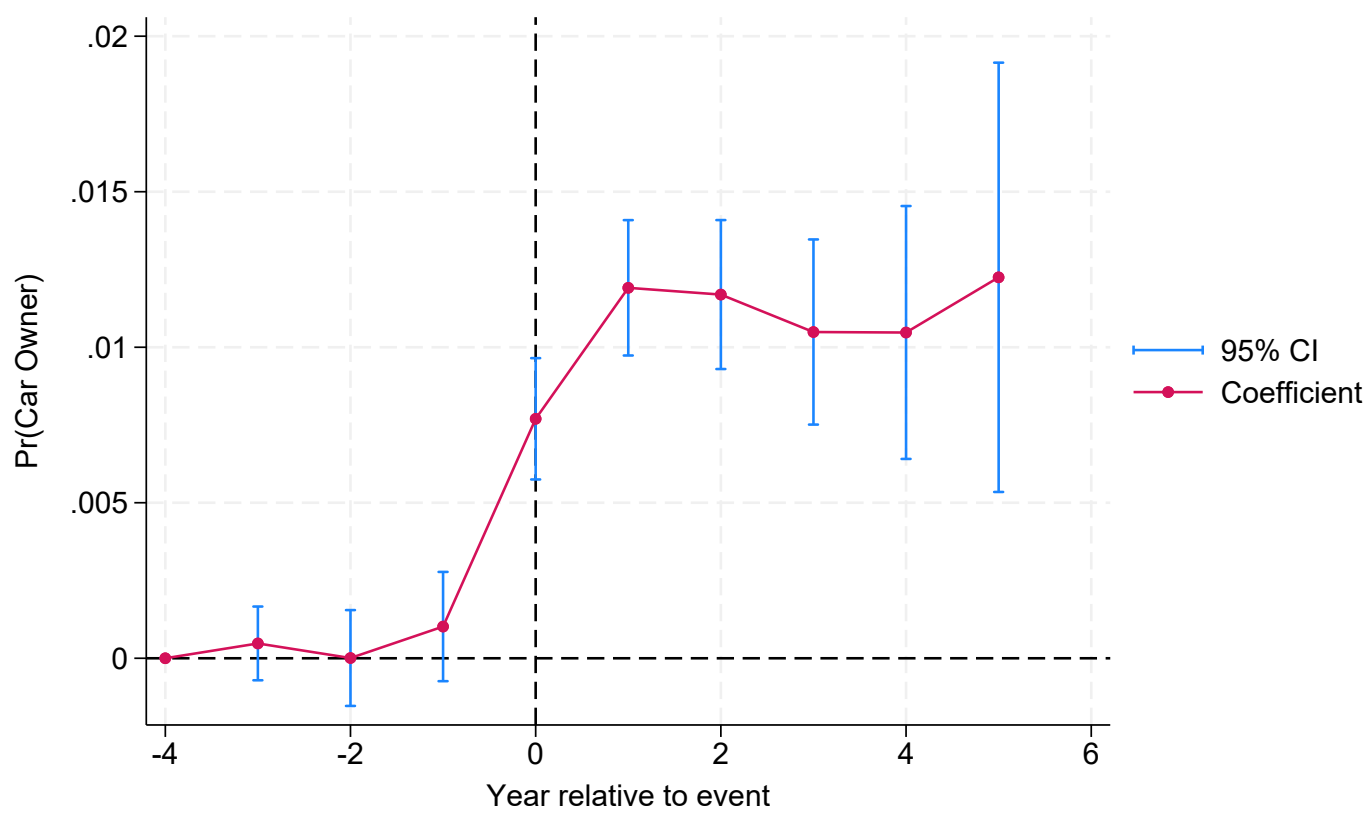


Figure 20: Event Study of Taxable Earned Income around Parental Death



Caption

Figure 21: Event Study of Car Ownership around Parental Death



Caption

Figure 22: Event Study of Speeding Rates around Parental Death: First Inheritance Quintile as Bottom Group

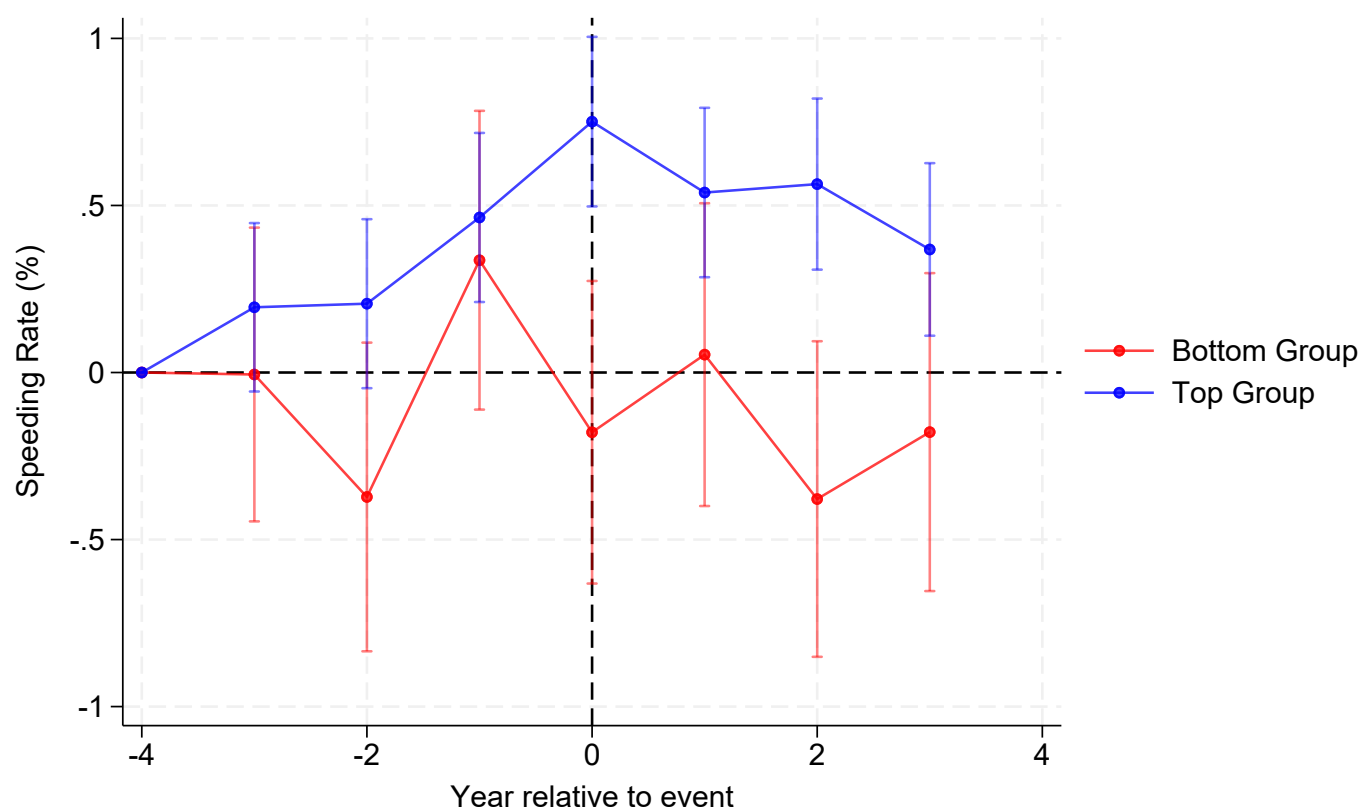


Figure 23: Triple-Difference Effects on Earnings

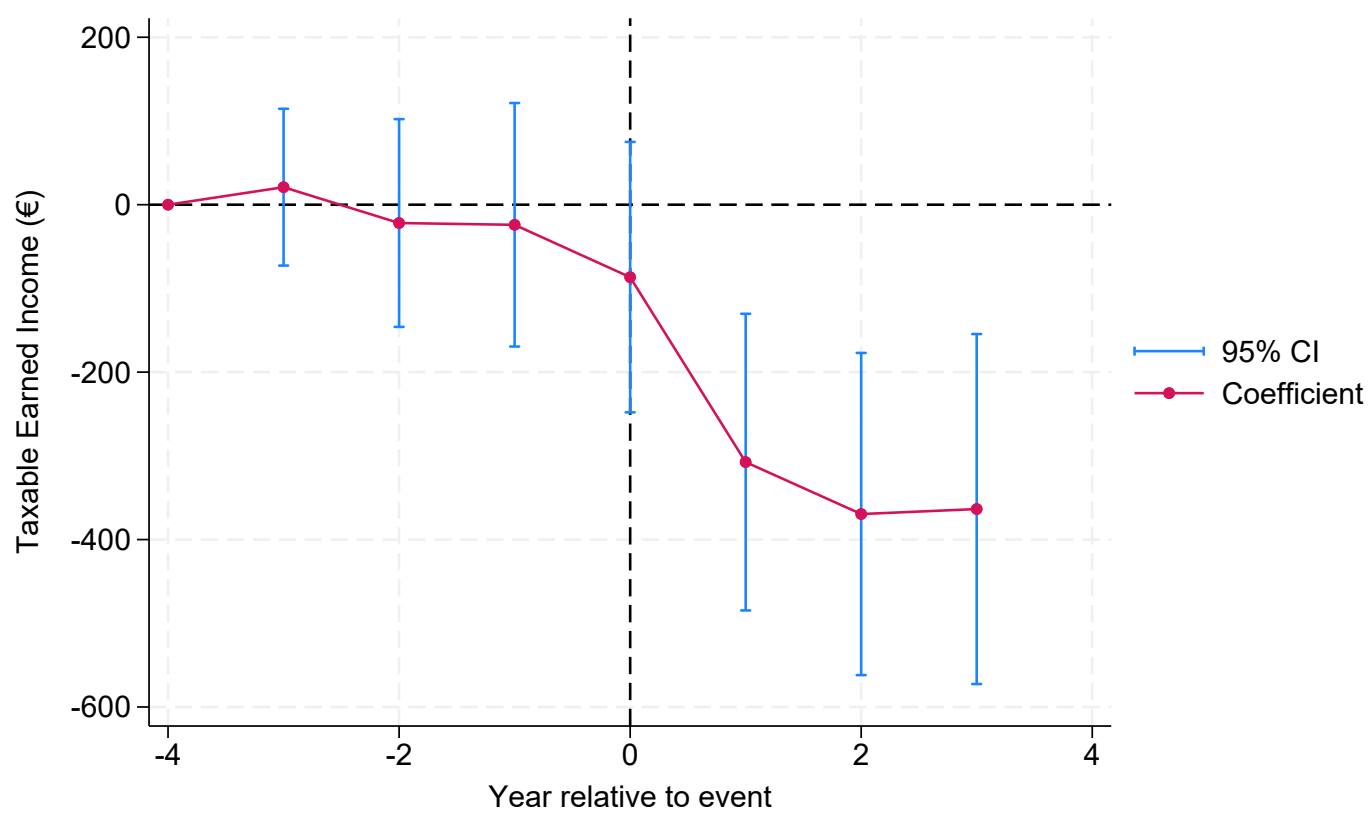


Figure 24: Triple-Difference Effects on Labor Supply

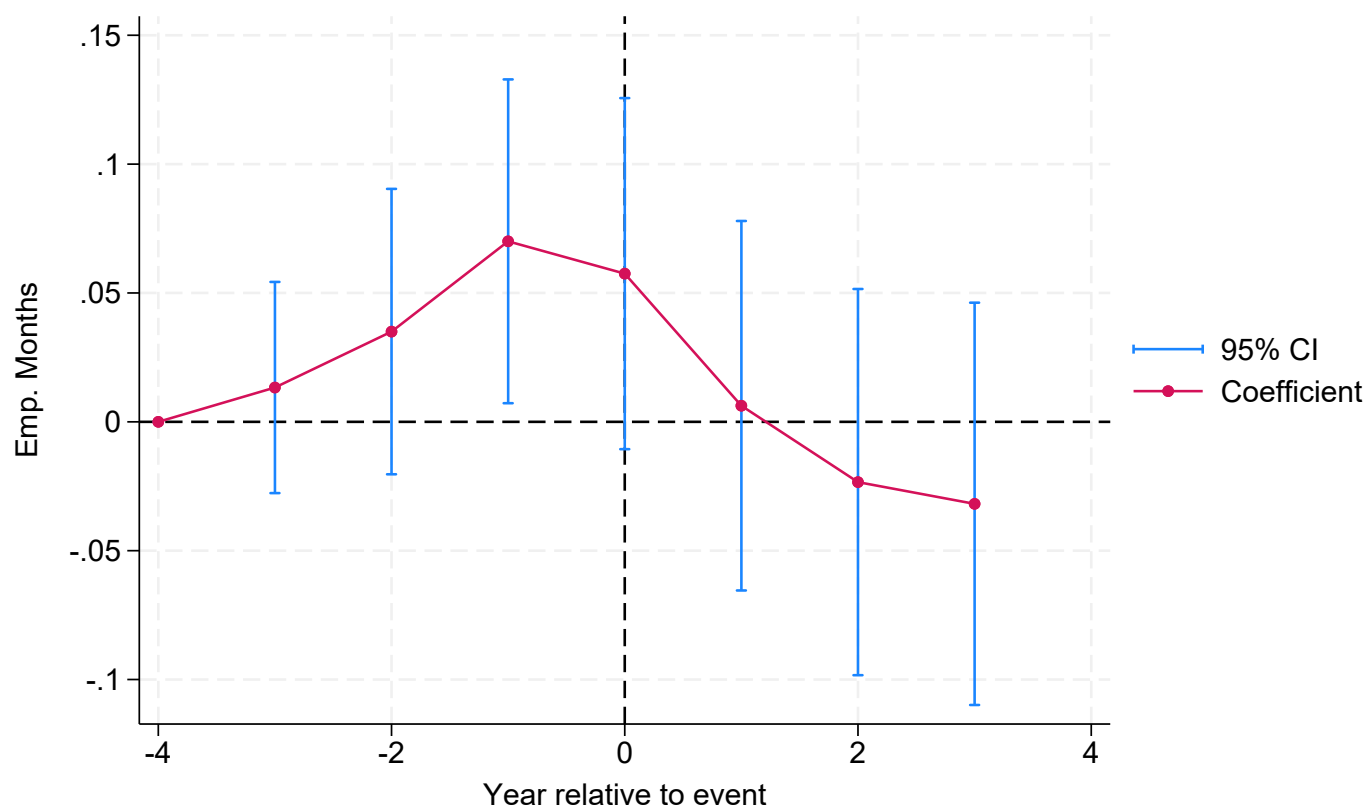
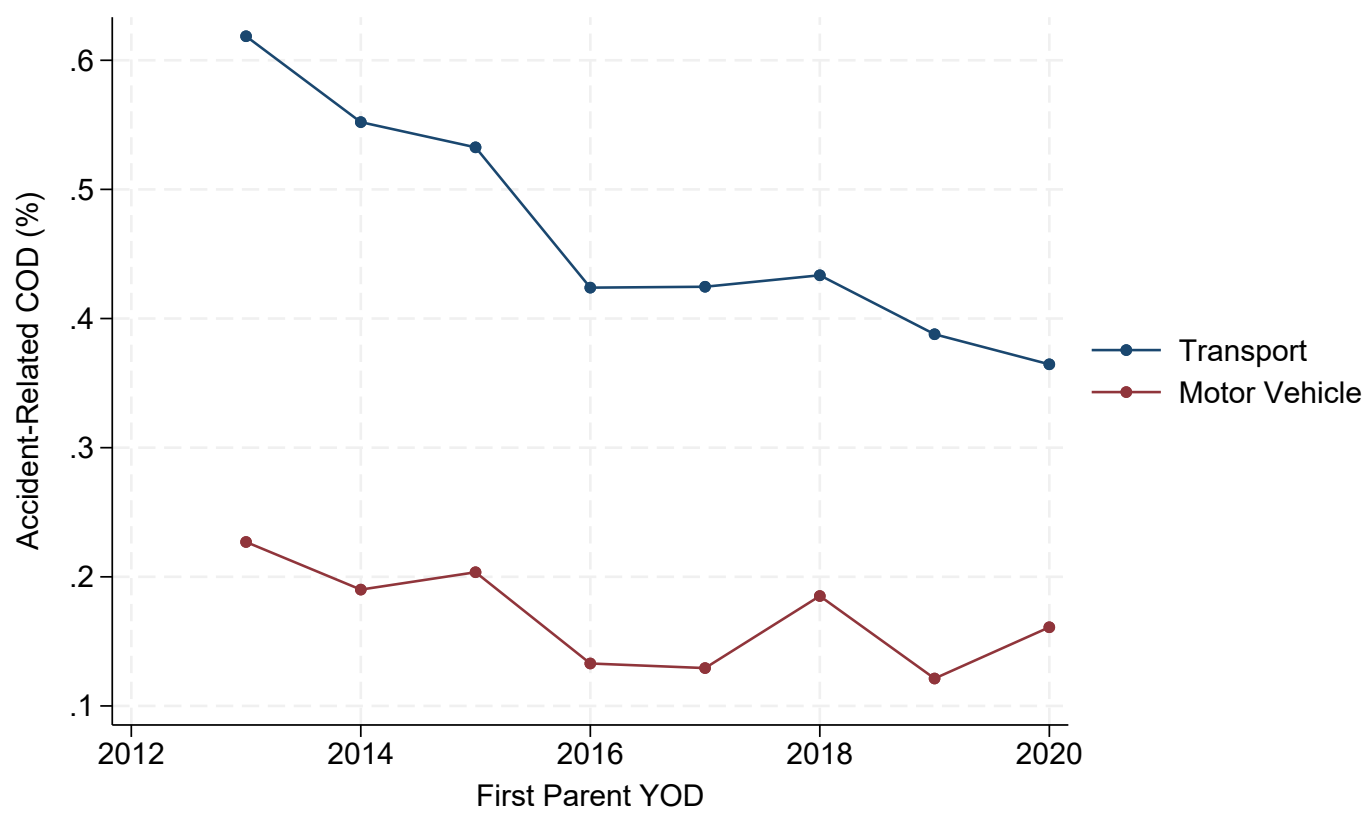
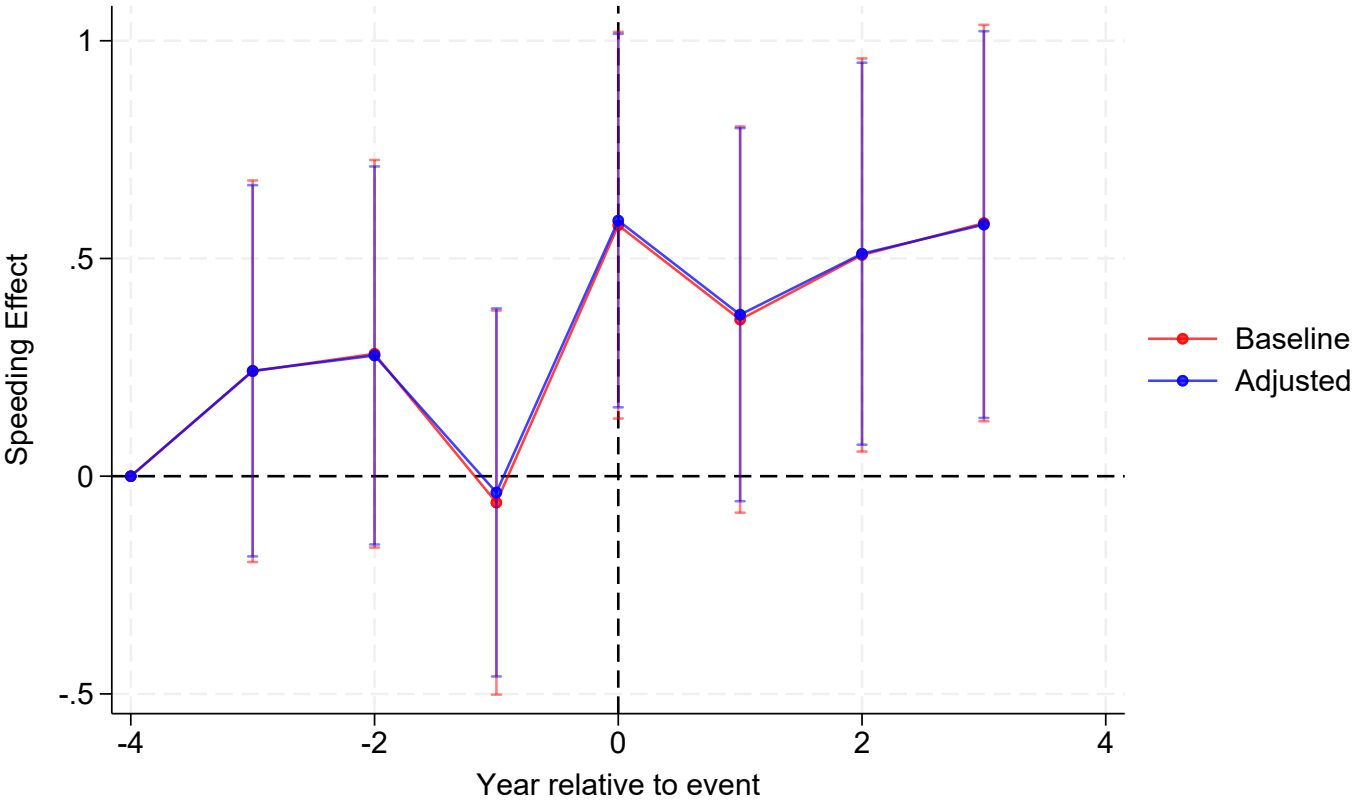


Figure 25: Cause-of-Death by Parental Death Cohort



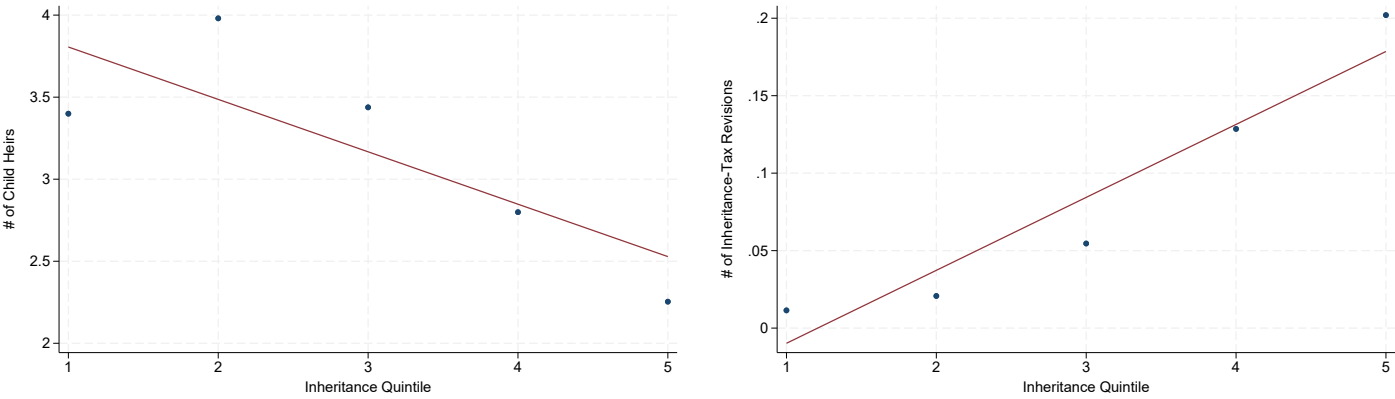
Caption

Figure 26: DDD Event Study of Speeding Rates around Parental Death: Relative vs. Level Effects



Caption

Figure 27: Potential Child Heirs and Tax Revisions by Inheritance Quintile



(a) Child Heirs

(b) Tax Revisions

Description

Figure 28: DDD Event Study of Speeding Rates around Parental Death: 5-year Event Horizon

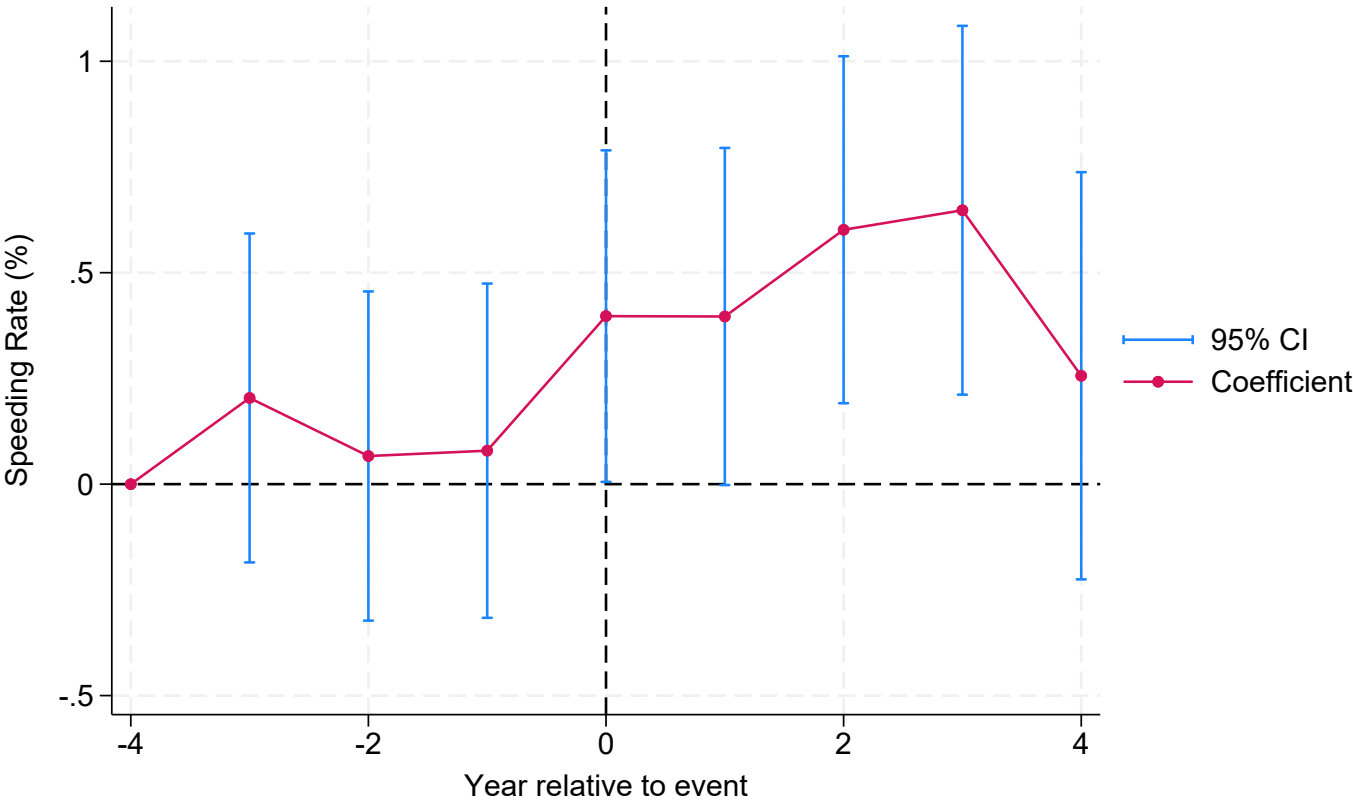


Figure 29: Relationship between Earnings and Speeding Rates (2012)

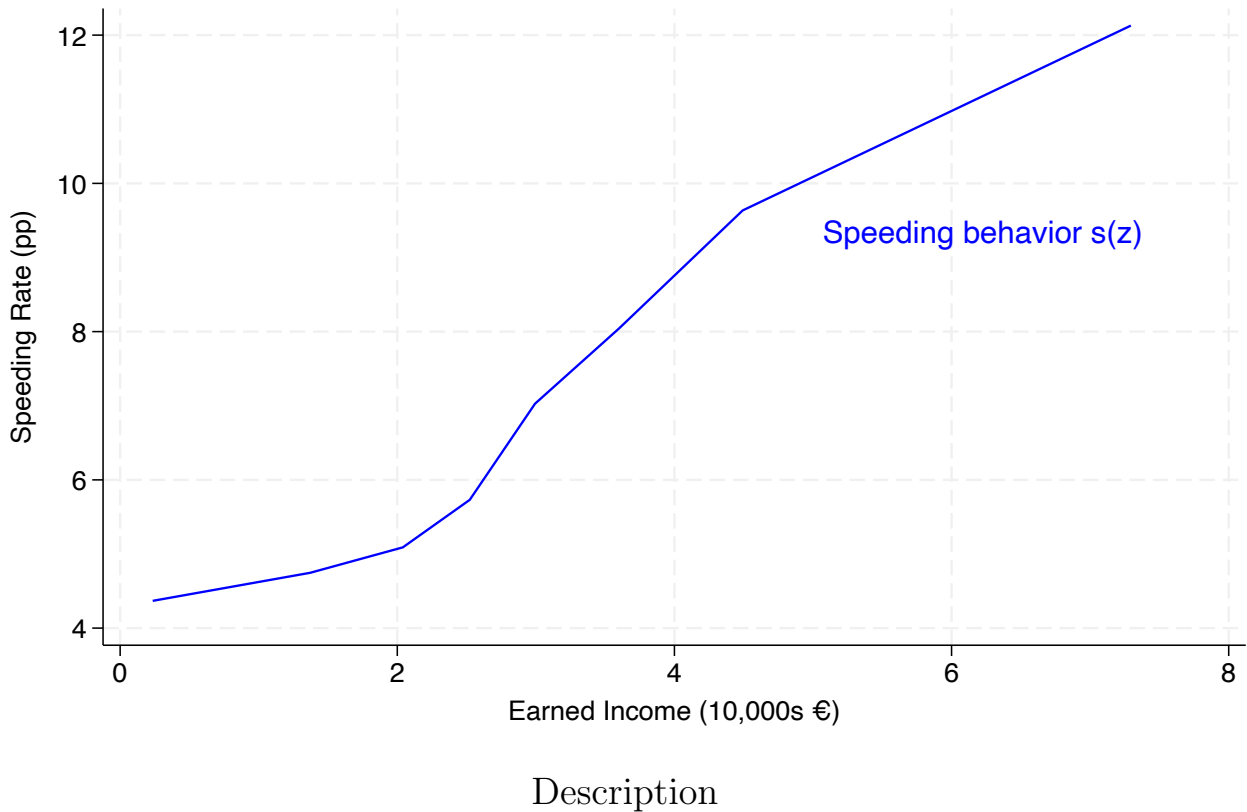


Figure 30: Heatmap of Optimal Redistributive Gradients

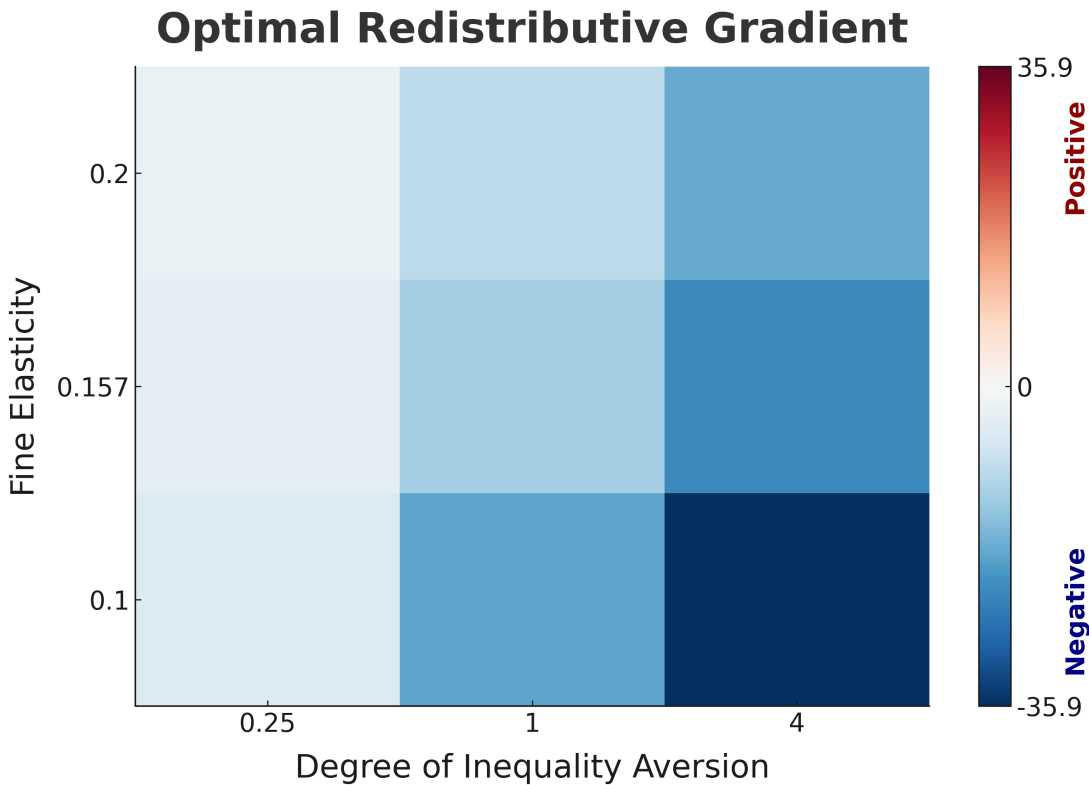
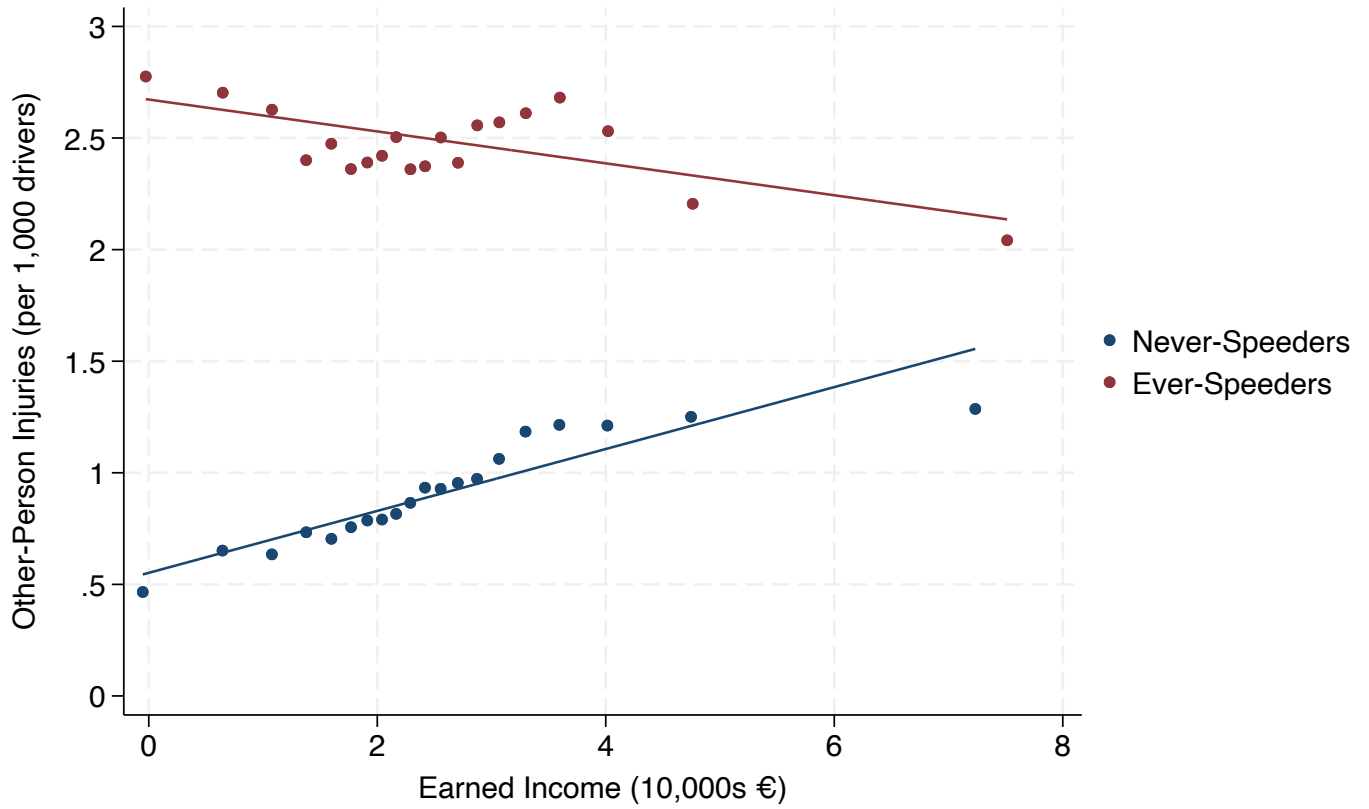
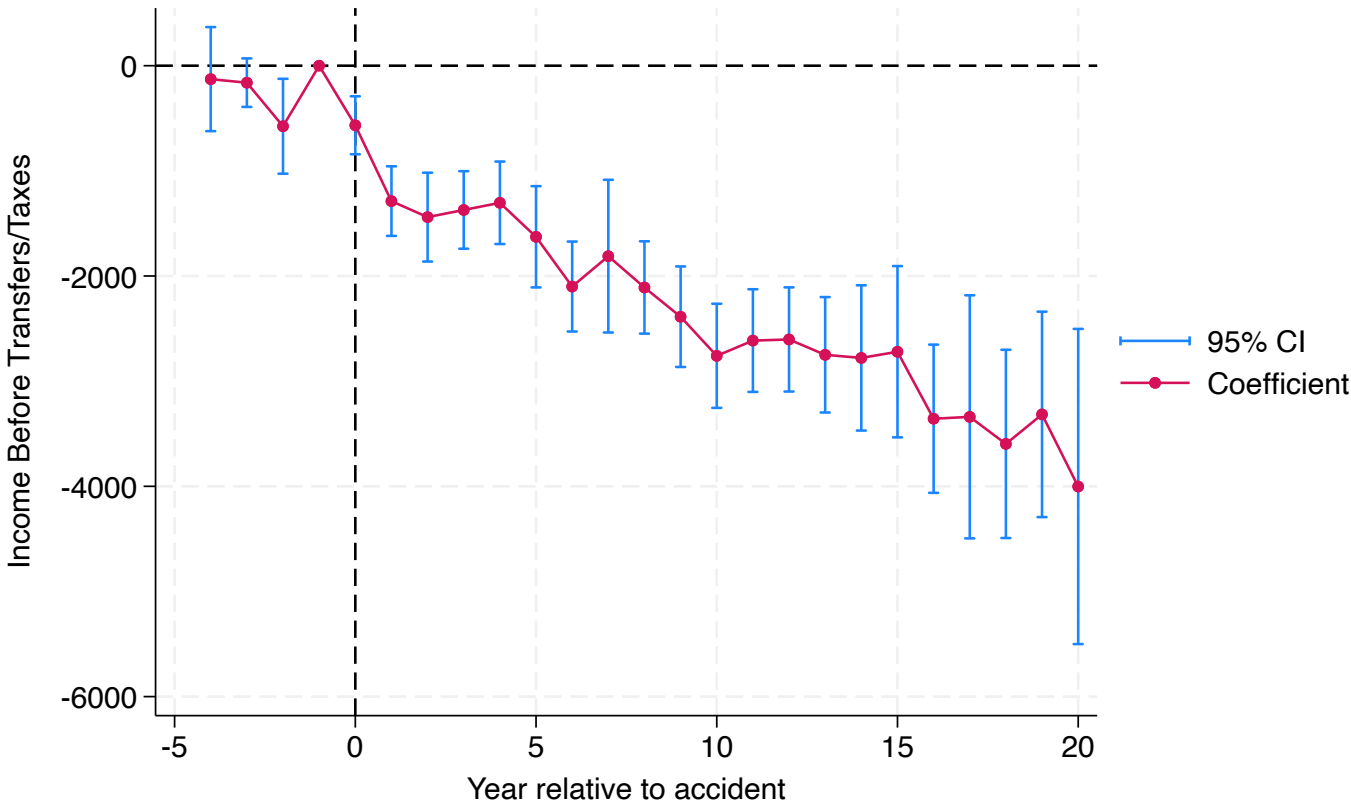


Figure 31: Expected Other-Person Injuries, by Income and Speeding History



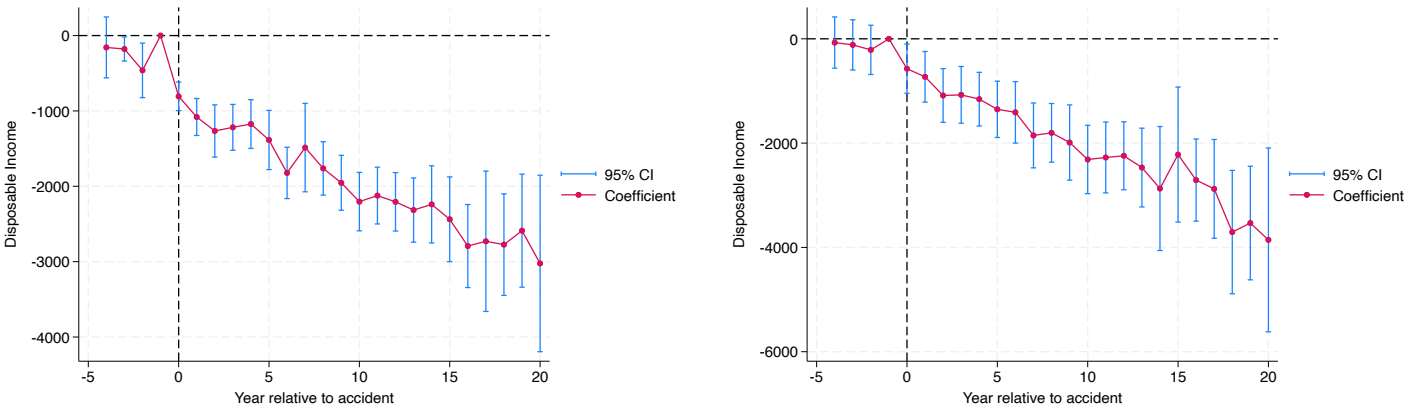
Binscatter relationships between average taxable earned income and average other-person injuries using the dataset described in section 5.1. Income and other-party injuries are residualized on birth-cohort fixed effects, and each individual is weighted by the number of years they appear in the population register from 2006-2018.

Figure 32: Effect of Accident Injury on Income Before Taxes and Transfers



Caption

Figure 33: Event Study Effects of Injury/Death on Disposable Income

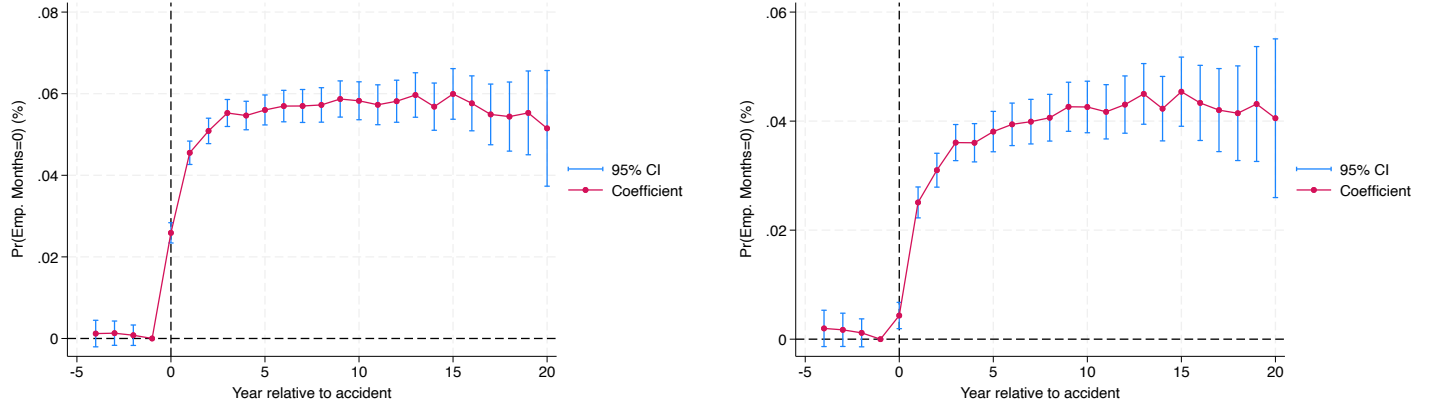


(a) Drivers

(b) Passengers

Description

Figure 34: Event Study Effects of Injury on Probability of Working

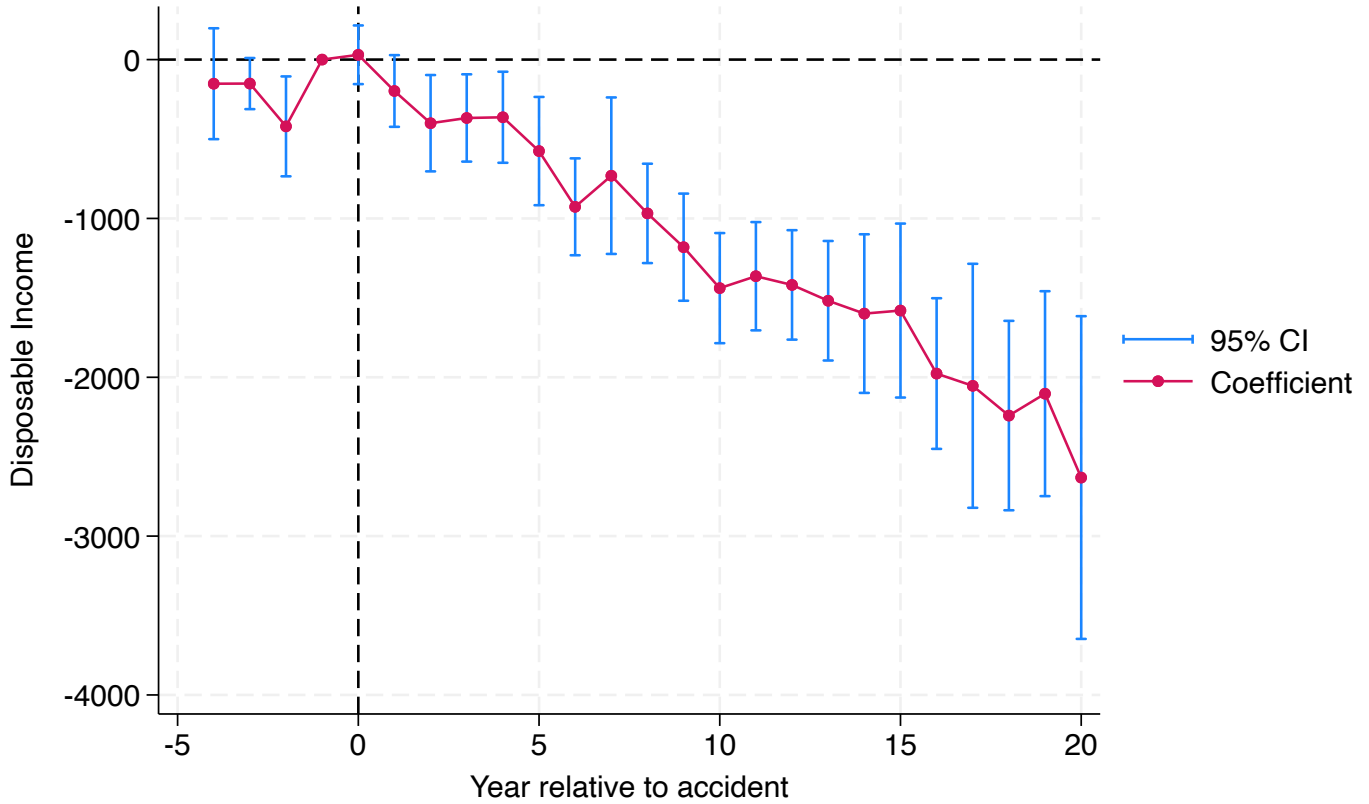


(a) Fatal and Non-Fatal Injuries

(b) Non-Fatal Injuries

Estimates of β_k from equation 5. The sample includes all accident-related injuries (panel a) or all non-fatal accident-related injuries (panel b) from 2003-2022, restricted to accidents that include a motor vehicle (cars, trucks, buses, vans, mopeds, motor bikes, and motorcycles). Injuries from accidents involving a snow mobile, trailer, tram, train, tractor, or other unidentified motor vehicles are excluded.

Figure 35: Event Study Effects of Non-Fatal Injury on Disposable Income



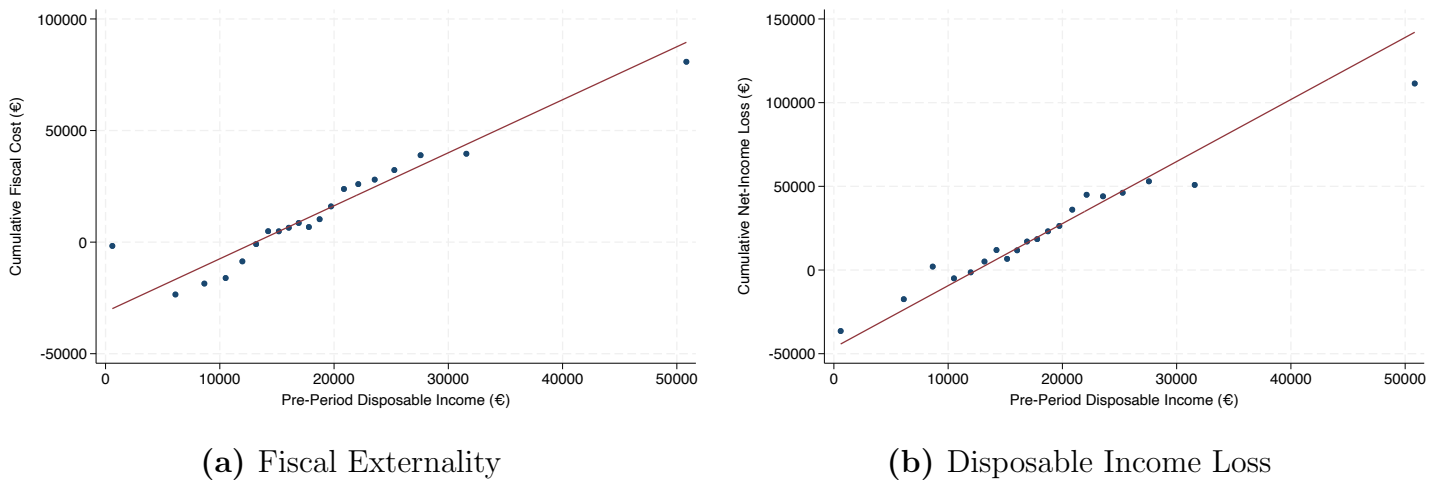
Estimates of β_k from equation 5. The sample includes all non-fatal accident-related injuries from 2003-2022, restricted to accidents that include a motor vehicle (cars, trucks, buses, vans, mopeds, motor bikes, and motorcycles). Injuries from accidents involving a snow mobile, trailer, tram, train, tractor, or other unidentified motor vehicles are excluded.

Figure 36: Event Study Effects of Non-Fatal Injuries on Social Costs



Description

Figure 37: Cumulative Social Costs of Injury/Death, by Pre-Period Income

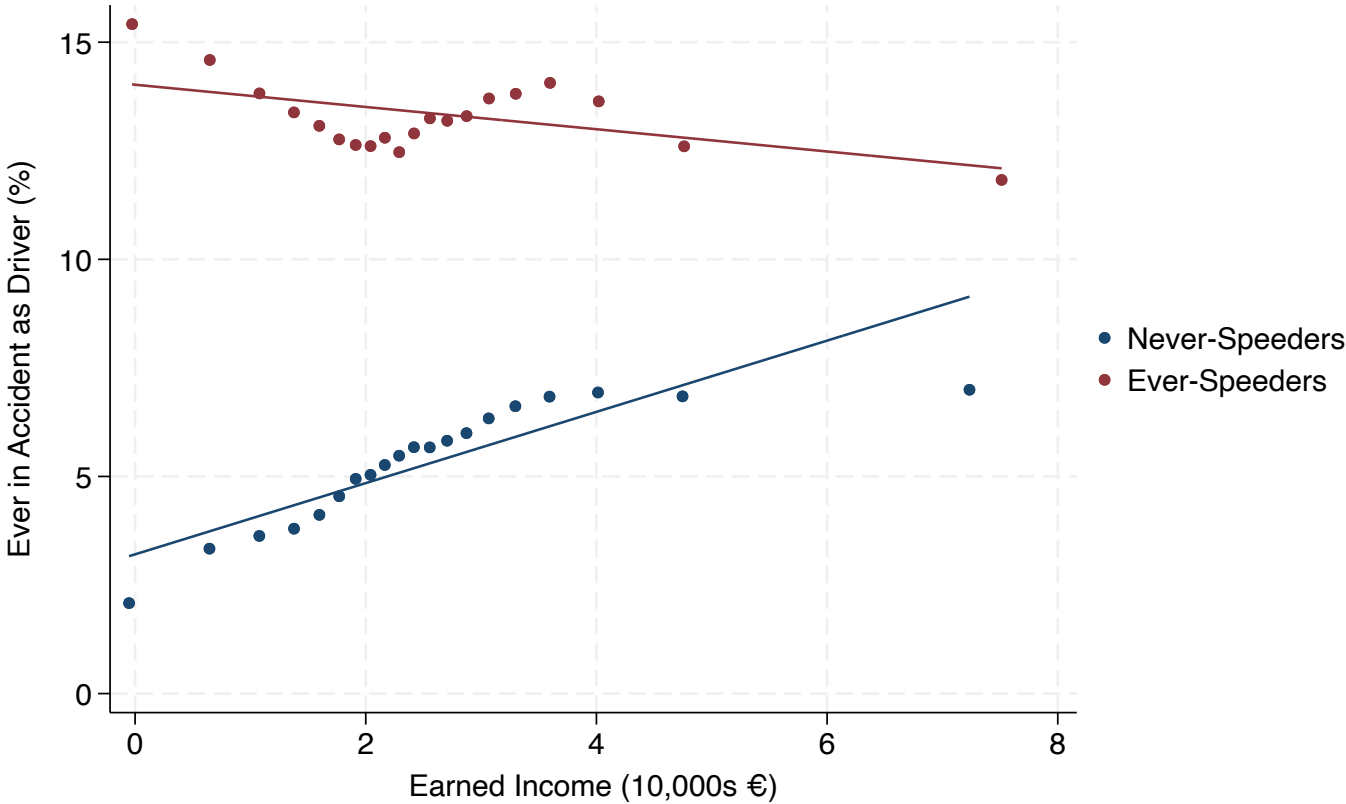


(a) Fiscal Externality

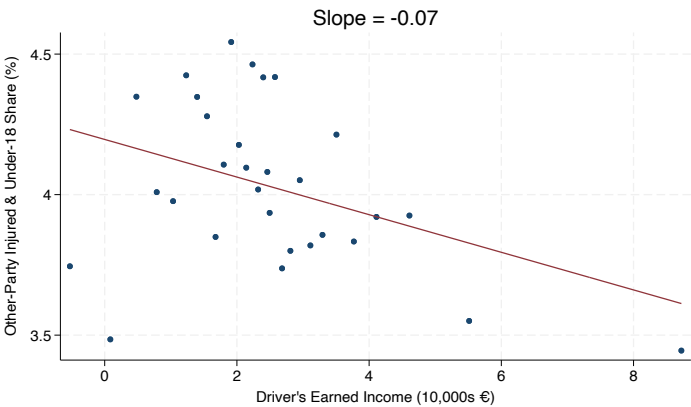
(b) Disposable Income Loss

Description

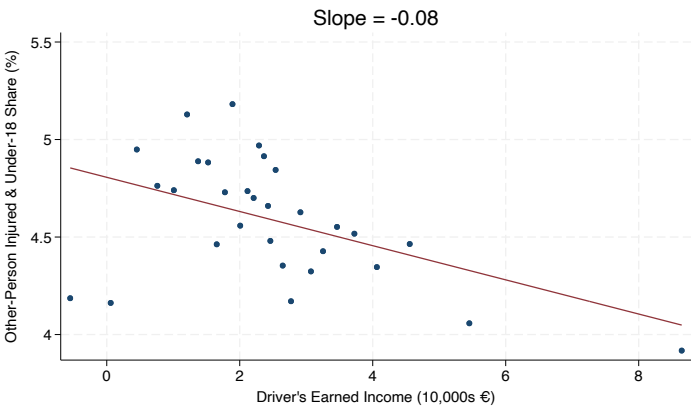
Figure 38: Percent of Individuals Ever in an Accident as Driver, by Speeding History



Description



(a) Other-Party



(b) Other-Person

Description

Figure 40: Slider for Fine Policy Comparison

Consider options A and B below.

<u>Option A</u>	<u>Option B</u>
Income-based fines are implemented	Fixed fines are implemented and Savings of Y € per adult (X million € in total) annually

The tables below present the two fine policies again. What would be the minimum annual savings required for you to choose option B?

I would choose option B if the savings were
at least Y € per adult (X million € in total) annually



*(Remember, the **fine YOU would have to pay is not determined by the policies described below.** If the smallest total savings that would make you switch is greater than the values shown on the slider, choose the highest possible value.)*

Figure 41: Transfer Comparison Question

Imagine that the Finnish government receives a **one-time** unexpected revenue increase. The government plans to distribute some of this extra revenue to the Finnish adult population in the form of cash transfers. There are two alternative systems for distributing the funds:

1. **Fixed cash transfers** - every adult living in Finland receives the same amount.
2. **Income-based cash transfers** - the amount distributed to a person depends on whether they belong to a low- or high-income group.

We ask you to choose **which system would be better for Finland overall**. Suppose that your choice between the two ways of distributing the funds **wouldn't affect you: YOU would not receive a cash transfer in either case**.

The table below shows the two ways of distributing the funds described above. **Each income group covers half of the adult Finnish population. The low-income group includes the half of the population with the lowest incomes; the high-income group includes the half with the highest incomes.**

	<u>Cash transfer to each individual in a group</u>	
<u>Income Group</u>	Fixed cash transfers	Income-based cash transfers
Low (<u>Average Income:</u> 15 000 €)	25 €	32 €
High (<u>Average Income:</u> 51 000 €)	25 €	20 €

Which system do you prefer?

Please note that the systems differ in how much cash they distribute to different income groups. In addition, the systems do not necessarily distribute the same amount of transfers in total. The systems are otherwise completely similar.

Figure 42: Frequency of Fine Awareness Beliefs

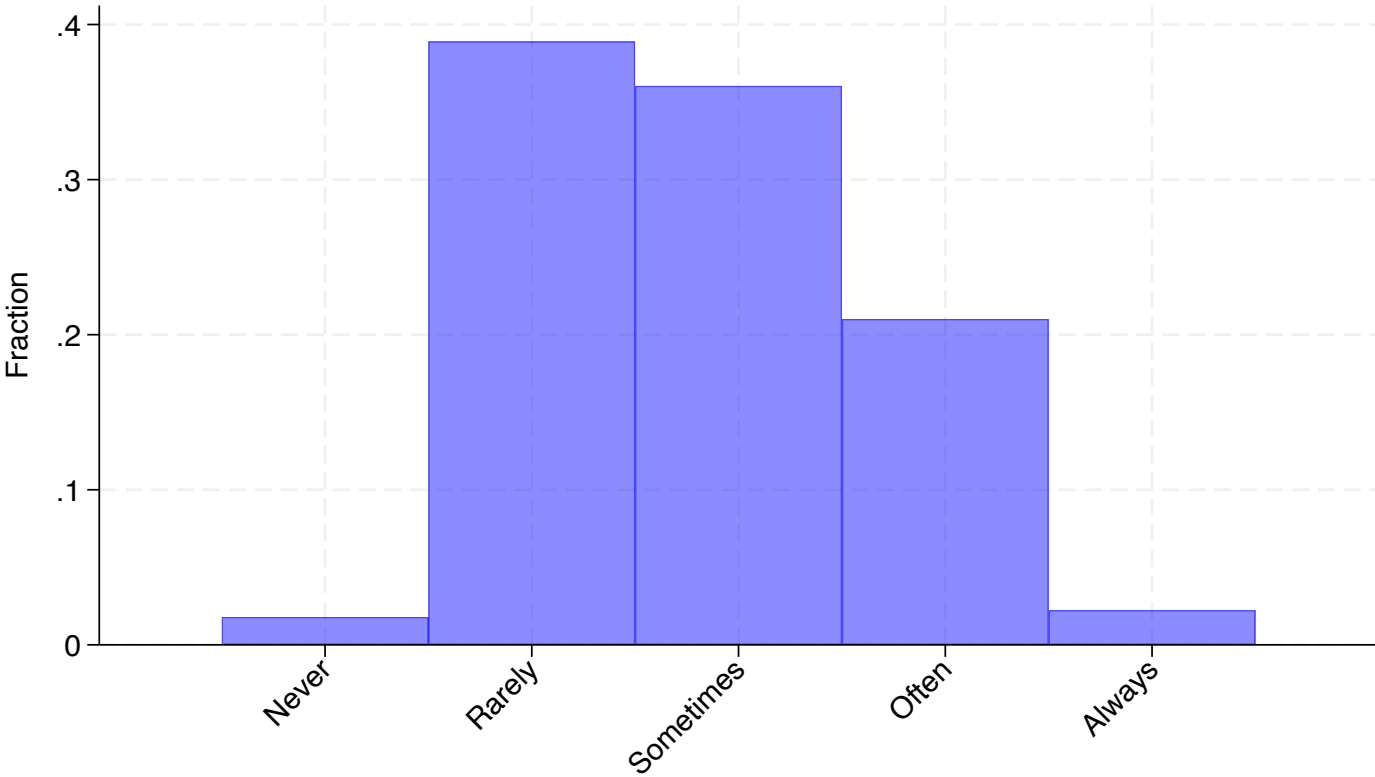


Figure 43: Frequency of Fine Responsiveness Beliefs

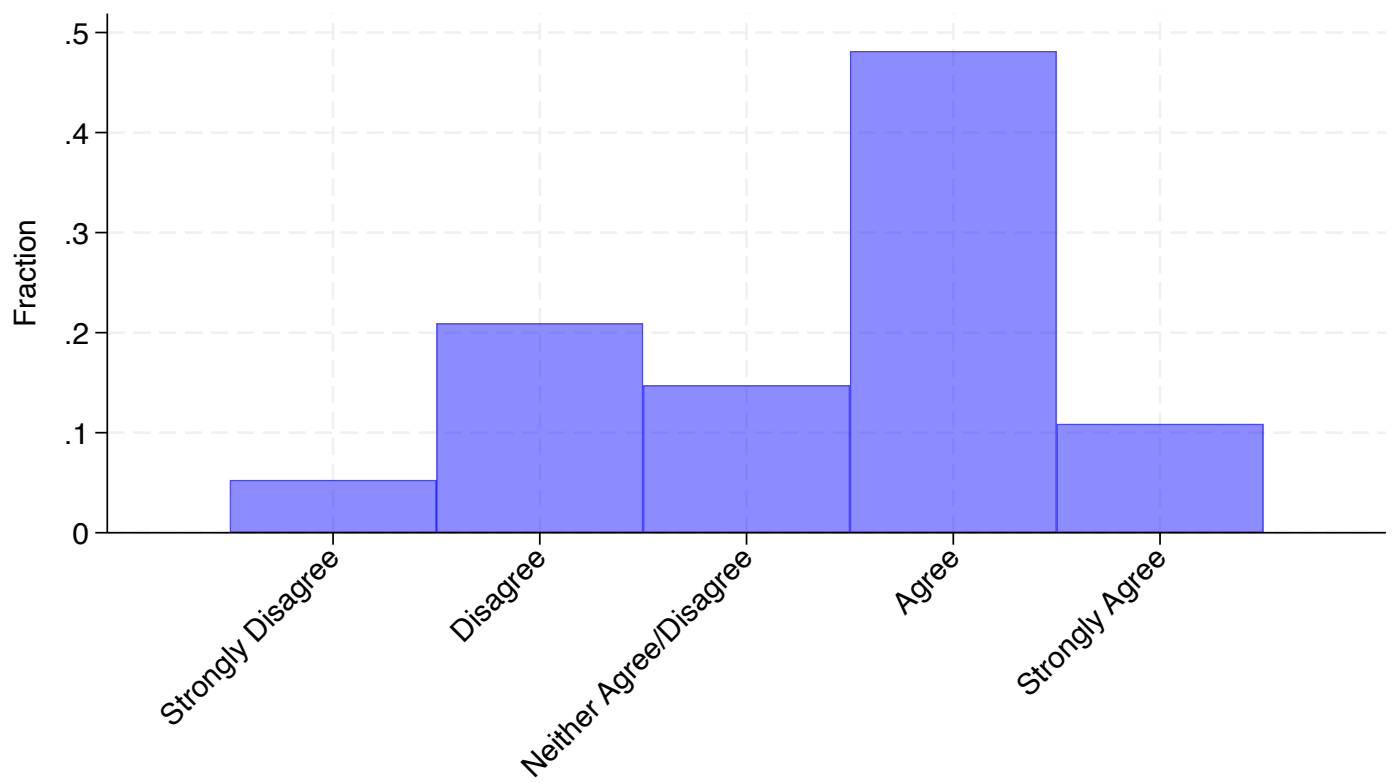


Figure 44: Assigned Fine, by Hypothetical Income and Excess Speed

